

REINFORCE Pro Max: Adaptive Advantage Normalization with Causal Prefix Filtering for LLM-RL

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Abstract

Critic-free reinforcement learning for large language models avoids a value network, but token-weighted advantage imbalance and inference–training policy mismatch can destabilize the sampled update. We present REINFORCE Pro Max, combining a leave-one-out (RLOO) baseline with adaptive asymmetric advantage normalization and causal prefix filtering. The ideal normalization preserves advantage signs while enforcing zero empirical mean and unit empirical variance on nonzero tokens; its implemented numerical safeguards admit explicit moment residuals. For binary RLOO rewards, its ideal scale ratio equals the class-conditional mean response-length ratio. Under stated deviation conditions, prefix filtering propagates early rejection and confines late rejection to a suffix, with additional bounds on accepted prefix mass. We analyze the sparse-terminal-reward core without KL shaping or post-RLOO reward clipping on a finite autoregressive model with bounded rewards and mutual policy support. An exact decomposition connects the implemented clipped objective to the raw critic-free surrogate, retaining the frozen snapshot ratios, prefix gate, normalization, and random response lengths. Combining this identity with the Adaptive trust-region bound yields a finite-sample sufficient condition for true reward improvement. An observed token-count rescaling removes the unknown mean response length, while an exact shared-baseline RLOO variance formula separates response-sampling uncertainty from prompt-sampling uncertainty and quantifies the benefit of additional responses per prompt. The guarantee supports selection from a fixed finite candidate family, and extends to policies fitted on the same evaluation data within a pre-specified class admitting a fixed finite cover, with an explicit approximation and complexity cost. Uniform logit-change bounds provide explicit energy, remainder, and cover-radius calibrations without a minimum token-probability assumption. These mathematical identities, variance formulas, and conditional reward guarantees are verified in Lean on the same finite autoregressive policy model used by the formal statements; this does not formalize a floating-point neural-network implementation.

1 Introduction

Reinforcement Learning (RL) has become a widely adopted paradigm for aligning Large Language Models (LLMs) with human preferences and improving their reasoning capabilities [Zeng et al., 2025, Guo et al., 2025]. As tasks shift from short-form chat to long-horizon reasoning—mathematical problem solving, code generation, and multi-step planning—the computational cost of maintaining a critic (value) network becomes prohibitive. Critic-free methods such as REINFORCE++ Baseline [Hu et al., 2025], GRPO [Shao et al., 2024], and RLOO [Ahmadian et al., 2024] have emerged as practical alternatives, eliminating the critic and saving approximately 25% of GPU memory in comparable settings [Hu et al., 2025].

Despite their efficiency, critic-free methods face two fundamental challenges that become increasingly severe as sequence lengths grow:

Challenge 1: Advantage scale and token-mass imbalance. Without a learned value function, the advantage must be estimated from reward signals alone. Including each reward in its

own mean baseline produces a deterministic $(n-1)/n$ scaling of the RLOO estimator (Appendix C.1). Separately, token expansion weights responses by their length, so zero-sum response advantages need not have zero token-weighted mean. A common positive scale preserves this imbalance, while subtracting a token mean can change advantage signs. This motivates sign-dependent normalization that meets the token-level moment constraints while preserving each advantage’s sign.

Challenge 2: Causal off-policy mismatch. PPO-style LLM-RL optimizes a policy π_θ using trajectories generated by a fixed rollout policy π_{roll} and a frozen PPO snapshot π_{old} . Distributed staleness, numerical effects, and architectural discontinuities (e.g., MoE routing flips [Liu et al., 2024]) can amplify the divergence between π_{roll} and π_θ , producing large per-token importance ratios $\rho_t = \pi_\theta(y_t | c_t) / \pi_{\text{roll}}(y_t | c_t)$. Because generation is autoregressive, an early ratio deviation shifts the future context distribution, so token-independent ratio interventions may fail to respect the causal structure of error accumulation.

These challenges interact in the sampled update: each contribution combines an advantage estimate with an importance ratio. The population surrogate error decomposes into a true advantage factor and a context-distribution shift factor (Equation (5)); the true advantage factor is distinct from the sampling variance of its estimator. RLOO supplies a baseline conditionally independent of each sample’s own response given the prompt, while TIS and IcePop operate on individual importance ratios. ProMax combines control of empirical advantage statistics with a prefix-based rule for retaining ratio-weighted contributions.

We present **REINFORCE Pro Max**, combining adaptive advantage normalization (Section 3) with prefix-level Causal Trust Region masking (Section 4). Under the finite conditional model stated in the preliminaries, Proposition 2.8 establishes the exact population connection of the raw critic-free surrogate to the per-step advantage surrogate. Proposition 5.3 then isolates the changes made by normalization, masking, and clipping on each finite batch. Theorem 5.4 combines these identities with independent-group sampling and the Adaptive bound to give a performance certificate for the complete ProMax objective. Our contributions are:

1. **Adaptive asymmetric normalization** (Section 3.2): We introduce a normalization scheme that applies different scaling factors α, β to positive and negative advantages. The scheme provably preserves the sign of each non-zero advantage (and the relative ordering within positive and negative groups). With both sign groups nonempty, the ideal closed-form factors enforce zero empirical mean and unit empirical variance on non-zero tokens (Definition 3.2); Proposition C.6 gives exact moment formulas and residual bounds for the implemented additive stabilizer and clamps. For actual binary-reward RLOO groups, Proposition 3.3 identifies the exact token amplitudes and shows that the ratio of the two scales is the ratio of the class-conditional mean response lengths. For the complete on-policy binary population update, including numerical safeguards and content-dependent lengths, Corollary C.12 gives a covariance-margin condition for local true-reward improvement at a fixed prompt with accepted gates and inactive clipping.
2. **Prefix Causal Trust Region masking with causal structure** (Section 4.3): We propose a prefix-cumulative masking strategy constructed theoretically from the per-token ratio $\rho_t = \pi_\theta / \pi_{\text{roll}}$ and implemented from the cached $\pi_{\text{old}} / \pi_{\text{roll}}$ factor. Motivated by long-horizon trust-region analyses for LLM-RL (e.g., TRM [Li et al., 2025]), we apply a per-position prefix gate to the token-level loss, respecting the autoregressive causal structure. This differs from TRM, which uses a sequence-level acceptance gate $M^{\text{TRM}}(x, y)$ to discard entire trajectories. Under the explicit conditions in Theorem 4.7, the mask propagates early-deviation rejection and confines late-deviation rejection to a suffix. A single-spike construction gives

prefix rejection with full-sequence acceptance; sustained violations make all three masks reject. These results characterize conditional comparisons rather than a single dominance ordering.

3. **A unified finite-sample reward guarantee** (Theorem 5.4 and Corollary 5.7): We connect the actual clipped ProMax objective to expected reward improvement through explicit gate, normalization, clipping and sampling terms, using the Adaptive bound from the trust-region framework [Li et al., 2025]. Its complete finite-tree proof retains history dependence (Theorem 2.10). The statistical proof uses complete RLOO groups as independent blocks, retains random response lengths and finite prompt averaging, and controls false improvement certificates after selecting from a finite family fixed before evaluation. Corollary 5.5 removes the unknown mean response length by observed-token rescaling and controls sampling error through additive conditional ratio energy. The exact shared-baseline variance in Proposition 5.6 quantifies the effect of responses per prompt while retaining the separate variance across prompts. A fixed finite cover extends the guarantee to policies fitted on the evaluation data within a pre-specified class, with an explicit approximation and complexity cost. Uniform logit-change bounds give explicit certificate inputs and a sufficient cover radius without requiring a minimum token probability (Corollary 5.8). The objective identities and statistical bounds are verified in Lean on the same finite autoregressive policy model used by the formal statements. This does not formalize a floating-point neural-network implementation. The statistical guarantee is stated for unconditioned iid evaluation groups; optional training-time whole-group score filtering changes that law and is covered by a separate finite counterexample.

Paper organization. Section 2 introduces notation, assumptions, and the surrogate objective error decomposition. Section 3 presents REINFORCE Max (leave-one-out baseline, token expansion, and adaptive normalization). Section 4 develops REINFORCE Pro (prefix-level Causal Trust Region masking and its connection to the Adaptive bound). Section 5 unifies both components and presents the full algorithm. Section 7 discusses related work, Section 6 describes experimental evaluation, and Section 8 concludes.

2 Preliminaries

2.1 Autoregressive Generation and Objective

For the finite-horizon identities below, T denotes a fixed deterministic horizon. Variable-length responses are represented by padding with an absorbing end-of-sequence token and by zeroing padded action terms with the corresponding action mask. The post-termination transition is deterministic and shared by both policies, so its likelihood ratio is 1; this is the convention used whenever a common horizon appears in a sum or product.

A (stochastic) policy π generates a response $y = (y_1, \dots, y_T)$ conditioned on a prompt x . The induced trajectory probability admits the autoregressive factorization

$$P^\pi(y \mid x) = \prod_{t=1}^T \pi(y_t \mid x, y_{<t}).$$

For each position t , we define the context

$$c_t := (x, y_{<t}) = (x, y_{<t}).$$

The induced *context visitation distribution* (the marginal distribution of length- $(t - 1)$ prefixes, i.e. contexts before token t) is

$$d_t^\pi(c_t) = P(x) P^\pi(y_{<t} | x) = P(x) \prod_{s=1}^{t-1} \pi(y_s | c_s).$$

Given a scalar reward $R(x, y)$, the objective of the current policy π_θ is

$$J(\pi_\theta) = \mathbb{E}_{x \sim P(x), y \sim \pi_\theta(\cdot | x)} [R(x, y)].$$

2.2 Notation

Policies. We consider:

- π_{roll} : the *behavior* (rollout) policy used to generate samples; it is fixed during each policy update;
- π_θ : the *training* policy being optimized by gradient steps.
- π_{old} : the frozen PPO snapshot used by the clipped current-policy ratio; it is an implementation object and need not equal π_{roll} .

This notation follows long-horizon trust-region analyses for LLM-RL (e.g., Li et al. [2025]).

Importance ratios. The per-token importance ratio and its log-ratio are

$$\rho_t := \frac{\pi_\theta(y_t | c_t)}{\pi_{\text{roll}}(y_t | c_t)}, \quad (1)$$

$$\log \rho_t = \log \pi_\theta(y_t | c_t) - \log \pi_{\text{roll}}(y_t | c_t). \quad (2)$$

Rollout-time anchoring. In practice, inference and training stacks can differ. The theoretical ρ_t uses rollout-time cached log-probabilities $\log \pi_{\text{roll}}(y_t | c_t)$. The implementation obtains the corresponding correction through the factorization $\rho_t^{\theta/\pi_{\text{roll}}} = r_t^{\theta/\pi_{\text{old}}} w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$; see Appendix B.1 for practical mismatch modes and implementation mappings.

2.3 Divergence Measures

We use token-level divergence measures between conditional token distributions, following the trust-region literature for long-horizon LLM-RL [Li et al., 2025]. For two policies π and π' , define the token-level total variation and KL divergences at context c_t :

Definition 2.1 (Token-Level Divergences). For a context $c_t = (x, y_{<t})$ and two policies π and π' ,

$$D_{\text{TV}}^{\text{tok}}(c_t; \pi, \pi') := D_{\text{TV}}(\pi(\cdot | c_t) \| \pi'(\cdot | c_t)) = \frac{1}{2} \sum_v |\pi(v | c_t) - \pi'(v | c_t)|,$$

$$D_{\text{KL}}^{\text{tok}}(c_t; \pi, \pi') := D_{\text{KL}}(\pi(\cdot | c_t) \| \pi'(\cdot | c_t)) = \sum_v \pi(v | c_t) \log \frac{\pi(v | c_t)}{\pi'(v | c_t)}.$$

When the policy pair is clear from context, we write $D_{\text{TV}}^{\text{tok}}(c_t)$ and $D_{\text{KL}}^{\text{tok}}(c_t)$.

Definition 2.2 (Max Token-Level Divergences and Expected Per-Position TV). For the policy pair $(\pi_{\text{roll}}, \pi_\theta)$, define:

$$\epsilon := D_{\text{TV}}^{\text{tok}, \max} := \max_{t, c_t} D_{\text{TV}}^{\text{tok}}(c_t),$$

$$\delta := D_{\text{KL}}^{\text{tok}, \max} := \max_{t, c_t} D_{\text{KL}}^{\text{tok}}(c_t),$$

$$\bar{D}_t := \mathbb{E}_{c_t \sim d_t^{\pi_{\text{roll}}}} [D_{\text{TV}}^{\text{tok}}(c_t)].$$

2.4 Assumptions

The following assumptions are used throughout the paper:

Assumption 2.3 (Mutual Policy Support). For all contexts c_t and tokens y_t in the vocabulary,

$$\pi_{\text{roll}}(y_t | c_t) > 0 \iff \pi_{\theta}(y_t | c_t) > 0.$$

That is, the two conditional policies have mutual support. This makes the per-token importance ratio ρ_t and its logarithm in Eqs. (1)–(2) well-defined on every sampled token, and permits the change-of-measure identities used below (in particular $\mathbb{E}_{\pi_{\text{roll}}}[\rho_t | c_t] = 1$ and the TV/KL ratio formulas). All ratio identities are interpreted on the rollout support $\{y_t : \pi_{\text{roll}}(y_t | c_t) > 0\}$. If both policies assign zero mass to a token, the ratio at that token is assigned an arbitrary harmless value (the guarded finite KL formalization uses 1), because such tokens contribute zero to rollout expectations and to the KL/TV sums. Proofs that work directly with token ratios restrict them to the positive rollout support. Thus the assumption does not require strict positivity on the whole vocabulary.

Assumption 2.4 (Bounded Rewards). There exists $R_{\max} < \infty$ such that $|R(x, y)| \leq R_{\max}$ for all prompt-response pairs (x, y) .

Assumption 2.5 (Finite Sequence Length). There is a fixed finite generation cap T . Every response has at most T active actions and is extended to this common horizon by the shared absorbing EOS convention above.

Remark 2.6 (Three-policy ratio decomposition in the implementation). The mathematical statements below use the behavior/training pair $(\pi_{\text{roll}}, \pi_{\theta})$. The PPO implementation also has a frozen snapshot π_{old} and therefore uses two ratios:

$$r_t^{\theta/\pi_{\text{old}}} := \frac{\pi_{\theta}(y_t | c_t)}{\pi_{\text{old}}(y_t | c_t)}, \quad w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} := \frac{\pi_{\text{old}}(y_t | c_t)}{\pi_{\text{roll}}(y_t | c_t)}.$$

On their common support,

$$\rho_t^{\theta/\pi_{\text{roll}}} := \frac{\pi_{\theta}(y_t | c_t)}{\pi_{\text{roll}}(y_t | c_t)} = r_t^{\theta/\pi_{\text{old}}} w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}.$$

The corresponding log-probability implementation identity is checked by `ImplementationBridge.current_rollout_ratio_factorization`; it is the exponentiated form of the three-policy decomposition used below. The prefix gate in the current implementation is computed from $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$, using cached rollout log-probabilities and the old-policy log-probabilities; the PPO surrogate is computed from $r_t^{\theta/\pi_{\text{old}}}$. When $\pi_{\text{old}} = \pi_{\text{roll}}$ at the beginning of an update, $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} = 1$ and the PPO ratio equals the theoretical behavior-to-current ratio. For stale snapshots, the prefix results below apply directly to the pair $(\pi_{\text{roll}}, \pi_{\text{old}})$; transferring them to the current pair additionally requires a bound on the PPO update.

2.5 The Surrogate Objective and Error Decomposition

Samples are generated from a behavior policy π_{roll} that generally differs from π_{θ} . Following Kakade and Langford [2002], the standard surrogate objective uses the per-step advantage under π_{roll} ,

$$A_t^{\pi_{\text{roll}}}(c_t, y_t) := \mathbb{E}_{\pi_{\text{roll}}}[R(x, y) | c_t, y_t] - \mathbb{E}_{\pi_{\text{roll}}}[R(x, y) | c_t].$$

This yields the per-step surrogate

$$\mathcal{L}_{\pi_{\text{roll}}}(\pi_{\theta}) = \sum_{t=1}^T \mathbb{E}_{c_t \sim d_t^{\pi_{\text{roll}}}} \left[m_t(c_t) \mathbb{E}_{y_t \sim \pi_{\text{roll}}(\cdot | c_t)} [\rho_t A_t^{\pi_{\text{roll}}}(c_t, y_t)] \right], \quad (3)$$

where $m_t(c_t) \in \{0, 1\}$ is the predictable action mask from Definition 2.7. In the unpadded fixed-horizon model $m_t \equiv 1$; with EOS padding it removes post-termination contexts, whose shared absorbing transition has zero advantage and unit ratio. The ratio ρ_t is defined in Eq. (1).

Definition 2.7 (Trajectory-Level Surrogate). Let b be a baseline that is independent of the sampled trajectory y conditional on the prompt x (e.g., a prompt-level baseline computed from other independent rollout samples in the same prompt group). The rollout policy and the baseline sampling law are fixed during the update. Let $m_t = m_t(c_t) \in \{0, 1\}$ denote the predictable action mask: it equals one while generation is active, including the action that emits EOS, and zero at subsequent padding positions. Define the trajectory-level advantage

$$A := R(x, y) - b.$$

The unclipped, unnormalized trajectory-level surrogate underlying critic-free implementations is

$$\tilde{\mathcal{L}}_{\pi_{\text{roll}}}(\pi_{\theta}) := \mathbb{E}_{\pi_{\text{roll}}} \left[A \cdot \sum_{t=1}^T m_t \rho_t \right].$$

Proposition 2.8 (Exact Centered Surrogate Equivalence). *For finite prompt, action, and continuation supports, under mutual policy support and the finite/EOS-padded horizon convention, assume that the expectations below exist and that b and m_t satisfy Definition 2.7. Then, for any current policy π_{θ} ,*

$$\tilde{\mathcal{L}}_{\pi_{\text{roll}}}(\pi_{\theta}) - \tilde{\mathcal{L}}_{\pi_{\text{roll}}}(\pi_{\text{roll}}) = \mathcal{L}_{\pi_{\text{roll}}}(\pi_{\theta}) = \mathbb{E}_{\pi_{\text{roll}}} \left[R(x, y) \sum_{t=1}^T m_t (\rho_t - 1) \right]. \quad (4)$$

The subtracted quantity is constant during the policy update; the equality does not require $\pi_{\theta} = \pi_{\text{roll}}$.

The proof is in Appendix E.1. This is a population identity; individual Monte Carlo estimates need not coincide. The random-length batch reductions used by ProMax are retained in Section 5.3.

Remark 2.9 (Gradient Interpretation and ProMax Modifications). Where differentiable, Eq. (4) implies equality of the gradients of the two surrogates while keeping the rollout distribution fixed. At the on-policy evaluation point, the REINFORCE identity gives

$$\mathbb{E}_{\pi_{\theta}} \left[A \cdot \sum_{t=1}^T m_t \nabla_{\theta} \log \pi_{\theta}(y_t | c_t) \right] = \nabla_{\theta} J(\pi_{\theta}),$$

under the usual score-function differentiation conditions. Away from that point, equality of the two surrogate gradients does not equate either gradient with $\nabla_{\theta} J(\pi_{\theta})$. The actual ProMax loss additionally applies adaptive normalization, an action-dependent prefix gate, PPO clipping, and the selected batch reduction. These operations require their own analysis; the exact finite-batch decomposition in Proposition 5.3 retains all of them. In particular, detaching a prefix gate does not make it independent of the sampled action.

The approximation error of the per-step surrogate is

$$\text{Error}(\pi_\theta) := J(\pi_\theta) - J(\pi_{\text{roll}}) - \mathcal{L}_{\pi_{\text{roll}}}(\pi_\theta).$$

Under the finite-horizon Performance Difference Identity [Kakade and Langford, 2002], and assuming finite context/token supports so that the conditional expectations are finite sums, this error decomposes as:

$$|\text{Error}(\pi_\theta)| \leq \sum_{t=1}^T 2\|g_t\|_\infty \cdot \|d_t^{\pi_\theta} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}}, \quad (5)$$

where

$$g_t(c_t) := m_t(c_t) \mathbb{E}_{y_t \sim \pi_\theta(\cdot|c_t)}[A_t^{\pi_{\text{roll}}}(c_t, y_t)]$$

is the expected advantage shift. Classical bounds on Eq. (5) scale as $O(T^2)$ [Kakade and Langford, 2002, Achiam et al., 2017], rendering them theoretically vacuous for long-horizon LLM tasks with responses of thousands of tokens. Recent work [Li et al., 2025] derives tighter bounds, including the Pinsker-Marginal bound ($O(T^{3/2})$), the Mixed bound ($O(T)$), and the data-dependent Adaptive bound.

The finite-sum error derivation and its formalization are recorded in Appendix E.1.

The Adaptive bound of Li et al. [2025] is particularly relevant to our work. We give a complete finite-tree proof, including the linear future-TV route, to fix its constants and connect it to the same surrogate used by the method.

Theorem 2.10 (Finite-tree Adaptive remainder bound). *For a finite prompt distribution and finite vocabulary, let both policies be normalized history-dependent kernels satisfying Assumption 2.3 and the fixed/EOS-padded horizon convention. Suppose $|R| \leq R_{\max}$ with $R_{\max} \geq 0$, and let ϵ, δ bound the local TV and reverse KL at every context before horizon T , as in Definition 2.2. Then*

$$B_{\text{Adap}}^* = 4R_{\max} \sum_{t=1}^T \bar{D}_t \cdot \min\left(1, (T-t)\epsilon, \sqrt{\frac{(T-t)\delta}{2}}\right), \quad (6)$$

satisfies

$$|J(\pi_\theta) - J(\pi_{\text{roll}}) - \mathcal{L}_{\pi_{\text{roll}}}(\pi_\theta)| \leq B_{\text{Adap}}^*, \quad J(\pi_\theta) - J(\pi_{\text{roll}}) \geq \mathcal{L}_{\pi_{\text{roll}}}(\pi_\theta) - B_{\text{Adap}}^*.$$

The same bound controls the absolute reward-weighted trajectory remainder in Definition E.7.

Proof. The exact history-tree error recursion separates the current token’s probability shift from the difference in future expected rewards. The latter is bounded by $2R_{\max}$ times the future trajectory TV, which is at most each of 1 , $(T-t)\epsilon$, and $\sqrt{(T-t)\delta/2}$. Summing the local interaction under rollout prefixes gives Eq. (6). The full recursion and finite prompt averaging are proved in Appendix D.2 and checked in `formal/AdaptiveBound.lean`. $\square$

The factor $R_{\max}$ is required for general bounded rewards; $R_{\max} = 1$ recovers the normalized-reward form. This result retains the per-position expected TV and the remaining future horizon. It motivates causal prefix filtering, while the sampled gate’s own guarantees are stated separately in Theorem 4.11; a sampled acceptance event does not establish the population TV/KL assumptions of Theorem 2.10.

Table 1: Existing trust-region and IS stabilization methods in critic-free LLM-RL.

Method	IS Statistic	Truncation	Granularity
TIS	ρ_t	$w_t^{\text{tok}} = \text{clip}(\rho_t, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}})$	Token
IcePop	ρ_t	$M_t^{\text{tok}} = \mathbb{I}[\rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]]$	Token
TRM	ρ_t	$M^{\text{TRM}}(x, y) = \mathbb{I}[\max_t D_{\text{KL}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta}) \leq \delta_{\text{TRM}}]$	Sequence
GSPO	ρ^{seq}	$w^{\text{seq}} = \text{clip}(\rho^{\text{seq}}, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}})$	Sequence

2.6 Existing Importance Sampling Correction Methods

Several methods have been proposed to stabilize policy-gradient updates by constraining the per-token importance ratio ρ_t between the training policy π_{θ} and the behavior policy π_{roll} . We use the per-token log-ratio $\log \rho_t$ defined in Eq. (2).

Table 1 summarizes these stabilization mechanisms under a unified notation.

The table uses the theoretical behavior-to-training ratio for a common notation. In the PPO implementation, the corresponding mismatch correction weights are evaluated from $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$, while the differentiable PPO ratio is $r_t^{\theta/\pi_{\text{old}}}$, as specified above.

Here ρ^{seq} is the length-normalized sequence likelihood ratio (a geometric mean of token ratios),

$$\begin{aligned} \rho^{\text{seq}} &:= \left(\frac{P^{\pi_{\theta}}(y | x)}{P^{\pi_{\text{roll}}}(y | x)} \right)^{\frac{1}{T}} \\ &= \left(\prod_{t=1}^T \rho_t \right)^{\frac{1}{T}} = \exp \left(\frac{1}{T} \sum_{t=1}^T \log \rho_t \right), \end{aligned}$$

where T is the fixed (or EOS-padded) horizon used for the comparison. For an unpadded response, this reduces to its actual length $T = |y|$. This definition matches the sequence-level ratio used in GSPO [Zheng et al., 2025] (denoted $s_i(\theta)$ in the GSPO paper) up to notation.

TRM (Trust Region Masking, Li et al. [2025]) uses a binary *sequence-level* mask $M^{\text{TRM}}(x, y)$ to discard trajectories that violate the trust region. A standard (max-based) criterion is

$$M^{\text{TRM}}(x, y) := \mathbb{I} \left[\max_t D_{\text{KL}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta}) \leq \delta_{\text{TRM}} \right].$$

A comparative discussion of these mechanisms and their limitations is provided in Section 5.

Remark 2.11 (Context Shift Propagation). Fix a prompt and a finite vocabulary, with normalized policies on complete prefix contexts. Let s be the first position of disagreement: the policies agree at every rollout-reachable context before s , and $D_{\text{TV}}^{\text{tok}}(c_s; \pi, \pi') > 0$ at some context with $d_s^{\pi}(c_s) > 0$. Then

$$\|d_{s+1}^{\pi} - d_{s+1}^{\pi'}\|_{\text{TV}} = \mathbb{E}_{c_s \sim d_s^{\pi}} [D_{\text{TV}}^{\text{tok}}(c_s; \pi, \pi')] > 0.$$

Moreover, full-prefix TV is nondecreasing with prefix length, so every later context distribution retains at least this discrepancy. Marginalizing the last token recovers the preceding complete prefix; the triangle inequality therefore prevents its TV from increasing under that marginalization. No support restriction on subsequent policy kernels is needed, including after a shared absorbing EOS transition. The equality, monotonicity, and first-disagreement implication are checked by `prefixTV_next_of_equal_prefix_mass`, `prefixTV_mono`, and `first_disagreement_propagates` in `formal/TreeTV.lean`. This statement concerns full histories; a representation that discards earlier tokens need not retain their discrepancy. See also Li et al. [2025].

3 REINFORCE Max: Adaptive Advantage Normalization

3.1 Token-Level Expansion

Given a prompt x , we sample n responses $\{y^{(1)}, \dots, y^{(n)}\}$ with rewards $\{r_1, \dots, r_n\}$. Each response $y^{(i)}$ consists of $T_i \geq 1$ active action tokens (the action that emits EOS counts as active when EOS is used). REINFORCE Max combines three ingredients: a leave-one-out baseline (Equation (7)), token-level expansion (Section 3.1), and adaptive normalization (Section 3.2).

We use the standard leave-one-out (RLOO) baseline [Ahmadian et al., 2024]; independence of each response and its baseline conditioned on the prompt supports the raw-surrogate analysis in Appendix C.1. The effect of subsequent normalization is analyzed separately below.

Definition 3.1 (Leave-One-Out Baseline). Assume $n \geq 2$. For sample i , the baseline and shaped reward are:

$$b_i = \frac{1}{n-1} \sum_{j \neq i} r_j, \quad \tilde{r}_i = r_i - b_i. \quad (7)$$

For an exact RLOO group with $n \geq 2$, the shaped rewards satisfy the deterministic conservation identity

$$\sum_{i=1}^n \tilde{r}_i = 0.$$

Consequently, if all shaped rewards have one weak sign, they are all zero; a non-degenerate group necessarily contains both a positive and a negative advantage. This identity is checked on finite groups by the Lean theorems `rloo_shaped_sum_zero` and `rloo_shaped_sum_zero_erase`.

Since the reward is a scalar per response and we use $\gamma = 1.0$ (see Appendix C.8 for a brief note), the token-level advantage is simply the shaped reward itself:

$$A_{i,t} = \tilde{r}_i = r_i - b_i, \quad \forall t \in \{1, \dots, T_i\}.$$

The theorem-level REINFORCE Max analysis in this paper uses the sparse extrinsic-reward configuration: the reference-model KL coefficient is set to 0, no separate KL loss is added, and the scalar reward is not clipped after RLOO shaping. A configured length penalty is included in the scalar reward before the RLOO baseline is formed. This scope is needed for the displayed identity $A_{i,t} = \tilde{r}_i$. If KL regularization or post-RLOO reward clipping is enabled in the implementation, the effective token return changes, so the same formulas require an explicitly modified return definition; the KL-free, unclipped statements below do not claim that extension automatically.

This is a theorem scope condition rather than the command-line default. In the current CLI, “`-init_kl_coef`” defaults to 0.01 and “`-reward_clip_range`” defaults to $[-10, 10]$. A run intended to instantiate the displayed sparse-reward theorems must set the KL coefficient to zero, disable the KL loss, and choose a reward-clip interval that is inactive on the stated reward support (or otherwise replace the return definition in the analysis). Runs using the CLI defaults are valid algorithm configurations, but their post-shaping returns are outside the exact $A_{i,t} = \tilde{r}_i$ theorem unless the clipping interval is provably inactive.

3.2 Adaptive Asymmetric Normalization

Standard normalization $\hat{A} = (A - \mu)/\sigma$ combines centering with a common positive scale. The scale preserves relative magnitudes; the centering can change signs when the token-weighted mean is nonzero. Because response lengths weight the token statistics, the RLOO zero-sum property over responses alone does not solve the token-level constraints. We therefore use separate positive scales to balance token mass while preserving the original signs.

Definition 3.2 (Adaptive Normalization). Fix a prompt group with token-level advantages $\{A_{i,t}\}$, and exclude zero-advantage positions from the normalization statistics. Define the index sets

$$\mathcal{P} := \{(i, t) : A_{i,t} > 0\}, \quad \mathcal{N} := \{(i, t) : A_{i,t} < 0\},$$

The normalized advantage is defined by a sign-dependent rescaling:

$$\hat{A}_{i,t} = \begin{cases} \alpha \cdot A_{i,t}, & \text{if } A_{i,t} > 0, \\ \beta \cdot A_{i,t}, & \text{if } A_{i,t} < 0, \\ 0, & \text{if } A_{i,t} = 0. \end{cases}$$

where $\alpha, \beta > 0$ are chosen to satisfy the following *empirical* constraints over non-zero tokens:

$$\begin{aligned} \frac{1}{|\mathcal{P}| + |\mathcal{N}|} \sum_{(i,t) \in \mathcal{P} \cup \mathcal{N}} \hat{A}_{i,t} &= 0, \\ \frac{1}{|\mathcal{P}| + |\mathcal{N}|} \sum_{(i,t) \in \mathcal{P} \cup \mathcal{N}} \hat{A}_{i,t}^2 &= 1. \end{aligned}$$

When $\mathcal{P} \neq \emptyset$ and $\mathcal{N} \neq \emptyset$, define the ratio

$$\kappa := \frac{\sum_{(i,t) \in \mathcal{P}} A_{i,t}}{-\sum_{(i,t) \in \mathcal{N}} A_{i,t}}.$$

Equivalently, these constraints can be written as

$$\begin{aligned} \beta &= \alpha \kappa, \\ \alpha^2 \sum_{(i,t) \in \mathcal{P}} A_{i,t}^2 + \beta^2 \sum_{(i,t) \in \mathcal{N}} A_{i,t}^2 &= |\mathcal{P}| + |\mathcal{N}|. \end{aligned}$$

When $\mathcal{P} \neq \emptyset$ and $\mathcal{N} \neq \emptyset$, solving the constraints in Definition 3.2 yields the closed-form scaling factors:

$$\begin{aligned} \alpha &= \sqrt{\frac{|\mathcal{P}| + |\mathcal{N}|}{\sum_{(i,t) \in \mathcal{P}} A_{i,t}^2 + \kappa^2 \sum_{(i,t) \in \mathcal{N}} A_{i,t}^2}}, \\ \beta &= \alpha \kappa. \end{aligned} \tag{8}$$

Advantage Sign Preservation The adaptive normalization preserves the sign of the advantage at every token: $\text{sign}(\hat{A}_{i,t}) = \text{sign}(A_{i,t})$ for all (i, t) with $A_{i,t} \neq 0$. Moreover, the relative ordering of magnitudes is preserved within the positive group $\mathcal{P}$ and within the negative group $\mathcal{N}$. This follows immediately because the normalization scales $A_{i,t}$ by $\alpha > 0$ on $\mathcal{P}$ and by $\beta > 0$ on $\mathcal{N}$ (Equation (8)), and multiplication by a positive constant preserves both sign and ordering. These are scalar advantage properties. Different positive and negative scales can change the direction of the summed policy-gradient vector; they do not imply preservation of the full batch gradient direction.

Connection to Response Length Our asymmetry comes from sign-dependent rescaling of advantages under the token-level expansion and the empirical constraints in Definition 3.2. Since Section 3.1 uses $A_{i,t} = \tilde{r}_i$ for all $t \in \{1, \dots, T_i\}$, the prompt-group statistics in Definition 3.2 are implicitly length-weighted. In particular, the positive/negative token index sets satisfy

$$|\mathcal{P}| = \sum_{i: \tilde{r}_i > 0} T_i, \quad |\mathcal{N}| = \sum_{i: \tilde{r}_i < 0} T_i,$$

and the ratio κ in Definition 3.2 can be written as

$$\kappa = \frac{\sum_{(i,t) \in \mathcal{P}} A_{i,t}}{-\sum_{(i,t) \in \mathcal{N}} A_{i,t}} = \frac{\sum_{i: \tilde{r}_i > 0} T_i \tilde{r}_i}{-\sum_{i: \tilde{r}_i < 0} T_i \tilde{r}_i}.$$

Therefore the closed-form scales (α, β) in Eq. (8) depend not only on reward magnitudes but also on the ratio of positive-token mass to negative-token mass within the same prompt group, which is directly affected by response lengths (T_i) . This is the sense in which the advantage scaling implicitly incorporates response-length information, rather than implementing any form of ratio clipping.

For binary correctness rewards, the role of length can be characterized exactly using the actual RLOO advantages, without replacing them by equal magnitudes of opposite signs.

Proposition 3.3 (Binary RLOO normalization and response-length balance). *Suppose a prompt group contains $k \geq 1$ responses with scalar reward r_+ and $m \geq 1$ with reward $r_- < r_+$, and write $n = k + m$ and $\Delta = r_+ - r_- > 0$. Work in the RLOO branch and use the ideal factors in Eq. (8). Let P and N be the total active-token counts in the higher- and lower-reward responses, respectively; assume $P, N > 0$. Then the raw RLOO advantages are a_+ and $-a_-$, where*

$$a_+ = \frac{m\Delta}{n-1}, \quad a_- = \frac{k\Delta}{n-1}.$$

Every active token in the two classes has normalized advantage

$$\hat{A}_+ = \alpha a_+ = \sqrt{N/P}, \quad \hat{A}_- = -\beta a_- = -\sqrt{P/N}. \quad (9)$$

Writing $\bar{T}_+ = P/k$ and $\bar{T}_- = N/m$ for the class-conditional mean response lengths, the factor ratio and aggregate coefficient balance satisfy

$$\frac{\beta}{\alpha} = \kappa = \frac{Pa_+}{Na_-} = \frac{\bar{T}_+}{\bar{T}_-}, \quad P\hat{A}_+ = -N\hat{A}_-. \quad (10)$$

These conclusions concern scalar advantages before importance weighting, prefix filtering, or multiplication by score vectors.

The proof in Appendix C.9 starts from the finite group reward sum and applies the normalization to its actual length-weighted RLOO moments. If lower-reward responses are longer on average, then $\beta < \alpha$: they receive a smaller scaling factor relative to the higher-reward responses. If the two average lengths are equal, the factors are equal even when successes are rare. Thus rarity alone does not imply that the negative factor is less than one. For example, one reward-1 response and nine reward-0 responses of equal length have raw advantages 1 and $-1/9$, and both ideal factors equal 3. The method balances the aggregate scalar coefficients; this does not assert cancellation or dominance of the corresponding gradient vectors.

Proposition 3.4 (Scale invariance of ideal normalization). *Fix a finite active-token set with both advantage signs present and use the ideal normalization of Definition 3.2. For every $c > 0$, replacing all advantages by cA leaves every normalized advantage unchanged. Consequently, for any nondegenerate group with $n \geq 2$, applying the same ideal asymmetric normalization to the mean-baseline and RLOO signals gives identical token coefficients, including with unequal response lengths:*

$$A_i^{\text{mean}} = \frac{n-1}{n} A_i^{\text{RLOO}} \implies \mathcal{N}(A^{\text{mean}}) = \mathcal{N}(A^{\text{RLOO}}), \quad (11)$$

where $\mathcal{N}$ denotes this ideal normalization with the same active positions. This excludes numerical safeguards and fallback branches.

The proof is in Appendix C.9. RLOO remains the independent baseline used in the raw-surrogate and group-variance analysis. Its common scale relative to a mean baseline does not itself distinguish the ideal normalized method; the normalization rule and its response-length weighting are the relevant choices. With only the additive stabilizer active, both ideal normalized amplitudes are multiplied by the common factor $\sqrt{D/(D + \varepsilon_{\text{num}})}$, where D is the denominator inside the ideal square root in Eq. (8). Product and factor clamps, or fallback decisions, require the residual analysis in Proposition C.6.

Corollary 3.5 (Equivalence to token-weighted standardization for binary rewards). *Under Proposition 3.3, let μ_{tok} and $\sigma_{\text{tok}} > 0$ be the empirical mean and population standard deviation of the raw advantages over the active tokens of this prompt group. Then the ideal asymmetric coefficients also satisfy $\hat{A}_{i,t} = (A_{i,t} - \mu_{\text{tok}})/\sigma_{\text{tok}}$.*

The proof is in Appendix C.9. For exact binary rewards, the distinction from response-weighted or cross-prompt standardization is the token weighting and prompt grouping. For general multivalued rewards, token centering can change advantage signs, whereas positive sign-dependent scaling preserves them. The binary equivalence is restricted to the same group, token weights, and ideal moment convention. A bounded three-reward RLOO example in Section C.9 exhibits a positive raw advantage that token standardization makes negative, while asymmetric normalization keeps it positive.

Uniform Scale Mode When the implementation’s reward-uniformity test is satisfied (exactly equal rewards are the ideal case, while its tolerance may also admit near-uniform rewards), the ordinary leave-one-out signal is replaced by an optional *uniform scale* rule $\tilde{r}_i = r_i/n$ and adaptive normalization is skipped. For exactly equal rewards the ordinary leave-one-out advantages are zero. This rule supplies nonzero coefficients when that common reward is nonzero; an all-zero group still contributes zero. For binary 0/1 rewards, its additional expected on-policy sequence-score update is a nonnegative multiple of the per-prompt reward gradient (Proposition C.9). Under the additional class-dependent-length condition, the complete token-reduced update, including the implemented stabilizer and clamps, is a strictly positive multiple of that gradient (Proposition C.10). With arbitrary content-dependent lengths, an explicit within-class covariance margin instead guarantees a positive reward directional derivative and strict improvement for sufficiently small steps along the population update (Corollary C.12). This statement retains the on-policy, accepted-gate and inactive-clipping premises and holds with uniform scale either enabled or disabled. The multiplier and its dependence on the reward zero point are explicit; the optional branch does not inherit RLOO’s invariance to reward translation. See Appendix C.11 for the conditions and an offset counterexample.

Numerical safeguards In implementation, we include simple numerical checks and clamping to prevent extreme or unstable scales; see Appendix C.10. Equations (8) and the two moment constraints describe the ideal factors, without the additive numerical stabilizer or clamps. The implementation always adds $\varepsilon_{\text{num}} > 0$ to the denominator of α^2 . Even when no clamp activates, it therefore has zero empirical mean but second moment $D/(D + \varepsilon_{\text{num}}) < 1$, where $D = \sum_{\mathcal{P}} A_{i,t}^2 + \kappa^2 \sum_{\mathcal{N}} A_{i,t}^2$. Proposition C.6 gives the exact stabilizer deficit, product-cap formula, and factor-clamp residual bounds. Sign preservation on active positions survives positive factor clamps and identity fallbacks; the finite-batch objective decomposition in Proposition 5.3 treats the resulting $\hat{A}$ directly.

4 REINFORCE Pro: Prefix Causal Trust Region Masking

4.1 Off-Policy Ratio Setup

In PPO-style LLM-RL, samples are generated by a fixed rollout policy π_{roll} , and gradients are computed under a training policy π_{θ} that is updated by gradient steps. This matches the trust-region setup in Li et al. [2025].

For the theoretical comparison, the per-token importance ratio between these two policies is

$$\rho_t := \frac{\pi_{\theta}(y_t \mid c_t)}{\pi_{\text{roll}}(y_t \mid c_t)},$$

as defined in Eq. (1). The finite-sequence comparison below uses only this selected policy pair. The implementation mapping to the intermediate PPO snapshot π_{old} is given in Section 2 and the implementation bridge after Lemma 4.6.

A Causal Trust Region Viewpoint Trust Region Masking (TRM; Li et al. [2025]) introduces a *sequence-level* rejection mechanism via a binary mask $M^{\text{TRM}}(x, y)$, yielding a masked surrogate objective of the form

$$\mathcal{L}_{\text{masked}}^{\text{TRM}}(\pi_{\theta}) := \mathbb{E}_{\pi_{\text{roll}}} \left[M^{\text{TRM}}(x, y) \cdot A(x, y) \cdot \sum_{t=1}^T \rho_t \right].$$

This matches the masked surrogate objective in Li et al. [2025] up to notation. Motivated by the same trust-region perspective for long-horizon generation, we introduce a prefix-level, per-position gate M_t^{pre} computed from a ratio-based *prefix* statistic and multiply it into the token-level loss. Unlike TRM, which accepts or rejects an entire trajectory via a single gate $M^{\text{TRM}}(x, y)$, the prefix gate can mask only the positions whose prefixes violate the threshold, thereby retaining positional granularity while respecting the autoregressive order. For comparison, a prefix-level masked surrogate can be written as

$$\mathcal{L}_{\text{masked}}^{\text{pre}}(\pi_{\theta}) := \mathbb{E}_{\pi_{\text{roll}}} \left[A(x, y) \cdot \sum_{t=1}^T M_t^{\text{pre}} \rho_t \right],$$

where the gate is applied at token granularity. Our implementation uses a per-token importance-weighted surrogate with normalized advantages; see Section 4.6 for the exact objective. Importantly, our prefix criterion is a tractable sample-level *proxy* for Causal Trust Region filtering and is not the sequence-level max-divergence criterion used in TRM (based on $\max_t D_{\text{KL}}(c_t)$).

In the implementation, M_t^{pre} is evaluated from $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$, while the differentiable PPO term uses $r_t^{\theta/\pi_{\text{old}}}$. Thus the abstract comparison applies directly to the cached rollout/snapshot pair, and its interpretation for the current policy uses the finite-prefix coupling condition stated above.

4.2 Limitations of Existing Methods

Remark 4.1 (Token-Level Importance Sampling Ignores Causal Structure). This is a standard limitation of token-level ratio truncation and masking. Fix thresholds $\epsilon_{\text{low}} \in (0, 1)$ and $\epsilon_{\text{high}} > 0$ and define the acceptance interval

$$[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}].$$

Consider a length- T trajectory such that

$$\rho_1 \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}], \quad \rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}] \text{ for all } t \in \{2, \dots, T\}.$$

Then the token-level IcePop mask,

$$M_t^{\text{tok}} := \mathbb{I}[\rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]],$$

masks only the first position. In particular, it does not prevent later positions $t \geq 2$ from contributing gradients even though the contexts $(c_t)_{t \geq 2}$ are sampled under the behavior distribution induced by π_{roll} , which differs from the context distribution that would be induced by the updated policy π_θ when $\rho_1 \neq 1$.

The conclusion follows immediately from the definition of M_t^{tok} and context shift propagation (Remark 2.11); see also Li et al. [2025], Yao et al. [2025].

Remark 4.2 (Sequence-Level Importance Sampling Loses Positional Information). This is a standard limitation of sequence-level aggregation and gating. Let ρ^{seq} denote the geometric-mean sequence likelihood ratio from Section 2.6. Then its log is the sequence-average log-ratio,

$$\log \rho^{\text{seq}} = \frac{1}{T} \sum_{t=1}^T \log \rho_t.$$

Assume there exists $t^* \in \{1, \dots, T\}$ such that $|\log \rho_t| \leq \varepsilon$ for all $t \neq t^*$. Then

$$\left| \log \rho^{\text{seq}} - \frac{1}{T} \log \rho_{t^*} \right| \leq \frac{T-1}{T} \varepsilon \leq \varepsilon.$$

In particular, in the pure single-spike case $\log \rho_t = 0$ for $t \neq t^*$, one has $\log \rho^{\text{seq}} = \log \rho_{t^*}/T$, so $\rho^{\text{seq}} = (\rho_{t^*})^{1/T} \rightarrow 1$ as $T \rightarrow \infty$ for any fixed $\rho_{t^*} > 0$.

This dilution is a direct consequence of averaging $\log \rho_t$ over the full horizon; see also Zheng et al. [2025].

Example 4.3 (Numerical Dilution at Long Horizons). Consider the pure single-spike case with $T = 4096$, $\log \rho_{t^*} = 3$, and $\log \rho_t = 0$ for $t \neq t^*$. Then

$$\rho^{\text{seq}} = \exp(3/4096) \approx 1.00073,$$

which can easily fall inside practical acceptance intervals $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ even though the local deviation at t^* is large in log-ratio magnitude.

4.3 Prefix Causal Trust Region Proxy: Definition and Properties

Definition 4.4 (Prefix Cumulative Ratio Proxy). For a response trajectory $y = (y_1, \dots, y_T)$ with per-token log-ratios $\log \rho_t$ defined in Eq. (2), define the prefix cumulative ratio statistic

$$\rho_t^{\text{pre}} := \exp\left(\frac{1}{t} \sum_{s=1}^t \log \rho_s\right) = \left(\prod_{s=1}^t \rho_s\right)^{\frac{1}{t}}, \quad t \in \{1, \dots, T\}. \quad (12)$$

The statistic ρ_t^{pre} is the geometric mean of per-token importance ratios over the length- t prefix. Here t counts active action positions. EOS-padding positions have $m_t = 0$ and are excluded from both the cumulative sum and the denominator; the mask is evaluated only on active positions. Equivalently, on a fixed padded horizon one may replace t in Eq. (12) by $\sum_{s \leq t} m_s$ and replace the sum by $\sum_{s \leq t} m_s \log \rho_s$. This convention matches the implementation, which computes ‘positions’ as the cumulative action-mask count. For any binary action mask, increasing enumeration of its active positions preserves the prefix average and the number of rejected active tokens. At each

active position the cumulative count is at least one, so the implementation’s denominator clamp has no effect there. These exact-real identities are verified in `formal/PaddedPrefix.lean`; they do not require contiguous active positions. In the abstract theory ρ_t refers to the selected policy pair; in the implementation it is instantiated by the cached $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$ ratio as explained below.

Definition 4.5 (Prefix Causal Trust Region Mask). Given thresholds $\epsilon_{\text{low}} \in (0, 1)$ and $\epsilon_{\text{high}} > 0$, define the per-position prefix mask

$$M_t^{\text{pre}} := \mathbb{I}[\rho_t^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]]. \quad (13)$$

Appendix D.5 collects several compatible variants (Variants I–IV) that modify the prefix gate while retaining the same ratio setup based on $(\pi_{\text{roll}}, \pi_{\theta})$.

Three structural properties distinguish the prefix-level mask from existing methods:

Causality ρ_t^{pre} depends only on the prefix ratios $(\rho_s)_{s \leq t}$, respecting the autoregressive order.

Stability under in-bound increments The following lemma formalizes a basic stability property of prefix geometric means.

Lemma 4.6 (Stability of Prefix Geometric Means). *Fix thresholds $\epsilon_{\text{low}} \in (0, 1)$ and $\epsilon_{\text{high}} > 0$, and define ρ_t^{pre} as in Eq. (12). If*

$$\rho_t^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}] \quad \text{and} \quad \rho_{t+1} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}],$$

then $\rho_{t+1}^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$.

Proof. Taking logarithms and using Eq. (12), we have

$$\log \rho_t^{\text{pre}} = \frac{1}{t} \sum_{s=1}^t \log \rho_s.$$

Therefore,

$$\begin{aligned} \log \rho_{t+1}^{\text{pre}} &= \frac{1}{t+1} \sum_{s=1}^{t+1} \log \rho_s \\ &= \frac{t}{t+1} \left(\frac{1}{t} \sum_{s=1}^t \log \rho_s \right) + \frac{1}{t+1} \log \rho_{t+1} \\ &= \frac{t}{t+1} \log \rho_t^{\text{pre}} + \frac{1}{t+1} \log \rho_{t+1}. \end{aligned}$$

Since $\rho_t^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ and $\rho_{t+1} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$, both $\log \rho_t^{\text{pre}}$ and $\log \rho_{t+1}$ lie in $[\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})]$. The last display is a convex combination, hence $\log \rho_{t+1}^{\text{pre}} \in [\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})]$, which implies $\rho_{t+1}^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$. $\square$

Implementation bridge for a stale PPO snapshot The prefix statistic used by the implementation is based on $w_s^{\pi_{\text{old}}/\pi_{\text{roll}}}$ from the three-policy decomposition in Section 2. Define $z_s := \log w_s^{\pi_{\text{old}}/\pi_{\text{roll}}}$ and $u_s := \log r_s^{\theta/\pi_{\text{old}}}$. Then the current-policy log-ratio satisfies

$$\log \rho_s^{\theta/\pi_{\text{roll}}} = z_s + u_s, \quad \frac{1}{t} \sum_{s=1}^t \log \rho_s^{\theta/\pi_{\text{roll}}} = \frac{1}{t} \sum_{s=1}^t z_s + \frac{1}{t} \sum_{s=1}^t u_s.$$

If $|u_s| \leq \eta$ for every active position, then

$$\left| \frac{1}{t} \sum_{s=1}^t \log \rho_s^{\theta/\pi_{\text{roll}}} - \frac{1}{t} \sum_{s=1}^t z_s \right| \leq \eta.$$

This finite-prefix perturbation bound is verified by the Lean theorem `prefix_average_additive_perturbation`. Consequently, a gate computed from cached $(\pi_{\text{old}}, \pi_{\text{roll}})$ ratios controls the current $(\pi_{\theta}, \pi_{\text{roll}})$ prefix statistic up to an explicit threshold margin whenever the PPO update has the stated per-token log-ratio bound. This is a coupling condition for transferring the sample-level proxy; it is not needed for the purely algebraic prefix comparisons below. The corresponding finite-active-set statement, which excludes EOS-padding positions from both the average and its denominator, is checked by `finset_average_additive_perturbation` in `formal/ProMax.lean`.

Granularity The mask is computed at *per-position* granularity: a single trajectory can contribute gradients at some positions while being masked at others. This can be viewed as a prefix-level refinement of the sequence-level rejection mechanism in TRM.

Notation To avoid overloading symbols, we reserve (ϵ, δ) for the divergence quantities in Section 2. The ratio-truncation thresholds are denoted by $(\epsilon_{\text{low}}, \epsilon_{\text{high}})$, defining the acceptance interval

$$[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}].$$

For convenience, we also use the equivalent log-thresholds

$$\tau_+ := \log(1 + \epsilon_{\text{high}}), \quad \tau_- := -\log(1 - \epsilon_{\text{low}}),$$

so that $\rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ is equivalent to $\log \rho_t \in [-\tau_-, \tau_+]$. The implementation performs this comparison in exponentiated space: it computes the prefix log-average, applies $\exp$, and compares with $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$. Under the explicit positivity conditions $1 - \epsilon_{\text{low}} > 0$ and $1 + \epsilon_{\text{high}} > 0$, the equivalence between these two forms is verified by `ImplementationBridge.prefix_exp_gate_iff_log_gate` and `ImplementationBridge.prefix_gate_thresholds_match`. In the deviation-pattern analysis below, we use ε to denote a uniform bound on the per-token log-ratios $\log \rho_t$.

Theorem 4.7 (Causality-Aware Masking with Conditional Dominance). *Let M_t^{pre} , M_t^{tok} , and M_t^{seq} denote the prefix-level mask (Definition 4.5), the token-level IcePop mask, and a sequence-level (GSPO-style) mask respectively, all with the same acceptance interval $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ for some $\epsilon_{\text{low}} \in (0, 1)$ and $\epsilon_{\text{high}} > 0$. Concretely, the sequence-level mask is defined by a single statistic*

$$\rho^{\text{seq}} := \exp\left(\frac{1}{T} \sum_{s=1}^T \log \rho_s\right) = \left(\prod_{s=1}^T \rho_s\right)^{\frac{1}{T}},$$

and $M_t^{\text{seq}} := \mathbb{I}[\rho^{\text{seq}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]]$ for all t . Assume mutual policy support (Assumption 2.3). Then:

- (a) **Early deviation: the prefix mask rejects a (possibly longer) prefix.** Assume $\rho_1 \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ and $|\log \rho_t| \leq \varepsilon$ for all $t > 1$, where $0 \leq \varepsilon \leq \min(\tau_+, \tau_-)$. Then token-level IcePop masks only position $t = 1$, while the prefix mask rejects at least position $t = 1$. Moreover, if $\log \rho_1 > \tau_+$ then for every position t satisfying

$$t < \frac{\log \rho_1 + \varepsilon}{\tau_+ + \varepsilon}, \tag{14}$$

we have $M_t^{\text{pre}} = 0$. Symmetrically, if $\log \rho_1 < -\tau_-$ then for every position t satisfying

$$t < \frac{-\log \rho_1 + \varepsilon}{\tau_- + \varepsilon}, \quad (15)$$

we have $M_t^{\text{pre}} = 0$. In particular, in the early-deviation regime the prefix mask rejects at least as many tokens as token-level IcePop.

- (b) **Late deviation: the prefix mask is more local than sequence masking.** Suppose there exists $t^* \in \{1, \dots, T\}$ such that $|\log \rho_t| \leq \varepsilon$ for all $t \neq t^*$ and $\rho_{t^*} \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$, where $0 \leq \varepsilon \leq \min(\tau_+, \tau_-)$. Then prefix masking can only occur at positions $t \geq t^*$, so

$$|\{t : M_t^{\text{pre}} = 0\}| \leq T - t^* + 1.$$

In contrast, sequence-level masking is all-or-nothing: $|\{t : M_t^{\text{seq}} = 0\}| \in \{0, T\}$. Moreover, in the pure single-spike case where $\rho_t = 1$ for $t \neq t^*$ and $\rho_{t^*} > 1$, one has

$$\log \rho_{t^*}^{\text{pre}} = \frac{1}{t^*} \log \rho_{t^*} \geq \frac{1}{T} \log \rho_{t^*} = \log \rho^{\text{seq}},$$

For every $t^* < T$, choosing $\log \rho_{t^*} = \tau_+(t^* + T)/2$ gives $M_{t^*}^{\text{pre}} = 0$ while $M_t^{\text{seq}} = 1$ for all t . The lower-spike choice $\log \rho_{t^*} = -\tau_-(t^* + T)/2$ gives the same separation at the lower threshold. At $t^* = T$, the prefix and sequence statistics coincide, so this strict separation at the spike is impossible.

- (c) **Sustained one-sided violation: all methods reject.** If $\log \rho_s > \tau_+$ for all s (upper violation), or $\log \rho_s < -\tau_-$ for all s (lower violation), then $M_t^{\text{tok}} = M_t^{\text{pre}} = M_t^{\text{seq}} = 0$ for all t , so each of the three methods rejects all T tokens. No monotonicity in s is required.

Proof sketch. The full proof is in Appendix D.1. We highlight the key steps.

Part (a). The ε -bounded condition implies $\log \rho_t \in [-\tau_-, \tau_+]$ for all $t > 1$, hence IcePop masks only $t = 1$. For the prefix statistic, note that

$$\log \rho_t^{\text{pre}} = \frac{1}{t} \sum_{s=1}^t \log \rho_s = \frac{1}{t} \left(\log \rho_1 + \sum_{s=2}^t \log \rho_s \right).$$

Using the bounds

$$\begin{aligned} \sum_{s=2}^t \log \rho_s &\geq -(t-1)\varepsilon, \\ \sum_{s=2}^t \log \rho_s &\leq (t-1)\varepsilon, \end{aligned}$$

which yield the sufficient conditions Equations (14) and (15).

Part (b). For $t < t^*$, all active log-ratios in the prefix satisfy $\log \rho_s \in [-\tau_-, \tau_+]$, so repeated application of Lemma 4.6 implies $\rho_t^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ (equivalently, $\log \rho_t^{\text{pre}} \in [-\tau_-, \tau_+]$) and hence $M_t^{\text{pre}} = 1$. Therefore masking can only occur at positions $t \geq t^*$, yielding the stated bound. The single-spike comparison follows by specializing to $\rho_t = 1$ for $t \neq t^*$ and $\rho_{t^*} > 1$, which gives $\log \rho^{\text{seq}} = \log \rho_{t^*}/T$ and $\log \rho_{t^*}^{\text{pre}} = \log \rho_{t^*}/t^*$, so $\log \rho_{t^*}^{\text{pre}} \geq \log \rho^{\text{seq}}$ since $t^* \leq T$.

Part (c). If each $\log \rho_s$ is strictly beyond the same threshold, each token mask rejects and both the running and full-sequence averages are strictly beyond that threshold. Thus all three masks reject every token. $\square$

The complete three-case statement, binary rejection counts, both explicit spike constructions, and the final-position equality are verified in `formal/PrefixMasking.lean`. In particular, the early and late cases retain the stated bounded background noise. The active-position identities above transfer the prefix mask and its rejection count to the padded implementation in exact real arithmetic.

Remark 4.8 (Same-Sign but In-Bound Deviations). When all $\log \rho_s$ share the same sign but do not all exceed the corresponding threshold, the prefix log-statistic $\log \rho_t^{\text{pre}}$ need not be monotone in t and may return within bounds after crossing a threshold. In this regime there is no unconditional dominance ordering between prefix masking and token-level masking. If every token log-ratio lies in the acceptance interval, every prefix average lies in it as well, so repeated in-bound deviations alone cannot trigger this geometric-mean gate. Prefix rejection records a running average outside the interval, which can persist after individual tokens have returned within bounds.

4.4 What the Accepted Prefix Controls

To connect prefix filtering to the context-shift factor in Eq. (5), it is convenient to make explicit the change of measure between prefix context visitation distributions under π_{roll} and π_θ . For each $t \in \{1, \dots, T\}$, define the prefix density ratio random variable

$$W_t := \prod_{s=1}^t \rho_s, \quad \log W_t = \sum_{s=1}^t \log \rho_s.$$

In the finite history-tree model, d_t^π denotes the normalized path-mass distribution on length- $(t-1)$ prefix contexts. The next two lemmas show that W_t is exactly the Radon–Nikodym derivative between the length- t prefix distributions (indexed as d_{t+1}) and that the resulting context shift admits a clean ratio-residual identity.

Lemma 4.9 (Prefix density-ratio factorization). *Under the mutual-support Assumption 2.3, a finite vocabulary, and the normalized autoregressive history-tree model, for any $t \in \{1, \dots, T\}$ and any reachable context $c_{t+1} = (x, y_{\leq t})$ such that $d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) > 0$,*

$$\frac{d_{t+1}^{\pi_\theta}(c_{t+1})}{d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})} = \prod_{s=1}^t \frac{\pi_\theta(y_s \mid c_s)}{\pi_{\text{roll}}(y_s \mid c_s)} = W_t.$$

Lemma 4.10 (Context-shift TV as an expectation of the prefix ratio residual). *Under the mutual-support Assumption 2.3, a finite vocabulary, and the same normalized history-tree model, for any $t \in \{1, \dots, T\}$,*

$$\|d_{t+1}^{\pi_\theta} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} = \frac{1}{2} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [|W_t - 1|].$$

The prefix gate further enforces a deterministic per-position envelope on this drift along the accepted prefixes.

Theorem 4.11 (Per-position drift control on accepted prefixes). *Fix a finite vocabulary and a normalized history-tree model satisfying the mutual-support Assumption 2.3. Fix $\epsilon_{\text{low}} \in (0, 1)$ and $\epsilon_{\text{high}} > 0$, and write $\tau_+ := \log(1 + \epsilon_{\text{high}})$ and $\tau_- := -\log(1 - \epsilon_{\text{low}})$. Let M_t^{pre} be defined as in Eq. (13). Then for any $t \in \{1, \dots, T\}$, on the event $\{M_t^{\text{pre}} = 1\}$,*

$$e^{-t\tau_-} \leq W_t \leq e^{t\tau_+}, \quad |W_t - 1| \leq \max\{e^{t\tau_+} - 1, 1 - e^{-t\tau_-}\}.$$

Consequently,

$$\begin{aligned}\mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}} | W_t - 1] &\leq \max\{e^{t\tau_+} - 1, 1 - e^{-t\tau_-}\} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}}] \\ &\leq \max\{e^{t\tau_+} - 1, 1 - e^{-t\tau_-}\}.\end{aligned}$$

Why this matters for our method. Lemma 4.9 makes explicit the causal object that prefix filtering controls: the cumulative product of token ratios is exactly the density ratio between prefix context distributions. Lemma 4.10 then rewrites the context-shift factor in Eq. (5) as an expectation of a *prefix* ratio residual. Finally, Theorem 4.11 shows that our prefix gate enforces a per-position bounded change-of-measure regime on the *accepted* prefixes. This is the sense in which the mask acts as a prefix-level Causal Trust Region proxy: updates are filtered to exclude positions whose prefixes induce extreme reweighting of the rollout context distribution. Proofs are in Appendix D.3.

The bound concerns the unnormalized mass of accepted prefixes. It permits an action-dependent gate and gives no unbiasedness claim for the masked policy gradient. For the implementation, the policy pair supplying the gate is $(\pi_{\text{roll}}, \pi_{\text{old}})$; the coupling condition in Section 4.3 controls its transfer to $(\pi_{\text{roll}}, \pi_{\theta})$ during a stale-snapshot update. Under an additional conditional log-ratio concentration assumption, Appendix D.4 gives accepted-mass and effective-horizon bounds, with all assumptions and proofs stated there.

4.5 Connection to the Adaptive Bound

We relate prefix masking to the per-position, causal structure that appears in trust-region error bounds for long-horizon LLM-RL [Li et al., 2025]. In our notation, these bounds analyze the policy pair $(\pi_{\text{roll}}, \pi_{\theta})$.

Remark 4.12 (KL Chain Rule for the Cumulative Log-Ratio). Under the mutual-support Assumption 2.3, with a fixed (or EOS-padded) finite horizon as in Definition E.7, consider the finite normalized history-tree model. Define the cumulative conditional reverse-KL cost recursively by $\text{CKL}_0(h) = 0$ and

$$\text{CKL}_{n+1}(h) := D_{\text{KL}}(\pi_{\text{roll}}(\cdot|h) \parallel \pi_{\theta}(\cdot|h)) + \sum_a \pi_{\text{roll}}(a|h) \text{CKL}_n(h \parallel a). \quad (16)$$

Then, for every $t \leq T$,

$$\mathbb{E}_{\pi_{\text{roll}}}\left[\sum_{s=1}^t \log \rho_s\right] = -\text{CKL}_t(\emptyset) = -D_{\text{KL}}(d_{t+1}^{\pi_{\text{roll}}} \parallel d_{t+1}^{\pi_{\theta}}), \quad (17)$$

where d_{t+1}^{π} is the length- t prefix law obtained by multiplying the normalized conditional token probabilities. The expectation, recursion, and joint-prefix KL are evaluated on these same finite history trees.

This identity is the KL chain rule under autoregressive factorization; see, e.g., Li et al. [2025]. Its finite conditional weighted-sum algebra is checked by `REINFORCEProMax.finite_conditional_kl_chain` in `formal/KLChain.lean`. The fully recursive history-tree statement is checked by `AutoregressiveTree.chain_log_ratio_eq_neg_chain_kl` in `formal/TreeKL.lean`. The same file verifies `trajectory_log_ratio_expectation` and `jointPrefixKL_eq_chainKL`: the sampled cumulative log-ratio has the stated expectation, and the independently constructed joint-prefix KL equals the recursive conditional cost.

Remark 4.13 (Prefix Filtering as a Sample-Level Proxy). Thresholding ρ_t^{pre} is equivalent to a per-position, sample-level constraint on the prefix average log-ratio:

$$\begin{aligned}\rho_t^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}] &\iff \log \rho_t^{\text{pre}} \in [\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})] \\ &\iff \frac{1}{t} \sum_{s=1}^t \log \rho_s \in [\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})].\end{aligned}$$

In contrast, Remark 4.12 is an *expectation* identity over the behavior distribution. Therefore, the threshold condition should be interpreted as a practical, sample-level proxy for Causal Trust Region filtering rather than a population-level guarantee without additional assumptions. For the finite joint-prefix distributions constructed above, Pinsker’s inequality gives:

$$\|d_{t+1}^{\pi_\theta} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \sqrt{\frac{D_{\text{KL}}(d_{t+1}^{\pi_{\text{roll}}} \| d_{t+1}^{\pi_\theta})}{2}}.$$

The finite inequality, including common zero-probability outcomes, is proved by `finite_pinsker` in `formal/Pinsker.lean`. Its application to the actual history-tree prefix laws and their KL chain rule is checked by `prefixTV_le_sqrt_jointPrefixKL` and `prefixTV_le_sqrt_chainKL` in `formal/TreePinsker.lean`.

Remark 4.14 (Per-Position Sample-Level Causal Trust Region Constraint). Under prefix Causal Trust Region masking with threshold $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$, for all accepted tokens ($M_t^{\text{pre}} = 1$), the sample-level cumulative log-ratio satisfies:

$$\log \rho_t^{\text{pre}} = \frac{1}{t} \sum_{s=1}^t \log \rho_s \in [\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})].$$

This provides a per-position, sample-level trust region constraint analogous to the per-position structure in the Adaptive bound. The constraint operates on individual trajectories; its connection to the population-level KL is through the expectation identity in Eq. (17).

4.6 The Combined Loss

The theoretical unclipped surrogate uses the total ratio $\rho_t^{\theta/\pi_{\text{roll}}} = \pi_\theta(y_t|c_t)/\pi_{\text{roll}}(y_t|c_t)$. The implementation keeps PPO’s frozen snapshot and its clipped objective. Define

$$\ell_t^{\text{PPO}} := -\min\left(r_t^{\theta/\pi_{\text{old}}} \hat{A}_t, \text{clip}\left(r_t^{\theta/\pi_{\text{old}}}, 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}\right) \hat{A}_t\right), \quad w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} := \frac{\pi_{\text{old}}(y_t|c_t)}{\pi_{\text{roll}}(y_t|c_t)}.$$

The implemented REINFORCE Pro Max loss is the masked, vLLM-corrected PPO loss

$$\mathcal{L}^{\text{ProMax}} := \text{MaskedMean}_t \left[m_t M_t^{\text{pre}} \text{stopgrad}\left(w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}\right) \ell_t^{\text{PPO}} \right]. \quad (18)$$

Here `MaskedMeant` is the same action-mask reduction used by the implementation, M_t^{pre} is computed from the prefix of $w_s^{\pi_{\text{old}}/\pi_{\text{roll}}}$, and $m_t \in \{0, 1\}$ removes EOS-padding positions. When the PPO clipping branch is inactive, the product of the retained weight and the PPO ratio is $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} r_t^{\theta/\pi_{\text{old}}} = \rho_t^{\theta/\pi_{\text{roll}}}$, even for a stale snapshot. Recovering the raw population surrogate additionally requires accounting for the prefix gate, adaptive normalization, and reduction. Proposition 5.3 gives the exact finite-batch decomposition with all these components present.

Gradient flow Gradients propagate through the PPO ratio $r_t^{\theta/\pi_{\text{old}}}$ only. The action mask, prefix mask M_t^{pre} , vLLM coefficient $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$, and normalized advantage $\hat{A}_t$ are detached from the computation graph. When either $m_t = 0$ or $M_t^{\text{pre}} = 0$, the loss at position t is zero and contributes no gradient.

Analyzed training configuration Equation (18) specifies the standard PPO policy term: `policy_loss_type=ppo`, with vLLM correction enabled and `vllm_is_correction_type` set to `reinforce_pro`. The optional `dual_clip` argument is unset. The analyzed complete loss also has no reference-KL, entropy, or MoE auxiliary loss contribution; the corresponding coefficients must be zero or the branches disabled. Adding these terms changes the optimizer’s update and requires retaining their contributions separately. The reduction qualifications in Section 5 still apply to microbatch accumulation.

5 The Unified Framework

5.1 Why the Two Components Are Complementary

The error decomposition (Equation (5)) has two factors at each position t :

$$|\text{Error}| \leq \sum_{t=1}^T \underbrace{2\|g_t\|_{\infty}}_{\text{advantage factor}} \cdot \underbrace{\|d_t^{\pi_{\theta}} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}}}_{\text{context shift factor}}.$$

Component 1 (REINFORCE Max) controls the sampled advantage statistics. The leave-one-out baseline provides conditional independence between each sample and its baseline given the prompt (Remark C.1), and the ideal adaptive normalization enforces zero empirical mean and unit empirical variance on non-zero tokens when both sign groups are nonempty and the ideal factors are used without the additive stabilizer or clamps (Definition 3.2). The implemented safeguards have explicit moment residuals (Proposition C.6). This controls empirical scale; it does not directly bound the true $\|g_t\|_{\infty}$ in the population error decomposition.

Component 2 (REINFORCE Pro) targets the context-shift factor by filtering updates that exhibit large prefix drift. In the implementation this drift is measured between the rollout policy π_{roll} and the frozen PPO snapshot π_{old} ; when the snapshot is current at the start of an update, this is the same pair as $(\pi_{\text{roll}}, \pi_{\theta})$. The prefix-level mask rejects tokens whose prefix average log-ratio leaves the threshold interval (Remark 4.14). The KL chain rule yields an expectation identity relating cumulative log-ratios to a joint KL divergence between prefix distributions (Remark 4.12); together with the threshold equivalence (Remark 4.13), this motivates viewing prefix masking as a sample-level proxy for Causal Trust Region filtering. For stale snapshots, the finite-prefix coupling bound in Section 4.3 states the extra condition needed to transfer the proxy to the current policy.

Remark 5.1 (Heuristic Complementarity). The discussion above should be interpreted as heuristic rather than as a formal inequality. It highlights that REINFORCE Max targets the advantage factor in Eq. (5), while REINFORCE Pro targets policy drift between π_{roll} and π_{θ} through sample-level prefix filtering. Together, they address two distinct sources of instability in the surrogate optimization pipeline.

Remark 5.2 (Moment Control and Gradient Variance). The RLOO second-moment decomposition (Remark C.1) and the normalization constraints (Equation (8)) quantify different parts of the estimator: a conditionally independent leave-one-out baseline contribution and realized token moments, respectively. Neither alone proves that ProMax has lower gradient variance than every

unnormalized estimator. Prefix masking changes the retained contributions, with the exact signed change characterized next.

5.2 Exact Connection to the Implemented Objective

The population equivalence in Proposition 2.8 provides the reference objective for critic-free learning. The following identity isolates the changes introduced by ProMax on an actual finite batch, preserving the prefix gate and asymmetric normalization.

Proposition 5.3 (Finite-Batch ProMax Objective Decomposition). *Let $k = (i, t)$ index a sampled response and position. Write $A_k = R_i - b_i$ for its raw RLOO signal, $\hat{A}_k$ for the actual normalized signal (including any configured fallback), $r_k = \pi_\theta / \pi_{\text{old}}$, $w_k = \pi_{\text{old}} / \pi_{\text{roll}}$, $\rho_k = r_k w_k$, and $M_k \in \{0, 1\}$ for the cached prefix gate. Let $\omega_k \geq 0$ include the action mask and the selected reduction: $\omega_{i,t} = m_{i,t} / \sum_{j,s} m_{j,s}$ for a token mean, or $\omega_{i,t} = m_{i,t} / (B \sum_s m_{i,s})$ for a mean of B response means, with positive denominators. Assume the ratio coefficients satisfy $r_k \geq 0$ and $w_k \geq 0$ on the represented support, as they do for probability ratios. Let c_k be the clipped value of r_k . For the standard PPO branch in Eq. (18), define*

$$\begin{aligned} S_{\text{PM}} &= -\mathcal{L}^{\text{ProMax}} = \sum_k \omega_k M_k w_k \min(r_k \hat{A}_k, c_k \hat{A}_k), & U &= \sum_k \omega_k \rho_k A_k, \\ G &= \sum_k \omega_k \rho_k (1 - M_k) A_k, & N &= \sum_k \omega_k \rho_k M_k (\hat{A}_k - A_k), \\ C_{\text{clip}} &= \sum_k \omega_k M_k w_k (r_k \hat{A}_k - \min(r_k \hat{A}_k, c_k \hat{A}_k)). \end{aligned}$$

Then

$$S_{\text{PM}} = U - G + N - C_{\text{clip}}, \quad C_{\text{clip}} \geq 0, \quad (19)$$

and

$$|S_{\text{PM}} - U| \leq \sum_k \omega_k \rho_k (1 - M_k) |A_k| + \sum_k \omega_k \rho_k M_k |\hat{A}_k - A_k| + C_{\text{clip}}.$$

Proof. Add and subtract $\sum_k \omega_k \rho_k M_k A_k$ and $\sum_k \omega_k \rho_k M_k \hat{A}_k$, and use $\rho_k = r_k w_k$. Each clipping deduction is nonnegative because $\min(r_k \hat{A}_k, c_k \hat{A}_k) \leq r_k \hat{A}_k$. The last inequality follows from the triangle inequality and nonnegative weights. These finite-batch identities and bounds are verified in `formal/ProMaxObjective.lean`; the theorem mapping is recorded in the proof audit. $\square$

The gate term G and normalization term N are signed: removing a negative raw contribution can increase the batch objective, and asymmetric scaling can move it in either direction. The identity therefore exposes exactly where the chosen stabilization changes the raw surrogate. It applies for every realized batch, including random response lengths and cached mismatch ratios. Taking an expectation of a randomly length-normalized batch is a separate weighting question; the identity does not identify $\mathbb{E}[U]$ with the unweighted population surrogate without checking that reduction.

The standard PPO restriction is essential for the nonnegative clipping deduction. With normalized advantage -1 , current-to-snapshot ratio 4 , upper clipped ratio $6/5$, and unit gate and IS weight, standard PPO gives the maximization contribution $\min(-4, -6/5) = -4$. The optional dual-clip coefficient 3 changes this to $\max(-4, -3) = -3$, so the deduction from the unclipped contribution is $-4 - (-3) = -1 < 0$. This counterexample is verified by `dual_clip_negative_penalty_counterexample` in `formal/ProMaxObjective.lean`. The proposition and certificates using its standard PPO correction do not directly describe a dual-clip run.

The reduction determines the population target. For one sampled response, let Z be its raw surrogate contribution and $L > 0$ its number of active action tokens. A response-level reduction targets $\mathbb{E}[Z/L]$, whereas a global token-level reduction targets the ratio $\mathbb{E}[Z]/\mathbb{E}[L]$. These are different under random response lengths: with $\kappa = \mathbb{E}[Z]/\mathbb{E}[L]$,

$$\mathbb{E}[Z/L] - \kappa = -\frac{\mathbb{E}[(L - \mathbb{E}[L])Z/L]}{\mathbb{E}[L]}.$$

Thus the difference is an explicit length–contribution covariance term. For concentration with RLOO, redefine Z and L as the sums over a full prompt group. The response-level statistic of that group is $n^{-1} \sum_i Z_i/L_i$, which generally differs from Z/L ; grouping for independence does not change the chosen reduction. For n i.i.d. blocks with $L \geq \ell > 0$, the self-normalized token estimator obeys

$$\mathbb{E} \left[\left(\frac{\bar{Z}_n}{\bar{L}_n} - \kappa \right)^2 \right] \leq \frac{\mathbb{E}[(Z - \kappa L)^2]}{n\ell^2}, \quad \Pr \left[\left| \frac{\bar{Z}_n}{\bar{L}_n} - \kappa \right| \geq \varepsilon \right] \leq \frac{\mathbb{E}[(Z - \kappa L)^2]}{n\ell^2\varepsilon^2}.$$

The corresponding finite identities and bounds are checked by `ratio_expectation_gap`, `iid_token_mean_mse_of_ratio`, and `iid_token_mean_tail` in `formal/BatchReduction.lean`. For the implemented PPO path, `PolicyLoss` uses the active-token mean within each loss call; GSPO is the separate path that switches to a sequence-level reduction. There is no independent CLI switch that makes the standard PPO path a response-mean estimator. The dynamic replay-buffer scale weights variable-size microbatches by response count, so it does not in general reconstruct a global token mean when response lengths differ. The finite reduction identities above therefore remain a qualification on how the implemented microbatch update relates to the population target; a token-level claim requires active-token weighting across the relevant accumulation.

There is also a useful population identity for the gate alone. At a fixed context, let $Q(a) = \mathbb{E}_{\pi_{\text{roll}}}[R \mid c, a]$, $V = \mathbb{E}_{\pi_{\text{roll}}}[R \mid c]$, and $\bar{b} = \mathbb{E}[b \mid c]$ for an independent baseline. For a fixed gate $M(c, a)$,

$$\mathbb{E}_{\pi_{\text{roll}}}[(R - b)\rho M \mid c] = \sum_a \pi_\theta(a \mid c)M(c, a)(Q(a) - V) + (V - \bar{b}) \sum_a \pi_\theta(a \mid c)M(c, a).$$

The finite Lean audit verifies this identity under the name `finite_masked_surrogate_identity`. When $M \equiv 1$, the last sum is one and the baseline offset is constant. For the actual prefix gate, which depends on the current sampled token, this sum is the current policy’s acceptance mass and can change during optimization even though the cached gate is detached. The formula identifies the resulting baseline term explicitly.

5.3 From the ProMax Objective to Reward Improvement

The preceding decomposition can be combined with the Adaptive bound to give a finite-sample performance certificate for the complete objective. The sampling unit must be a full RLOO group: sibling contributions share a baseline, whereas independently sampled groups provide independent blocks. This also permits random response lengths and an explicit finite prompt law.

Theorem 5.4 (A finite-sample ProMax performance certificate). *Let $p = \pi_{\text{roll}}$ and let $\{q_j : j \in \mathcal{J}\}$ be a finite family of candidate policies fixed before the evaluation data are sampled. Assume the finite prompt and vocabulary model, shared absorbing EOS padding to T , and the bounded-reward and mutual-support conditions of Theorem 2.10 for every candidate. Let B_j denote its Adaptive bound from Eq. (6). Use the sparse-reward, KL-free, reward-unclipped core mode of Section 3.*

Sample $K \geq 1$ independent blocks. Each block samples a prompt from the same prompt law and then $n \geq 2$ conditionally i.i.d. responses from p . Write $A_i = R_i - b_i$ for the within-block RLOO signal and assume each response has at least $\ell > 0$ active tokens. For a block, define

$$Z_j = \sum_{i=1}^n A_i \sum_{t=1}^T (\rho_{j,i,t} - 1), \quad L = \sum_{i=1}^n T_i, \quad \rho_{j,i,t} = \frac{q_j(y_{i,t} \mid c_{i,t})}{p(y_{i,t} \mid c_{i,t})}.$$

Here T_i counts active tokens, and post-EOS ratio residuals are zero. Let $\mu_L = \mathbb{E}[L]/n > 0$, $\mathcal{S}_j = \mathcal{L}_p(q_j)$, and

$$\kappa_j = \frac{\mathcal{S}_j}{\mu_L}, \quad v_j = \mathbb{E}[(Z_j - \kappa_j L)^2], \quad \hat{\kappa}_j = \frac{\sum_{b=1}^K Z_{j,b}}{\sum_{b=1}^K L_b}.$$

Then $\mathbb{E}[Z_j] = n\mathcal{S}_j$, and for any fixed $\eta_j > 0$,

$$\Pr[\exists j \in \mathcal{J} : |\hat{\kappa}_j - \kappa_j| \geq \eta_j] \leq \sum_{j \in \mathcal{J}} \frac{v_j}{K(n\ell)^2 \eta_j^2}. \quad (20)$$

Evaluate the standard PPO ProMax objective at q_j and at p on this same batch, using a common frozen snapshot, prefix gate, advantages, and global active-token reduction. The snapshot has positive probability on the rollout support, and all three policies share the absorbing EOS transition. For any quantity F in Proposition 5.3, write $\Delta_j F = F(q_j) - F(p)$. Then

$$\hat{\kappa}_j = \Delta_j S_{\text{PM}} + \Delta_j G - \Delta_j N + \Delta_j C_{\text{clip}}. \quad (21)$$

Outside the event in Eq. (20), simultaneously for every j ,

$$J(q_j) - J(p) \geq \mu_L(\Delta_j S_{\text{PM}} + \Delta_j G - \Delta_j N + \Delta_j C_{\text{clip}} - \eta_j) - B_j. \quad (22)$$

Consequently, any selector $\hat{j}$ based on all these evaluation blocks has probability at most the right-hand side of Eq. (20) of both satisfying $\hat{\kappa}_{\hat{j}} > B_{\hat{j}}/\mu_L + \eta_{\hat{j}}$ and failing to improve true expected reward.

Proof. Within each prompt group, the RLOO baseline cancels because it is independent of its own response and the response ratio-residual sum has rollout mean zero. Averaging over prompts gives $\mathbb{E}[Z_j] = n\mathcal{S}_j$. Center Z_j by $\kappa_j L$, use $L \geq n\ell$, and apply the independent-block second-moment bound and a finite union bound. The exact batch decomposition at the two policies gives Eq. (21); the Adaptive bound then proves Eq. (22). The full finite model, objective conversion and selection event are checked in `formal/ProMaxBatch.lean` and `formal/ProMaxCertificate.lean`; see Appendix C.3. $\square$

The certificate retains the method’s prefix gate, asymmetric normalization, configured advantage fallback and PPO clipping. It quantifies the observed objective gain needed to overcome their signed changes, sampling uncertainty and policy-distribution shift. It does not require the gate or normalization to be independent of the batch. At the reference evaluation p , the PPO ratio is p/π_{old} , which need not be one; hence the reference clipping penalty cannot generally be discarded, and its difference can have either sign. The frozen gate remains the implemented π_{old}/p gate.

Evaluation sampling and group filtering. The independent-block assumption refers to the unconditioned group law $P(x) \prod_i p(y_i | x)$. The implementation’s optional `dynamic_filtering` rejects whole groups according to their average score. Conditioning on acceptance can couple sibling responses and reweight prompts, so the retained groups do not automatically satisfy this theorem’s sampling assumptions. This is distinct from the prefix gate inside the loss. For a concrete counterexample, fix one prompt, take two independent Bernoulli(1/2) responses with reward equal to score, and retain only groups with mean score strictly between zero and one. Only mixed pairs survive: the retained rewards have covariance $-1/4$, although their marginal means remain 1/2. With the score derivative taken with respect to the Bernoulli success probability, the raw group RLOO estimator has expectation 1 before filtering and 2 after filtering, while the reward derivative is 1. These exact finite identities are verified in `formal/GroupFiltering.lean`. Thus a policy trained with group filtering may still be assessed by this certificate on a separate, unfiltered conditionally i.i.d. evaluation batch, subject to the other hypotheses; reusing accepted training groups requires a separate sampling-law analysis.

The family may comprise policies trained on separate data, with the theorem applied conditionally on that training data. Retaining finitely many checkpoints fitted on the evaluation blocks does not by itself meet the fixed-family condition. Corollary 5.7 below extends the guarantee to fitting within a pre-specified, finitely covered policy class, with an explicit approximation and complexity cost. The quantities v_j , μ_L , and B_j are population quantities; a numerical use of the certificate needs valid values or conservative bounds, rather than replacing them by uncalibrated estimates from the same batch. In particular, the prefix gate alone does not establish the local TV/KL premises of B_j .

The unknown mean length can be removed by an exact rescaling with the observed token count. The resulting sampling bound exploits the additive ratio residual’s conditional mean-zero structure.

Corollary 5.5 (An observed-length certificate with additive drift energy). *Under the model and evaluation conditions of Theorem 5.4, let $D = \sum_b L_b$ and define*

$$\hat{\mathcal{S}}_j := \frac{D}{Kn} \hat{\kappa}_j = \frac{1}{K} \sum_{b=1}^K \frac{Z_{j,b}}{n}, \quad (23)$$

$$\mathcal{E}_j := \mathbb{E}_x \sum_{t=1}^T \mathbb{E}_{c_t \sim p(\cdot | x)} \sum_a p(a | c_t) \left(\frac{q_j(a | c_t)}{p(a | c_t)} - 1 \right)^2. \quad (24)$$

Zero-probability actions contribute zero. Then $\mathbb{E}[\hat{\mathcal{S}}_j] = \mathcal{S}_j$, and, for fixed $\xi_j > 0$,

$$\Pr \left[\exists j : |\hat{\mathcal{S}}_j - \mathcal{S}_j| \geq \xi_j \right] \leq \sum_j \frac{4R_{\max}^2 \mathcal{E}_j}{K \xi_j^2}. \quad (25)$$

Outside this event, simultaneously for all candidates,

$$J(q_j) - J(p) \geq \frac{D}{Kn} (\Delta_j S_{\text{PM}} + \Delta_j G - \Delta_j N + \Delta_j C_{\text{clip}}) - \xi_j - B_j. \quad (26)$$

Any data-dependent selector from the fixed family therefore has probability at most the right-hand side of Eq. (25) of passing $\hat{\mathcal{S}}_j > \xi_j + B_j$ with nonpositive true reward gain.

Proof. Write $H_j(y) = \sum_t (\rho_{j,t}(y) - 1)$. Normalization and support give $\mathbb{E}_p[\rho_{j,t} - 1 | c_t] = 0$. Expanding the square along the history tree therefore yields $\mathbb{E}_{x,y \sim p}[H_j(y)^2] = \mathcal{E}_j$. Since $|A_i| \leq 2R_{\max}$, the squared group mean satisfies $(Z_j/n)^2 \leq n^{-1} \sum_i A_i^2 H_j(y_i)^2$ and thus $\mathbb{E}[(Z_j/n)^2] \leq 4R_{\max}^2 \mathcal{E}_j$. The exact rescaling gives an unbiased average of independent blocks. Variance is bounded by the second

moment, so Chebyshev’s inequality and a union bound prove the stated probability bound. Apply the Adaptive performance bound and Eq. (21). The history-tree identity and full RLOO/objective composition are checked in `formal/SurrogateEnergy.lean` and `formal/EmpiricalCertificate.lean`; details appear in Appendix C.4. $\square$

This rescaling uses the observed D as part of a new estimator; it does not substitute a sample mean into the population-length certificate. The original objective, including all stabilization terms, remains explicit in Eq. (26). If $|q_j(a \mid c)/p(a \mid c) - 1| \leq d_j$ on positive- p actions at every history within the horizon, then $\mathcal{E}_j \leq Td_j^2$. Consequently, the sampling margin for an individual fixed failure budget $\alpha_j > 0$ can be controlled on the scale $R_{\max}d_j\sqrt{T/(K\alpha_j)}$. The exact energy may be smaller, and shared absorbing EOS transitions add zero energy after termination. This is a statement about sampling error; the separate Adaptive cost B_j retains its own horizon dependence. The group-square argument alone does not quantify the n dependence; Proposition 5.6 computes it with shared baselines retained. Numerical use still requires a valid bound on $\mathcal{E}_j$ (or d_j) and B_j ; a maximum ratio observed only in the evaluation batch is insufficient.

Proposition 5.6 (Exact RLOO group variance and prompt sampling). *Retain the finite model and sampling scheme of Corollary 5.5. At a fixed prompt x , let $X = R(x, Y) - \mathbb{E}_p[R(x, Y) \mid x]$, $H_j = \sum_t (\rho_{j,t} - 1)$, and define*

$$\theta_{j,x} = \mathbb{E}_p[XH_j \mid x] = \mathcal{S}_{j,x}, \quad M_{j,x} = \mathbb{E}_p[X^2H_j^2 \mid x], \quad \sigma_{R,x}^2 = \mathbb{E}_p[X^2 \mid x], \quad \mathcal{E}_{j,x} = \mathbb{E}_p[H_j^2 \mid x].$$

For the actual RLOO group mean $U_j = Z_j/n$, including its shared baselines,

$$V_{j,x}(n) := \text{Var}(U_j \mid x) = \frac{M_{j,x} - \theta_{j,x}^2}{n} + \frac{\sigma_{R,x}^2 \mathcal{E}_{j,x} + \theta_{j,x}^2}{n(n-1)}. \quad (27)$$

Let $\tau_j^2 = \mathbb{E}_x[(\mathcal{S}_{j,x} - \mathcal{S}_j)^2]$ and $V_j(n) = \tau_j^2 + \mathbb{E}_x[V_{j,x}(n)]$. Then

$$\mathbb{E}[(\hat{\mathcal{S}}_j - \mathcal{S}_j)^2] = \frac{V_j(n)}{K}, \quad \Pr[\exists j : |\hat{\mathcal{S}}_j - \mathcal{S}_j| \geq \xi_j] \leq \sum_j \frac{V_j(n)}{K\xi_j^2}. \quad (28)$$

With $c_n = 4/n + 1/[n(n-1)] \leq 5/n$, bounded rewards imply

$$V_j(n) \leq \min\{4R_{\max}^2 \mathcal{E}_j, \tau_j^2 + c_n R_{\max}^2 \mathcal{E}_j\}. \quad (29)$$

The reward certificate in Eq. (26) and the guarantee for selection from a fixed candidate family therefore hold with the failure probability in Eq. (28).

Proof. RLOO is invariant to a common reward translation. For centered rewards, its group mean is the unbiased sample covariance $U_j = (n \sum_i X_i H_{j,i} - (\sum_i X_i)(\sum_i H_{j,i}))/[n(n-1)]$. Expanding its square under the actual i.i.d. response law gives Eq. (27); the mixed moments retain all shared sample terms. The law of total variance gives $V_j(n)$, and independent blocks give the exact $1/K$ mean-square factor. Chebyshev’s inequality and a union bound yield the confidence result. Finally, $M_{j,x} \leq 4R_{\max}^2 \mathcal{E}_{j,x}$ and $\sigma_{R,x}^2 \leq R_{\max}^2$; the total coefficient of $\theta_{j,x}^2$ in Eq. (27) is nonpositive for $n \geq 2$. Combine this bound with the previous group second-moment bound. The full finite-law calculation and its connection to the objective are checked in `formal/RLOOCovariance.lean` and `formal/GroupCertificate.lean`; see Appendix C.5. $\square$

The first term of Eq. (27) is the variance with the population mean reward as baseline, divided by n . The second term quantifies the additional variance of estimating that baseline from the other responses in the same group. Both coefficients are nonnegative. For a single fixed prompt, $\tau_j^2 = 0$, so the sampling margin can scale as $R_{\max} \sqrt{\mathcal{E}_j / (Kn\alpha_j)}$ for a fixed failure budget $\alpha_j > 0$. For heterogeneous prompts, increasing n reduces the conditional response-sampling component, while τ_j^2/K remains a separate component under this prompt-group sampling scheme. The estimator is still the exact corrected ProMax objective in Eq. (23); its gate, normalization and clipping terms remain present in the resulting reward guarantee.

Corollary 5.7 (Policy fitting with a fixed finite cover). *Retain the finite model and group sampling scheme of Proposition 5.6. Let $\{q_\theta : \theta \in \Theta\}$ be a policy class fixed before evaluation, where Θ may be continuous. Assume each member satisfies the reward and support conditions, and write $H_\theta(x, y) = \sum_t (\rho_{\theta,t} - 1)$. Fix finitely many centers q_{θ_j} , $j \in \mathcal{J}$, a map $\nu : \Theta \rightarrow \mathcal{J}$, and a radius $\zeta \geq 0$, all before evaluation, such that*

$$|H_\theta(x, y) - H_{\theta_{\nu(\theta)}}(x, y)| \leq \zeta \quad \text{for every } \theta, x, y. \quad (30)$$

For fixed $\xi_j > 0$, the exact corrected statistic from Eq. (23) satisfies

$$\Pr[\exists \theta \in \Theta : |\hat{\mathcal{S}}_\theta - \mathcal{S}_\theta| \geq \xi_{\nu(\theta)} + 3R_{\max}\zeta] \leq \sum_{j \in \mathcal{J}} \frac{V_j(n)}{K\xi_j^2}, \quad (31)$$

where $V_j(n)$ is the exact group variance at center q_{θ_j} . If the Adaptive bound applies to each member with cost B_θ , then, outside the same event, simultaneously for all θ ,

$$J(q_\theta) - J(p) \geq \hat{\mathcal{S}}_\theta - B_\theta - \xi_{\nu(\theta)} - 3R_{\max}\zeta. \quad (32)$$

Thus an arbitrary fit $\hat{\theta}$ based on all evaluation blocks has probability at most the right-hand side of Eq. (31) of passing the positive-improvement threshold while failing to improve expected reward. The fit need not be a cover center.

Proof. Since $|R_i - b_i| \leq 2R_{\max}$, the cover condition gives $|\hat{\mathcal{S}}_\theta - \hat{\mathcal{S}}_{\theta_{\nu(\theta)}}| \leq 2R_{\max}\zeta$ on every realized batch. The population representation $\mathcal{S}_\theta = \mathbb{E}_p[RH_\theta]$ gives $|\mathcal{S}_\theta - \mathcal{S}_{\theta_{\nu(\theta)}}| \leq R_{\max}\zeta$. The triangle inequality transfers the simultaneous center bound in Proposition 5.6 to the whole class. The Adaptive lower bound and the exact objective identity hold pointwise for the fitted policy. The finite-law transfer and false-improvement event are checked in `formal/PolicyCover.lean`; see Appendix C.6. $\square$

A sufficient construction for Eq. (30) is a uniform token-ratio cover of radius d : if $|q_\theta(a | h)/p(a | h) - q_{\theta_{\nu(\theta)}}(a | h)/p(a | h)| \leq d$ at every prompt and represented history of length less than T , then $\zeta = Td$ suffices. Shared zero-probability actions use the same ratio convention as the finite model. A cover of the additive residual itself can be tighter. The fixed cover costs a finite sum of center error budgets; finer covers reduce the approximation term but can increase this sum. This result allows data reuse within the specified class without changing ProMax’s gate, advantages, or clipping. It does not establish a small cover for an unrestricted neural policy class, and checking the cover only on the observed responses is insufficient.

Corollary 5.8 (Calibrating the certificate from logit changes). *Retain the finite model and evaluation scheme of Corollary 5.5. Suppose that, for each candidate j , every context within the horizon admits positive weights satisfying*

$$q_j(a | c) = \frac{p(a | c)v_j(a | c)}{\sum_b p(b | c)v_j(b | c)}, \quad 1 \leq v_j(a | c) \leq e^{\omega_j}, \quad \omega_j \geq 0. \quad (33)$$

Set $C_j = (e^{\omega_j} - 1)^2 / (4e^{\omega_j})$ and $\epsilon_j = \sqrt{C_j}/2$, and instantiate the Adaptive bound with local TV/KL bounds (ϵ_j, C_j) . Then

$$\begin{aligned} \mathcal{E}_j &\leq TC_j, & B_j &\leq \bar{B}_j, \\ \bar{B}_j &:= 4R_{\max}\epsilon_j \sum_{t=1}^T \min\left\{1, (T-t)\epsilon_j, \sqrt{(T-t)C_j/2}\right\}. \end{aligned} \tag{34}$$

Consequently, the corrected statistic in Eq. (23) satisfies

$$\Pr\left[\exists j : |\hat{\mathcal{S}}_j - \mathcal{S}_j| \geq \xi_j\right] \leq \sum_j \frac{4R_{\max}^2 TC_j}{K\xi_j^2}. \tag{35}$$

Outside this event, $J(q_j) - J(p) \geq \hat{\mathcal{S}}_j - \xi_j - \bar{B}_j$ simultaneously for all j . Thus a positive right-hand side certifies reward improvement, including after selection from the fixed family.

Proof. For a normalized tilt $q_a = p_a v_a / \mathbb{E}_p[v]$ with $v_a \in [\ell, u]$, $\ell > 0$, finite weighted-moment inequalities give $\chi^2(q\|p), D_{\text{KL}}(p\|q) \leq (u - \ell)^2 / (4\ell u)$ and $\text{TV}(p, q) \leq \frac{1}{2} \sqrt{(u - \ell)^2 / (4\ell u)}$. Use $(\ell, u) = (1, e^{\omega_j})$, sum the conditional energy over the history tree, and bound each rollout-averaged local TV by ϵ_j . Substitute these bounds into Corollary 5.5. The complete finite-policy composition is checked in `formal/LogitBounds.lean`; see Appendix C.7. $\square$

For softmax rows, let $\Delta z_{j,a}(c)$ be the change from rollout to candidate logits. If its span over actions is at most ω_j , set $v_j(a | c) = \exp(\Delta z_{j,a}(c) - \min_b \Delta z_{j,b}(c))$. Softmax normalization gives Eq. (33) exactly; no lower bound on individual positive token probabilities is needed. Common masked supports use the same construction on their allowed actions, and common absorbing EOS rows take $v_j = 1$. Similarly, a fixed logit cover whose centers have rollout-relative span at most ω , and whose members have center-relative span at most r , gives the response-residual radius $\zeta = Te^{\omega}(e^r - 1)$ in Corollary 5.7. These are optional, uniform conditions over all represented contexts; prefix gating or PPO clipping alone does not ensure them, and observed batch maxima do not replace them. The exact energy, occupancy-weighted Adaptive cost, and shared-baseline variance remain available when tighter.

5.4 Algorithm

Algorithm 1 presents the complete REINFORCE Pro Max training procedure. It unifies adaptive advantage normalization (REINFORCE Max, Section 3) and prefix-level Causal Trust Region masking (REINFORCE Pro, Section 4) into a single training loop. The algorithm processes one prompt at a time: it generates n rollout samples from the behavior policy, computes advantages with the RLOO baseline and adaptive normalization, then applies the prefix-level mask during the loss computation. We highlight the key design choices below.

Step-by-step walkthrough.

Rollout (Line 1). Samples are generated by a rollout engine (e.g., vLLM) running the behavior policy π_{roll} . The per-token log-probabilities $\log \pi_{\text{roll}}(y_t | c_t)$ are stored alongside the generated tokens. Together with the frozen snapshot log-probabilities, these cached values form $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$ and the total ratio factorization in Section 2. This design follows the trust-region anchoring choice analyzed in Qi et al. [2026]; see Appendix B.1.

Baseline and Token Expansion (Lines 3–4). The leave-one-out baseline (Definition 3.1) ensures that, under conditionally i.i.d. response sampling, each sample’s baseline b_i is independent of its own response and reward given the prompt, avoiding the correlated baseline artifacts of the mean

Algorithm 1 REINFORCE Pro Max: minimizing $\mathcal{L}^{\text{ProMax}}$ in Eq. (18).

Require: Prompt x ; n samples; thresholds $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$; KL-free, reward-unclipped core mode (`init_kl_coef` = 0, no KL loss, and no post-RLOO reward clipping)

- 1: **Rollout:** Generate $\{y^{(1)}, \dots, y^{(n)}\} \sim \pi_{\text{roll}}(\cdot|x)$; store $\log \pi_{\text{roll}}(y_t|c_t)$ and action masks m_t
- 2: **Reward:** Compute bounded scalar rewards $\{r_1, \dots, r_n\}$ (any length penalty is included in r_i)
- 3: **Baseline:** $\tilde{r}_i \leftarrow r_i - \frac{1}{n-1} \sum_{j \neq i} r_j$ {leave-one-out}
- 4: **Token Expansion:** $A_{i,t} \leftarrow \tilde{r}_i$ {sparse reward, $\gamma=1$ }
- 5: **Adaptive Normalization:** If the reward group is selected by `uniform_scale` as homogeneous (the implementation test is $\text{std}(r) < 10^{-8}$), set $\hat{A}_{i,t} \leftarrow r_i/n$ and skip normalization; otherwise compute α, β via Eq. (8)
- 6: and set $\hat{A}_{i,t} \leftarrow \alpha A_{i,t}$ or $\beta A_{i,t}$. If $\mathcal{P} = \emptyset$, $\mathcal{N} = \emptyset$, a signed sum is below ϵ , or a factor is nonfinite, keep $\hat{A} = A$.
- 7: Apply the implementation’s product cap, ϵ additive stabilizer, and factor clamp $[\epsilon, 10]$; these are not the ideal unit-variance branch.
- 8: **Forward Pass:** Compute $\log \pi_{\theta}(y_t|c_t)$ and $\log \pi_{\text{old}}(y_t|c_t)$; use cached $\log \pi_{\text{roll}}(y_t|c_t)$
- 9: **Ratios:** $r_t^{\theta/\pi_{\text{old}}} \leftarrow \exp(\log \pi_{\theta} - \log \pi_{\text{old}})$; $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} \leftarrow \exp(\log \pi_{\text{old}} - \log \pi_{\text{roll}})$
- 10: **Prefix Proxy:** Compute $\log w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$
- 11: Compute ρ_t^{pre} from the prefix of $w_s^{\pi_{\text{old}}/\pi_{\text{roll}}}$ as in Eq. (12)
- 12: Compute M_t^{pre} as in Eq. (13)
- 13: **Loss:** $\mathcal{L} \leftarrow \mathcal{L}^{\text{ProMax}}$ (Eq. (18)) {negative surrogate}
- 14: **Update:** $\theta \leftarrow \theta - \alpha_{\text{lr}} \nabla_{\theta} \mathcal{L}$

baseline (Appendix C.1). Under sparse reward with $\gamma = 1$ (Appendix C.8), the token-level advantage reduces to the shaped reward: $A_{i,t} = \tilde{r}_i$ for all positions t , eliminating the need for per-token reward assignment or discounted cumulative sums.

Adaptive Normalization (Lines 5–6). The scaling factors α and β are computed in closed form from Eq. (8). In the nondegenerate ideal form, without the additive stabilizer or clamps, they enforce the empirical constraints in Definition 3.2 on non-zero tokens. The actual safeguarded factors have the moment residuals in Proposition C.6 while preserving the sign and within-group ordering of advantages (Section 3.2). When all n samples receive exactly the same reward, the RLOO baseline yields zero advantages; in this degenerate case, normalization is skipped and the optional uniform-scale mode (Appendix C.11) supplies nonzero coefficients for nonzero common rewards. All-zero groups remain zero; Proposition C.9 quantifies the added expected score term and its reward-origin dependence. When active response length is determined by the binary reward class, Proposition C.10 further shows that the complete token-reduced update retains a strictly positive population multiplier after the implemented stabilizer, cap, clamp and fallback rules. The implementation also permits a configured near-uniformity tolerance; that branch is defined directly by $\tilde{r}_i = r_i/n$ and is covered by the scalar coefficient statement in Appendix C.11, without treating the rewards as exactly equal.

Forward Pass (Line 7). The training framework computes log-probabilities under the current policy π_{θ} and the frozen PPO snapshot π_{old} . The rollout-policy log-probabilities $\log \pi_{\text{roll}}(y_t|c_t)$ are reused from the rollout cache.

Prefix Causal Trust Region Masking (Lines 9–11). The implementation log-ratio $\log w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$ captures rollout-to-snapshot drift. The prefix cumulative statistic ρ_t^{pre} in Eq. (12) aggregates this drift causally: it depends only on positions $s \leq t$, respecting the autoregressive order (Definition 4.4). The corresponding binary mask M_t^{pre} in Eq. (13) filters out positions where the prefix average log-ratio

leaves $[\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})]$, enforcing a per-position, sample-level constraint (Remark 4.14).

Loss Computation (Line 12). The loss in Eq. (18) multiplies the clipped PPO term by the action mask, the detached vLLM coefficient, and the detached prefix gate. Concretely, m_t removes padded positions, M_t^{pre} is a binary mask (detached), and the normalized advantage $\hat{A}_t$ is detached, so gradients propagate only through $r_t^{\theta/\pi_{\text{old}}}$. The negative sign ensures that minimizing $\mathcal{L}$ maximizes the surrogate objective.

Computational overhead. Compared to standard PPO implementations that cache rollout-time log-probabilities, REINFORCE Pro Max adds only lightweight arithmetic overhead. The RLOO baseline requires $O(n)$ arithmetic operations per prompt group. Adaptive normalization computes the required first- and second-moment aggregates over positive and negative advantage tokens in a single pass. The prefix-level mask requires one cumulative sum over the sequence, which is $O(T)$ per sample. All components are computed from cached rollout log-probabilities and current-policy log-probabilities, requiring no additional model evaluations beyond those already used to compute $\log \pi_\theta$.

5.5 Positioning the Method

REINFORCE Pro Max combines three choices whose effects are separated by the preceding results. The standard RLOO baseline removes self-dependence from each response’s baseline; Proposition 5.6 quantifies the sampling benefit without assuming independence among sibling contributions. The asymmetric normalization solves the token-weighted empirical moment constraints while preserving advantage signs, including with unequal response lengths (Definition 3.2 and Section 3.2). The prefix gate supplies a causal decision at each position, with the conditional pattern comparisons in Theorem 4.7 and the accepted-prefix density bound in Theorem 4.11.

Unlike token-only filters, a prefix decision retains information about earlier deviations; unlike a sequence-wide gate, it can retain a prefix before a late deviation. These are conditional mechanism guarantees. The joint reward guarantee follows from Section 5.3, which retains the signed objective corrections and sampling costs of the complete method. Appendix E collects the supporting TV/KL identities, finite-horizon remainder derivation, and detailed comparisons with token, sequence, and coarse-grained alternatives.

6 Experiments

This section reports controlled checks for two structural claims of REINFORCE Pro Max: (i) the leave-one-out baseline and adaptive asymmetric normalization have the stated advantage-statistics properties; and (ii) prefix-level masking behaves as predicted under the deviation patterns analyzed in Theorem 4.7. The checks are deliberately synthetic and do not constitute a language-model benchmark. The executable protocol is in `reinforce_pro_max/structural_experiment.py`; its fixed-seed output is saved in `reinforce_pro_max/structural_experiment_20260918.json`. Benchmark results (e.g., AIME/MATH with a specified backbone) remain future work.

6.1 Experimental Setup

We specify a simulation protocol with $n = 8$ samples per prompt. Rewards are drawn from a skewed mixture that mimics sparse-success reasoning settings: with probability $p = 0.15$ a sample is drawn from a “success” cluster centered near 1, otherwise from a “failure” cluster centered near 0. We use bounded perturbations $\eta_i \sim \text{Unif}[-0.10, 0.10]$ around these centers and clip the resulting rewards to $[-0.10, 1.10]$, so the simulation satisfies the bounded-reward assumption used by the finite-sample

guarantees. To reflect token-level weighting effects after expansion (Section 3.1), we assign longer sequences to lower-quality samples (failure trajectories), setting lengths to

$$T_i = \begin{cases} 256 + u_i, & \text{success sample,} \\ 512 + u_i, & \text{failure sample,} \end{cases} \quad u_i \sim \text{Unif}\{-32, \dots, 32\}.$$

For IS masking, we use the acceptance interval $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}] = [0.9, 1.1]$ unless otherwise stated (i.e., $\epsilon_{\text{low}} = \epsilon_{\text{high}} = 0.1$). We also define the corresponding log-thresholds

$$\tau_+ := \log(1 + \epsilon_{\text{high}}), \quad \tau_- := -\log(1 - \epsilon_{\text{low}}),$$

so that $\rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ is equivalent to $\log \rho_t \in [-\tau_-, \tau_+]$. Finally, we construct synthetic log-ratio patterns with $T = 32$ to directly instantiate the three regimes in Theorem 4.7.

6.2 Baselines

Baselines and normalization. We compare the mean baseline (REINFORCE++/GRPO-style) to the leave-one-out (RLOO) baseline (Definition 3.1), and we compare global normalization $(A - \mu)/\sigma$ to adaptive asymmetric normalization (Definition 3.2).

IS masking. We compare token-level masking (IcePop), sequence-level masking (GSPO-style, based on ρ^{seq}), and prefix-level masking based on the prefix proxy in Definition 4.4.

6.3 Metrics

Baseline self-correlation. We measure the pooled Pearson correlation over all 8,000 sampled responses, $\text{corr}(r_i, b_i)$ between each sample reward r_i and its baseline b_i . A correlated baseline can couple a sample with its own control variate, complicating the clean second-moment analysis enabled by RLOO (see Appendix C.1).

Sign flip rate under normalization. After token expansion, we compute the token-weighted fraction of non-zero tokens whose normalized advantage changes sign, i.e., $\mathbb{I}[\text{sign}(\hat{A}_{i,t}) \neq \text{sign}(A_{i,t})]$. Sign flips are undesirable because they can turn punishment into reinforcement (or vice versa).

Masked-token coverage. For each synthetic log-ratio pattern, we report the number of masked tokens and the first/last masked positions for IcePop, sequence-level IS masking, and prefix masking.

6.4 Formal and Deterministic Checks

Table 2: Deterministic properties checked by the Lean formalization.

Property	Mean / global	RLOO / adaptive
Raw baseline scaling	$(n - 1)/n$	1
Within-group shaped-reward sum	0	0
Sign preservation under normalization	not guaranteed	exact for non-zero active advantages

The first row describes the raw estimators before normalization, using the exact deterministic proportionality between the mean-baseline and leave-one-out signals (`rloo_scaling`); the second row follows from the mean-baseline and RLOO conservation identities (`mean_baseline_shaped_sum_zero` and `rloo_shaped_sum_zero`); and the third row follows from positive sign-dependent scaling (`positive_scaling_preserves_sign`). The sign statement concerns active tokens; padding

positions are removed by the action mask before normalization. The zero-sum row concerns response-level rewards before token expansion or normalization; it holds for both baselines. Unequal response lengths can produce a nonzero token-weighted mean for either baseline. Finite-iid population baseline moments and the RLOO score cross-moment are also checked in `formal/RLOOMoments.lean`; these probability identities are distinct from numerical simulation results.

Table 3: Fixed-seed structural simulation (1,000 prompt groups, $n = 8$). These are synthetic checks, not RLVR benchmark results.

Measured quantity	Value
Mean-baseline/reward correlation	0.3547
RLOO-baseline/reward correlation	0.00235
Adaptive-normalization sign-flip rate	0
Global-normalization sign-flip rate	0.0465
Adaptive non-zero-token mean	1.8×10^{-9}
Adaptive non-zero-token population variance	1.0000
Maximum absolute RLOO group-sum residual	6.3×10^{-7}

The correlations use the ordinary within-group mean baseline and the leave-one-out baseline on the same sampled rewards. The latter is nearly uncorrelated with the held-out reward in this iid simulation, whereas the ordinary mean baseline includes that reward by construction. The adaptive statistics are computed on non-zero active tokens, matching the ideal normalization definition; implementation safeguards are excluded from this simulation and their exact-real formulas are formalized as residual bounds. These measurements do not validate the full training implementation. The sign-flip denominator contains 3,785,796 nonzero active tokens; adaptive normalization flips 0 and global normalization flips 176,197. The small nonzero moment and group-sum residuals reflect floating-point arithmetic, not exact mathematical equalities. These measurements support the structural claims only for this synthetic protocol.

Using the same ideal asymmetric normalization for both baselines removes their common scale and produces identical coefficients, as proved in Proposition 3.4. A baseline ablation must therefore record the normalization rule and active numerical safeguards. For exact two-level rewards, Proposition 3.3 gives the normalized amplitudes $+\sqrt{N/P}$ and $-\sqrt{P/N}$ and the factor ratio $\bar{T}_+/\bar{T}_-$. These formulas provide deterministic checks for reward-frequency and response-length sweeps; they do not constitute measured improvements on an LLM benchmark. For exact binary rewards, token-weighted standardization within the same prompt is an equivalent ideal operation (Corollary 3.5), so it cannot be treated as a different ideal normalization baseline. The proposed perturbed-reward protocol is multivalued; the general sign-preservation comparison can differ in that setting. The finite RLOO example in Section C.9 verifies such a difference exactly.

The optional uniform-scale ablation must also record the reward zero point. Proposition C.9 gives its added expected score term for binary rewards, and Eq. (39) provides an exact two-response check: shifting 0/1 rewards to 3/4 preserves their ordering and reward gradient but can reverse the update with uniform scale enabled. All-zero reward groups remain zero in either mode.

The early and late patterns instantiate Theorem 4.7(a,b) in the simplest form with $\varepsilon = 0$ (all non-spike tokens satisfy $\rho_t = 1$ and hence lie within the trust-region interval).

Early deviation. IcePop reacts only to the local outlier at $t = 1$ and therefore masks a single token. Sequence-level masking aggregates over the full horizon and dilutes the violation. Prefix

Table 4: Masking behavior under synthetic deviation patterns. We use $T = 32$ and $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}] = [0.9, 1.1]$. We report the number of masked tokens and the masked position range.

Pattern	IcePop (token)	Seq (GSPO-style)	Prefix (ours)
Early ($\log \rho_1 = 8\tau_+$; else 0)	1 ($[1, 1]$)	0 ($—$)	7 ($[1, 7]$)
Late ($\log \rho_{24} = 32\tau_+$; else 0)	1 ($[24, 24]$)	0 ($—$)	8 ($[24, 31]$)
Drift ($\log \rho_t = 2\tau_+$, $\forall t$)	32 ($[1, 32]$)	32 ($[1, 32]$)	32 ($[1, 32]$)

masking, in contrast, propagates the early mismatch forward by masking a short prefix:

$$\log \rho_t^{\text{pre}} = \frac{1}{t} \log \rho_1 = \frac{8}{t} \tau_+, \quad \log \rho^{\text{seq}} = \frac{1}{T} \log \rho_1 = \frac{1}{4} \tau_+, \quad (T = 32).$$

Thus $\log \rho_t^{\text{pre}} > \tau_+$ for $t \in \{1, \dots, 7\}$, while $\log \rho^{\text{seq}}$ remains in-bound.

Late deviation. For a single spike at $t^* = 24$, prefix masking is strictly suffix-local (it can only start at $t \geq t^*$), while sequence-level masking remains all-or-nothing. With $\log \rho_{24} = 32\tau_+$ and $\log \rho_t = 0$ otherwise, the sequence statistic satisfies $\log \rho^{\text{seq}} = \tau_+$ and is accepted, but the prefix average exceeds τ_+ for $t \in \{24, \dots, 31\}$, yielding a localized masked suffix. These behaviors align with the regimes in Theorem 4.7. The Lean certificates `early_table_counts`, `late_table_counts`, and `drift_table_counts` verify all three rows for any positive upper log-threshold and nonpositive lower log-threshold. The position certificates use zero-based indices, corresponding to the one-based positions in the table.

6.5 Ablation Studies

Without adaptive normalization. Global normalization can flip the sign of token-expanded advantages when token-length weighting makes the token-level mean non-zero, while adaptive normalization preserves sign by using separate positive/negative scalings (Section 3.2).

Without prefix masking. Token-level masking only reacts to per-token outliers and may leave future tokens unfiltered even when the context has already drifted; prefix masking filters based on the prefix statistic, which is aligned with the causal accumulation structure (Remark 2.11, Theorem 4.7).

6.6 Sensitivity to Thresholds

Masking behavior depends on $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$. Stricter thresholds can cause sequence-level IS masking to reject an entire trajectory, while prefix masking remains localized in position. For example, in the late-deviation pattern above, the sequence statistic satisfies $\rho^{\text{seq}} = 1 + \epsilon_{\text{high}} = 1.1$ under the default thresholds. If we instead use the stricter symmetric interval $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}] = [0.95, 1.05]$ (i.e., $\epsilon_{\text{low}} = \epsilon_{\text{high}} = 0.05$), then $\rho^{\text{seq}} > 1.05$ and the sequence-level mask rejects all 32 tokens, while prefix masking still rejects only suffix positions starting at $t = 24$.

6.7 Reproducibility Checklist

- The Monte Carlo protocol uses 1,000 groups, $n = 8$ samples per prompt, and seed 20260918, with CPU PyTorch 2.4.0 float32 arithmetic and three intra-op threads. Correlations pool responses; sign-flip rates divide by the number of active tokens with nonzero raw advantage.

Normalization is performed within each prompt group; reported moments pool its expanded tokens and use population variance.

- Reward distribution: mixture with $p = 0.15$, bounded perturbation $\eta_i \sim \text{Unif}[-0.10, 0.10]$, and clipping to $[-0.10, 1.10]$.
- Token lengths: success $T_i = 256 + u_i$, failure $T_i = 512 + u_i$, $u_i \sim \text{Unif}\{-32, \dots, 32\}$.
- Thresholds: default $(\epsilon_{\text{low}}, \epsilon_{\text{high}}) = (0.1, 0.1)$ (i.e., $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}] = [0.9, 1.1]$); sensitivity example $(\epsilon_{\text{low}}, \epsilon_{\text{high}}) = (0.05, 0.05)$ (i.e., $[0.95, 1.05]$).
- The deterministic masking table can be reproduced directly from the formulas in Theorem 4.7. Exact-real counts and rejection positions are also checked in `formal/StructuralExamples.lean`; the executable script reproduces both tables. No language-model benchmark checkpoint is included.

7 Related Work

7.1 Critic-Free RL for LLMs

The high memory cost of maintaining a critic network has motivated a family of critic-free reinforcement learning methods for large language models. REINFORCE++ Baseline [Hu et al., 2025] uses the batch mean reward as a baseline and normalizes advantages globally, providing a simple and memory-efficient alternative to PPO. GRPO [Shao et al., 2024] extends this idea by normalizing advantages with both the mean and standard deviation within each prompt group, achieving strong results on mathematical reasoning tasks. RLOO [Ahmadian et al., 2024] revisits the classical leave-one-out control variate, computing each sample’s baseline from the remaining samples in the group, which eliminates the correlation between a sample and its own baseline. All three methods avoid the need for a value network, saving approximately a quarter of GPU memory compared to standard PPO [Schulman et al., 2017] in comparable settings¹.

7.2 Off-Policy Correction in LLM-RL

Modern LLM-RL pipelines decouple inference from training: high-throughput engines such as vLLM [Kwon et al., 2023] generate rollouts, while gradient updates are performed in separate training frameworks. This architectural separation introduces a policy mismatch between the rollout engine and the training actor [Yao et al., 2025]. Recent empirical analyses further argue that the trust region must be defined with respect to the *original* rollout distribution (the behavior policy that generated the data) rather than a recomputed anchor evaluated inside the training stack; anchoring to a recomputed on-policy distribution can lead to instability under training–inference mismatch [Qi et al., 2026].

Several importance sampling correction methods have been proposed to address this mismatch. Truncated Importance Sampling (TIS) [Yao et al., 2025] clamps the per-token importance weight to a fixed interval, preventing extreme corrections. Its limitation is that it ignores the causal structure of autoregressive generation: an early deviation affects all future tokens, but TIS treats each token independently. IcePop [Team et al., 2025] takes a harder approach: it zeroes out tokens whose

¹The 25% memory saving is reported for single-model-size configurations where the critic has the same architecture as the actor; the actual saving depends on model size, parallel strategy, and batch configuration.

importance weight falls outside the acceptable range. Like TIS, it operates at the token-level and does not account for how deviations propagate through the sequence.

At the sequence-level, Trust Region Masking (TRM) [Li et al., 2025] discards entire trajectories that violate a trust region criterion defined by the maximum token-level divergence (e.g., $\max_t D_{\text{KL}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_\theta)$). GSPO [Zheng et al., 2025] instead computes a single sequence-level importance ratio coefficient (the geometric-mean ratio ρ^{seq}) and applies it uniformly across the sequence. **Prefix-level refinement.** Our REINFORCE Pro component can be viewed as a prefix-level refinement of TRM’s rejection mechanism: instead of a single sequence gate $M^{\text{TRM}}(x, y)$, we compute a per-position mask M_t^{pre} from a ratio-based prefix statistic and multiply it into the token-level loss, masking only the positions whose prefixes violate a proxy trust region. This retains per-position granularity while preserving the causal order of autoregressive generation.

These methods share a common limitation: token-level methods ignore that an early deviation affects the entire future trajectory, while sequence-level methods either reject the entire sequence or lose the ability to distinguish where within a sequence the divergence occurs.

7.3 Trust Region Methods

Trust region methods constrain the policy update to remain close to the current policy, ensuring stable optimization. TRPO [Schulman et al., 2015] enforces a hard KL divergence constraint, while PPO [Schulman et al., 2017] relaxes this to a clipped surrogate objective that is simpler to implement. Recent work on Trust Region Masking [Li et al., 2025] derives tighter error bounds for long-horizon LLM reinforcement learning, showing that classical $O(T^2)$ trust region bounds become theoretically vacuous for sequences of thousands of tokens. Their Adaptive bound achieves per-position granularity and preserves the causal structure of error accumulation, providing the theoretical foundation that motivates our prefix-level Causal Trust Region masking. Our work borrows the per-position error decomposition perspective from TRM but does not directly reuse its optimization objective.

7.4 Summary of Differences

Compared with existing critic-free methods, REINFORCE Pro Max differs along four dimensions: (1) *baseline*: RLOO, whose leave-one-out baseline is conditionally independent of the sample’s own response given the prompt, vs. mean (which includes that response); (2) *normalization*: adaptive asymmetric α/β vs. global μ/σ ; (3) *IS granularity*: prefix cumulative (per-position) vs. token-level or sequence-level; (4) *causal structure*: prefix masking respects autoregressive ordering, while the ratio-based stabilization methods considered in this paper do not. The baseline distinction concerns the raw estimator. With the same ideal asymmetric normalization and active-token weights, the mean-baseline and RLOO coefficients coincide exactly (Proposition 3.4). The normalization rule and prefix gate, rather than this removable common scale, distinguish the normalized method.

8 Conclusion

We presented REINFORCE Pro Max, a unified framework for critic-free LLM reinforcement learning that addresses two fundamental challenges: high-variance advantage estimation and causal off-policy mismatch between behavior and training policies.

The first component, REINFORCE Max, combines a leave-one-out baseline (conditionally independent of each sample’s own response given the prompt, enabling clean second-moment analysis, Remark C.1) with adaptive asymmetric normalization, which preserves advantage sign

and enforces the empirical constraints in Definition 3.2, Equation (8) in its nondegenerate ideal form. The implemented stabilizer and clamps have explicit moment formulas and residual bounds (Proposition C.6). For binary RLOO groups, the ideal normalized amplitudes depend only on the two token counts, while the factor ratio equals the class-conditional mean length ratio (Proposition 3.3). Exact scale invariance also isolates this normalization mechanism from the removable common mean-baseline/RLOO scale (Proposition 3.4). The second component, REINFORCE Pro, introduces prefix-level Causal Trust Region masking (a ratio-based prefix proxy) that preserves the causal structure of autoregressive generation. The implementation computes this proxy from the cached $(\pi_{\text{old}}, \pi_{\text{roll}})$ ratio while the PPO gradient uses the current-to-snapshot ratio. We showed that the resulting prefix mask propagates early-deviation rejection and localizes late-deviation rejection under the conditions in Theorem 4.7. The exact centered-surrogate equivalence and finite-batch objective decomposition connect the critic-free estimator to the per-step surrogate and separate the effects of the ProMax gate, normalization, and clipping. The complete finite-tree Adaptive remainder bound (Theorem 2.10), attributed to the trust-region framework, connects rollout surrogate gain to actual reward improvement with an explicit error allowance. The prefix mask has its own accepted-prefix density-ratio and expectation bounds (Theorem 4.11). Theorem 5.4 further connects the complete ProMax objective to a finite-sample sufficient condition for reward improvement. It retains random response lengths and RLOO group dependence, and controls selection from a finite candidate family fixed before evaluation. Corollary 5.5 removes the unknown mean response length through exact observed-token rescaling. Its sampling error is controlled by additive conditional ratio energy on the history tree, retaining dependence between tokens and between sibling RLOO terms. Proposition 5.6 sharpens this result with the exact RLOO group-average variance, exhibiting the response-count benefit within each prompt and the separate contribution of prompt sampling. Corollary 5.7 extends the guarantee to an arbitrary fit on the evaluation data within a fixed policy class with a finite cover. The extra margin $3R_{\max}\zeta$ and the finite sum of center error budgets make the approximation and complexity costs explicit. Corollary 5.8 supplies explicit energy, Adaptive-cost, and cover-radius bounds from uniform logit-change envelopes, without requiring a minimum positive token probability. The KL chain rule yields an expectation identity for the cumulative log-ratio (Remark 4.12), while the prefix thresholding rule enforces a per-position sample constraint (Remark 4.13). For stale snapshots, the finite-prefix coupling condition is required to transfer this diagnostic to the current policy.

Limitations and future work. The prefix-mask thresholds $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ are hyperparameters that require tuning. The sample-level prefix constraint controls accepted, unnormalized prefix mass; it does not certify the maximum population TV/KL assumptions needed by the Adaptive bound. The exact objective decomposition and sampling margins retain the effects of filtering, normalization, clipping, and reduction. The selection guarantee uses either a fixed finite candidate family or a pre-specified policy class with a fixed finite cover. A useful cover size for unrestricted neural policy fitting is not established; fitting or checking the cover only on the evaluation batch does not meet its premises. The logit-based calibration supplies one such numerical instantiation; establishing its uniform envelope for a trained neural policy remains a separate requirement, and batch-only measurements cannot establish it. The optional uniform-scale branch adds an on-policy per-prompt score term aligned with the 0/1 reward gradient, but depends on the reward zero point; a common reward offset can reverse its update (Proposition C.9). Under the explicit sufficient condition that active response length depends only on the binary reward class, the complete token-reduced update remains a strictly positive multiple of the success-probability gradient even with the implemented stabilizer, cap, clamps and identity fallback (Proposition C.10). If within-class length variation is correlated with response scores, this alignment needs a separate covariance bound and is not claimed by the theorem. The exact within-class covariance decomposition and its finite

Cauchy–Schwarz margin are given in Proposition C.11. When this covariance margin is dominated by the positive gradient component, Corollary C.12 proves strict true-reward improvement for every sufficiently small step along the fixed-prompt population update. The result permits content-dependent lengths and retains numerical safeguards; its on-policy, accepted-gate and inactive-clipping premises remain explicit. The theoretical analysis assumes bounded rewards and mutual policy support (Section 2.4), which may not hold in all practical settings. The finite-sample certificate also applies to the unconditioned iid evaluation law; the optional training-time `dynamic_filtering` branch changes the retained group law, as shown by the exact counterexample in Section 5. A filtered training run therefore needs a separate independent evaluation batch, or a new analysis of the accepted-group law. The current artifact contains only the synthetic structural checks in Section 6; a large-scale language-model benchmark has not yet been run. The paper therefore does not make an empirical performance claim. An ICLR submission requires matched AIME/MATH or comparable reasoning-task experiments, independent evaluation, multiple training seeds, and ablations for each component under the protocol in `reinforce_pro_max/EXPERIMENT_PROTOCOL.md`.

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A Appendix Roadmap

This appendix gives the complete mathematical arguments and their implementation correspondence.

- **Implementation and objective.** Appendices B.1 and B.2 explain rollout–training mismatch and PPO clipping. Appendices E and E.1 give the population surrogate proof and supporting TV/KL identities.
- **Reward certificates and sampling.** Appendices C.2 to C.4 derive the random-length reduction and finite-sample reward guarantees. Appendices C.5 and C.6 prove the shared-baseline variance formula and the extension to data-dependent fitting in a pre-specified covered policy class. Appendix C.7 derives explicit certificate inputs from uniform logit-change bounds.
- **Advantage construction.** Appendices C.1 and C.8 give the RLOO and sparse-reward arguments. Appendices C.10 and C.11 analyze numerical safeguards and the optional uniform-scale branch.
- **Prefix filtering.** Appendices D.1 and D.3 prove the conditional masking comparisons and accepted-prefix mechanism bound. Appendix D.4 collects the additional concentration assumption, its consequences, and their proofs. Appendix D.5 covers alternative prefix gates.

B Implementation Divergence and Trust-Region Anchoring

B.1 Details on Off-Policy Mismatch in LLM-RL

Modern LLM-RL pipelines decouple inference from training: rollouts are produced by an inference engine (e.g., vLLM or SGLang) under the rollout policy π_{roll} , while gradient updates are computed in a separate training stack under the training policy π_{θ} . This architectural separation can introduce off-policy mismatch even when parameters are nominally aligned.

Mismatch Sources Common sources of mismatch include backend discrepancies (different attention kernels, precision formats, or reduction order), MoE routing discontinuities (Top- K routing can flip under small perturbations), and distributed staleness (actor–learner lag between rollout and consumption). These effects can compound autoregressively over long horizons and lead to volatile per-token ratios.

Anchoring and the Apparent Three-Distribution Structure Practical pipelines can exhibit an apparent three-distribution structure: (i) a rollout engine generates tokens according to an inference-time distribution, (ii) a training framework computes gradients under π_{θ} , and (iii) some implementations recompute the behavior log-probabilities inside the training stack, yielding a recomputed anchor that can differ from the true rollout distribution even when parameters are identical. Importantly, this third object is not an additional policy in our theory; it is an alternative implementation-time anchor for the denominator of ρ_t .

Trust Region Setup Qi et al. [2026] argue that the trust region (and hence the denominator of the importance ratio) must be defined with respect to the *original* behavior policy that generated the data, rather than a recomputed policy distribution evaluated inside the training stack. Accordingly, we denote this rollout-time behavior policy by π_{roll} , define $\rho_t := \pi_{\theta}(y_t \mid c_t) / \pi_{\text{roll}}(y_t \mid c_t)$, and assume access to rollout-time cached log-probabilities $\log \pi_{\text{roll}}(y_t \mid c_t)$. The theoretical development

therefore involves only the policy pair $(\pi_{\text{roll}}, \pi_{\theta})$, consistent with long-horizon trust-region analyses for LLM-RL (e.g., Li et al. [2025]).

PPO-Style π_{old} Mismatch The implementation has a frozen PPO snapshot π_{old} in addition to the data-generating rollout policy π_{roll} . Its clipped PPO ratio and its vLLM correction ratio are respectively

$$r_t^{\theta/\pi_{\text{old}}} = \frac{\pi_{\theta}(y_t \mid c_t)}{\pi_{\text{old}}(y_t \mid c_t)}, \quad w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} = \frac{\pi_{\text{old}}(y_t \mid c_t)}{\pi_{\text{roll}}(y_t \mid c_t)}.$$

Their product is the behavior-to-current ratio $\rho_t^{\theta/\pi_{\text{roll}}} = r_t^{\theta/\pi_{\text{old}}} w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$. The prefix mask is computed from $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$ and is detached, while gradients flow through the PPO ratio. The abstract prefix theorems apply to whichever policy pair supplies the ratio; applying them to the current policy during stale updates requires the finite-prefix coupling condition in Section 4.3.

B.2 Why PPO Clipping Can Leak Gradients Under Mismatch

PPO Clipped Surrogate PPO [Schulman et al., 2017] constrains policy updates by clipping the current-to-snapshot ratio. In the notation of this paper, the ratio is

$$r_t^{\theta/\pi_{\text{old}}} := \frac{\pi_{\theta}(y_t \mid c_t)}{\pi_{\text{old}}(y_t \mid c_t)}.$$

The clipped surrogate takes the form

$$\mathcal{L}^{\text{CLIP}}(\pi_{\theta}) = \mathbb{E}_{\pi_{\text{roll}}} \left[\sum_{t=1}^T w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} \min \left(r_t^{\theta/\pi_{\text{old}}} \hat{A}_t, \text{clip}(r_t^{\theta/\pi_{\text{old}}}, 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_t \right) \right],$$

where $\hat{A}_t$ is an advantage estimator and $\epsilon_{\text{clip}} > 0$ is the clipping threshold.

Asymmetry and Gradient Leakage The clipping operator is asymmetric in the sign of $\hat{A}_t$. When $\hat{A}_t < 0$ and $r_t^{\theta/\pi_{\text{old}}} \gg 1 + \epsilon_{\text{clip}}$, the unclipped term can still be selected:

$$r_t^{\theta/\pi_{\text{old}}} > 1 + \epsilon_{\text{clip}}, \hat{A}_t < 0 \implies r_t^{\theta/\pi_{\text{old}}} \hat{A}_t < (1 + \epsilon_{\text{clip}}) \hat{A}_t = \text{clip}(r_t^{\theta/\pi_{\text{old}}}, 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_t,$$

hence $\min(\cdot)$ returns $r_t^{\theta/\pi_{\text{old}}} \hat{A}_t$. In this regime, the gradient magnitude still scales with $r_t^{\theta/\pi_{\text{old}}}$, so extremely large ratios can produce large updates even though they lie far outside the nominal trust region.

Why This Is Benign in Standard RL In classical on-policy settings, large ratios typically reflect genuine policy changes that over-emphasize low-quality actions; keeping the unclipped penalty for $\hat{A}_t < 0$ can be desirable as it discourages such actions.

Failure Mode Under Implementation Divergence In LLM-RL pipelines with inference-training mismatch, a large rollout-to-snapshot coefficient can arise from numerical or backend discrepancies rather than meaningful policy improvements. For example, a routing discontinuity in a Mixture-of-Experts model can yield a token v such that

$$\pi_{\text{roll}}(v \mid c_t) \approx 0.001, \quad \pi_{\text{old}}(v \mid c_t) \approx 0.9, \quad w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} \approx 900.$$

Such spikes reflect implementation-level divergence and can multiply the PPO loss by a massive, erroneous detached coefficient. This motivates using a detached prefix trust-region mask to discard mismatched tokens rather than relying on PPO clipping, which acts on the separate current-to-snapshot ratio.

C REINFORCE Max: Supplementary Details

C.1 Leave-One-Out Baseline Properties

We summarize standard properties of the leave-one-out baseline estimator used in Section 3; see Ahmadian et al. [2024] for a detailed treatment.

Remark C.1 (Properties of the Leave-One-Out Baseline). Fix a prompt x and define its conditional reward objective by

$$J_x(\theta) := \mathbb{E}_{y \sim \pi_\theta(\cdot | x)}[R(x, y)].$$

Draw $n \geq 2$ i.i.d. responses $\{y^{(1)}, \dots, y^{(n)}\} \sim \pi_\theta(\cdot | x)$ with rewards $r_i = R(x, y^{(i)})$. Let

$$f_i = \nabla_\theta \log P^{\pi_\theta}(y^{(i)} | x)$$

denote the trajectory score function for sample i , and define the per-sample policy gradient contribution with baseline b as $g_i(b) = (r_i - b)f_i$. With the leave-one-out baseline $b_i = \frac{1}{n-1} \sum_{j \neq i} r_j$, the estimator

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g_i(b_i)$$

satisfies the following statements, provided the usual score-function regularity conditions hold: the n responses are conditionally i.i.d. given x ; r_i and f_i are integrable, f_i has a finite second moment, the products needed below are integrable (in particular $r_i f_i$ and $(r_i - \mu(x))^2 f_i f_i^\top$), and differentiation may be interchanged with the expectation defining $J_x(\theta)$. In particular, the normalized policy has support independent of θ (or obeys the corresponding boundary conditions), so that $\mathbb{E}[f_i | x] = 0$. Under these conditions:

1. **Conditional unbiasedness:** $\mathbb{E}[\hat{g} | x] = \nabla_\theta J_x(\theta)$.
2. **Conditional baseline independence:** For each i , b_i is independent of (r_i, f_i) given x ; in particular, $\mathbb{E}[b_i f_i | x] = 0$ and $\text{Cov}(f_i, b_i | x) = 0$. The moment assumptions ensure that these products and covariances are well-defined.
3. **Mean-baseline scaling:** Let $b^{\text{mean}} = \frac{1}{n} \sum_{j=1}^n r_j$ and $\hat{g}^{\text{mean}} = \frac{1}{n} \sum_{i=1}^n (r_i - b^{\text{mean}}) f_i$. Then deterministically $\hat{g}^{\text{mean}} = \frac{n-1}{n} \hat{g}$, and hence $\mathbb{E}[\hat{g}^{\text{mean}} | x] = \frac{n-1}{n} \nabla_\theta J_x(\theta)$.
4. **Second-moment decomposition:** Let $\mu(x) = \mathbb{E}[r_i | x]$ and $\sigma^2(x) = \text{Var}(r_i | x)$. Then $\text{Var}(b_i | x) = \sigma^2(x)/(n-1)$, and

$$\mathbb{E}[(r_i - b_i)^2 f_i f_i^\top | x] = \mathbb{E}[(r_i - \mu(x))^2 f_i f_i^\top | x] + \text{Var}(b_i | x) \mathbb{E}[f_i f_i^\top | x].$$

For a prompt distribution P independent of θ , the paper’s global objective is $J(\pi_\theta) = \mathbb{E}_{x \sim P}[J_x(\theta)]$. If differentiation also commutes with this outer expectation, then

$$\mathbb{E}_{x, Y_{1:n}}[\hat{g}] = \mathbb{E}_{x \sim P}[\nabla_\theta J_x(\theta)] = \nabla_\theta J(\pi_\theta).$$

For a fixed finite prompt law, this follows by differentiating a finite weighted sum; the joint prompt/response sampling law and the complete RLOO derivative are checked by `joint_prompt_response_mass_normalized`, `joint_prompt_response_mass_nonneg` and `finite_prompt_rloo_policy_gradient` in `formal/PromptGradient.lean`. Fixing one prompt cannot substitute for this averaging step: two equally likely prompts with Bernoulli success probabilities θ and $1 - \theta$ have a constant global reward $1/2$, while the first prompt’s reward derivative is 1. Their valid policies on $0 < \theta < 1$ and the derivative separation are checked by `two_prompt_policy_valid`, `two_prompt_reward` and `fixed_prompt_gradient_ne_global`.

The same statement does not require a finite prompt set. Let μ be an arbitrary prompt measure and interpret the global objective as the Bochner integral $\int J_x(\theta) d\mu(x)$. If the conditional policy regularity above holds μ -almost everywhere, the conditional RLOO differential is almost-everywhere measurable, and the prompt integrand is locally Lipschitz in θ with an integrable dominating bound (together with the corresponding measurability and integrability conditions), then differentiation under the prompt integral gives

$$DJ(\theta) = \int \mathbb{E}[\hat{g} \mid x] d\mu(x).$$

These are the standard dominated differentiation conditions; they make the outer exchange explicit rather than treating it as an automatic consequence of conditional unbiasedness. The finite-response RLOO calculation and this measure-level extension are checked by the following two Lean theorems:

`ae_prompt_rloo_policy_gradient`
`measure_prompt_rloo_policy_gradient.`

They are both defined in `formal/PromptGradientMeasure.lean`.

Remark C.2 (Relationship Between the Two Estimators). The deterministic proportionality $\hat{g}^{\text{mean}} = \frac{n-1}{n} \hat{g}$ implies that the mean-baseline estimator is a rescaled version of the leave-one-out estimator. Its smaller raw variance is entirely due to the constant scaling factor. In practice, this shrinkage can be absorbed into the learning rate, but it breaks exact unbiasedness whenever the target gradient is nonzero. The main advantage of the leave-one-out baseline is that b_i is independent of sample i given the prompt, which simplifies moment calculations and avoids self-correlation terms that arise when the baseline depends on r_i .

Remark C.3. The leave-one-out baseline is equivalent to the leave-one-out control variate from the classical REINFORCE literature [Williams, 1992]. In the LLM-RL setting with n samples per prompt, each sample’s baseline is conditionally independent of its own trajectory given the prompt, making it particularly well-suited for group-based sampling strategies.

The stochastic statements in this remark are conditional on the stated independence, integrability, and differentiation assumptions. The finite Lean audit verifies the deterministic RLOO scaling, zero-sum, and same-sign degeneracy identities, as well as the independent product-space control-variate identity and scalar and cross-coordinate second-moment decompositions. The finite i.i.d. leave-one-out construction is checked directly, including $\text{Var}(b_i) = \sigma^2/(n-1)$ and every entry of the displayed outer-product identity. Extensions to general probability spaces and conditional expectations retain the stated regularity assumptions. One finite product-space building block is now checked explicitly: if p is the trajectory distribution, u is an independent baseline-sample distribution, and the score f has zero p -mean, then

$$\sum_{i,j} p_i u_j (r_i - b_j) f_i = \sum_i p_i r_i f_i.$$

This is the finite form of the control-variate step $\mathbb{E}[b_i f_i] = \mathbb{E}[b_i] \mathbb{E}[f_i] = 0$ and is verified by `finite_independent_baseline_score` in `formal/RL00Probability.lean`. The theorem does not by itself discharge the general probability-space and differentiation–expectation obligations. As concrete finite checks, `finite_iid_rloo_unbiased_two` proves the two-sample leave-one-out estimator exactly, while `finite_iid_rloo_unbiased` proves the arbitrary finite group-size case directly: for $m + 1$ independent draws from a finite mass function p (with $m \neq 0$), the average leave-one-out estimator using the other m rewards has expectation $\sum_y p(y) r(y) f(y)$ whenever $\sum_y p(y) f(y) = 0$. The abstract interface `finite_rloo_independent_baselines` remains useful when baseline laws are specified independently of this concrete i.i.d. construction. The finite-dimensional score version is checked componentwise by `finite_iid_rloo_unbiased_vector`; hence the displayed vector estimator’s expectation equals the target score expectation on finite response support. The derivative connection is now also checked on finite response support. Let $p_\theta(y)$ be locally nonnegative and normalized, differentiable at θ , and let $R(y)$ be independent of θ . Define its score differential as $s_\theta(y) = p_\theta(y)^{-1} D p_\theta(y)$ on positive-mass responses and as zero otherwise. Local nonnegativity implies $D p_\theta(y) = 0$ at a zero-mass response. Thus

$$\sum_y p_\theta(y) s_\theta(y) = 0, \quad D \left(\sum_y p_\theta(y) R(y) \right) = \sum_y p_\theta(y) R(y) s_\theta(y) = \mathbb{E}[\hat{g}].$$

In a finite-dimensional Euclidean parameter space this differential is identified with the usual gradient. The certificate `finite_iid_rloo_policy_gradient` in `formal/PolicyGradient.lean` proves these identities from local normalization, nonnegativity, and differentiability, rather than assuming a zero-mean score or an exchange of derivative and finite expectation. On positive-mass responses, `policyScore_is_log_derivative` identifies $s_\theta(y)$ with $D \log p_\theta(y)$. The general probability-space statement above retains the listed integrability and differentiation assumptions. This certificate concerns the raw on-policy RLOO estimator; the changes introduced by normalization, prefix gates, off-policy weights, and clipping are treated separately in Proposition 5.3. The connection to the paper’s autoregressive model is explicit: `tree_iid_rloo_policy_gradient` in `formal/TreePolicyGradient.lean` constructs the response masses by multiplying history-dependent token probabilities and proves the result for the recursively defined terminal-reward objective used in the performance-difference theorems. Differentiability of the response masses follows from differentiability of the conditional token probabilities. This permits common absorbing EOS transitions and makes no independence assumption across positions within a response.

For a scalar score coordinate, the corresponding finite second-moment calculation is also checked. If p is the trajectory mass, u is an independent baseline mass with mean μ , then

$$\sum_{i,j} p_i u_j (r_i - b_j)^2 f_i^2 = \sum_i p_i (r_i - \mu)^2 f_i^2 + \left(\sum_j u_j (b_j - \mu)^2 \right) \left(\sum_i p_i f_i^2 \right).$$

This is the scalar finite product-space form of the conditional second-moment decomposition in the statement above and is verified by `finite_independent_baseline_second_moment`. The cross-coordinate version is verified by `finite_independent_baseline_cross_moment` and gives each entry of a finite-dimensional outer-product decomposition. In `formal/RL00Moments.lean`, `finite_loo_baseline_marginal` derives the baseline law by separating sample i from the other $n - 1$ draws in the actual i.i.d. product mass. From this construction, `finite_loo_baseline_variance` proves $\text{Var}(b_i) = \sigma^2 / (n - 1)$, and `finite_loo_score_cross_moment` proves

$$\mathbb{E}[(r_i - b_i)^2 f_{i,k} f_{i,l}] = \mathbb{E}[(r_i - \mu)^2 f_{i,k} f_{i,l}] + \frac{\sigma^2}{n - 1} \mathbb{E}[f_{i,k} f_{i,l}]$$

for any two score coordinates k, l . No independent-baseline variance formula is assumed in this final certificate. This is a per-sample second-moment identity; the covariance of the group-averaged estimator also involves cross-sample terms because its leave-one-out baselines share samples. The exact scalar group-average variance is now given in Proposition 5.6 and Appendix C.5. Its finite-law identity applies to any zero-mean scalar signal, including each score coordinate here and the trajectory ratio residual used in the reward certificate.

The finite proofs are supplied in the accompanying Lean files; see also Ahmadian et al. [2024] for background.

C.2 Autoregressive RLOO Population Bridge and Random Reduction

The finite-tree formalization also connects the estimator actually used by RLOO to the population surrogate in the performance-difference analysis. Let p be the rollout tree policy, q a candidate policy, and let $\rho_t(y)$ be the candidate-to-rollout token ratio at the t th history on a finite horizon. The rollout trajectory residual used in the formalization is $h(y) = \sum_t (\rho_t(y) - 1)$ and has zero p -mean. Consequently, for a group of $m + 1$ i.i.d. trajectories, the concrete leave-one-out estimator satisfies

$$\mathbb{E}_p[(m + 1) \hat{\mathcal{L}}_{\text{RLOO}}] = (m + 1) \mathcal{S}_p(q),$$

and its group-size-normalized expectation is exactly the raw population surrogate $\mathcal{S}_p(q)$. This allows arbitrary history dependence across response tokens; only the trajectories in the group are i.i.d. It is proved by `response_ratio_residual_mean_zero`, `iid_rloo_surrogate_increment_expectation`, and `group_surrogate_ratio` in `formal/RLOOSurrogate.lean`.

The bridge concerns the raw, unclipped surrogate. Prefix gating, adaptive normalization, off-policy weighting, and PPO clipping remain the explicit correction terms in Proposition 5.3; they are not silently absorbed into an unbiasedness claim. The reduction itself must also be stated precisely. If a response contributes Z over $L > 0$ active tokens, response-level averaging estimates $\mathbb{E}[Z/L]$, while global token-level averaging estimates $\kappa = \mathbb{E}[Z]/\mathbb{E}[L]$. The exact bias decomposition is

$$\mathbb{E}[Z/L] - \kappa = -\frac{\mathbb{E}[(L - \mathbb{E}[L])Z/L]}{\mathbb{E}[L]}.$$

For the independent-block analysis, Z and L instead denote full-group sums. A group response mean remains the mean of its individual Z_i/L_i , not the ratio of group sums: for contributions $(1, -1)$ and lengths $(1, 3)$, these quantities are $1/3$ and 0 , respectively. This arithmetic distinction is checked by `group_ratio_ne_response_mean` in `formal/BatchReduction.lean`. For i.i.d. blocks with $L \geq \ell > 0$, the self-normalized token estimator has the finite bounds

$$\mathbb{E}[(\bar{Z}_n/\bar{L}_n - \kappa)^2] \leq \frac{\mathbb{E}[(Z - \kappa L)^2]}{n\ell^2}, \quad \Pr(|\bar{Z}_n/\bar{L}_n - \kappa| \geq \varepsilon) \leq \frac{\mathbb{E}[(Z - \kappa L)^2]}{n\ell^2\varepsilon^2}.$$

These are checked by `ratio_expectation_gap`, `iid_token_mean_mse_of_ratio`, and `iid_token_mean_tail` in `formal/BatchReduction.lean`.

The finite performance certificate in `formal/BatchPerformance.lean` composes this estimation error with a valid population lower bound and explicitly states that it is for a fixed candidate policy evaluated on independent blocks; it is not a uniform guarantee after selecting the candidate from the same data.

C.3 Proof of the ProMax Performance Certificate

Fix the candidate family before drawing evaluation blocks. A block is the pair $(x, (y_1, \dots, y_n))$, with mass $P(x) \prod_{i=1}^n P^p(y_i | x)$. The finite history tree constructs each trajectory mass from its conditional token kernels; summing over responses and prompts gives a normalized block law. This is checked by `block_mass_normalized`. Conditional on x , the response residual $h_j(y) = \sum_t (\rho_{j,t}(y) - 1)$ has zero rollout mean. Each b_i is a function of the other $n - 1$ responses and is independent of y_i given x . Therefore

$$\mathbb{E}[Z_j | x] = n \mathbb{E}_{y \sim p(\cdot | x)}[R(x, y) h_j(y)] = n \mathcal{S}_{j,x}.$$

The equalities use the actual leave-one-out baseline; shared baselines are not replaced by independent copies. Prompt averaging proves $\mathbb{E}[Z_j] = n \mathcal{S}_j$, while $\mathbb{E}[L] = n \mu_L$. These steps are checked by `block_increment_expectation`, `block_length_expectation`, and `block_ratio_target`.

For the full objective, let $D = \sum_b L_b$ and use weights $m_{b,i,t}/D$. At the comparison policy p , the frozen-snapshot PPO ratio is $r_0 = p/\pi_{\text{old}}$ and satisfies $r_0 w = 1$. At q_j it is $r_j = q_j/\pi_{\text{old}}$ and satisfies $r_j w = \rho_j$. Subtracting the two exact objective decompositions gives

$$\begin{aligned} \Delta_j S_{\text{PM}} + \Delta_j G - \Delta_j N + \Delta_j C_{\text{clip}} &= \frac{1}{D} \sum_{b,i,t} m_{b,i,t} A_{b,i} (\rho_{j,b,i,t} - 1) \\ &= \frac{\sum_b Z_{j,b}}{\sum_b L_b}. \end{aligned}$$

The second equality uses the zero ratio residual on shared absorbing padding. The gate, normalized signal, snapshot coefficients and clipping terms remain in the first expression without independence requirements. `response_residual_eq_token_sum` and `group_increment_eq_token_sum` connect the token indices to the history-tree ratio trace; `corrected_batch_change_eq_rloo_ratio` and `corrected_block_objective_eq` verify the complete conversion. Configured uniform-scale signals are included in the actual normalized signal, so their difference from raw RLOO is retained in N .

Write $F_j = Z_j - \kappa_j L$. The population identities give $\mathbb{E}[F_j] = 0$. The blocks are independent, so $\mathbb{E}[\bar{F}_j^2] = v_j/K$, even though siblings within a block are dependent. Since $\bar{L} \geq n\ell$, the mean-square error of $\hat{\kappa}_j$ is at most $v_j/(K(n\ell)^2)$. Markov's inequality on its square gives the individual tail bound, and a finite union bound gives Eq. (20). No independence between candidate estimates is needed. `rloo_uniform_estimation_bound` verifies this result with the block law and its population means constructed above.

Finally, Theorem 2.10 yields $J(q_j) - J(p) \geq \mathcal{S}_j - B_j = \mu_L \kappa_j - B_j$. On the simultaneous estimation event, $\kappa_j \geq \hat{\kappa}_j - \eta_j$ for every candidate, proving the performance bound. If the selected candidate has nonpositive true gain but passes the strict threshold, its estimation error must exceed its margin. That event is a subset of the union event already bounded above. `rloo_performance_from_estimate` and `rloo_selected_false_improvement_bound` check these conclusions, including the actual Adaptive cost and true reward values on the same finite prompt/history model.

C.4 Proof of the Observed-Length and Drift-Energy Certificate

For one prompt, define $h(y) = \sum_{t=1}^{|y|} (\rho_t(y) - 1)$ and $e(c) = \sum_a p(a | c)(q(a | c)/p(a | c) - 1)^2$. Normalization and $q \ll p$ imply

$$\sum_a p(a | c)(q(a | c)/p(a | c) - 1) = 0.$$

Zero-probability actions are omitted, or assigned zero weighted contribution. Hence for every real u ,

$$\sum_a p(a | c)(u + q(a | c)/p(a | c) - 1)^2 = u^2 + e(c).$$

Induction over continuation length on the normalized history tree yields

$$\mathbb{E}_p[h(Y)^2 \mid \text{prefix } c] = h(c)^2 + \mathbb{E}_p \left[\sum_{t>|c|} e(c_t) \mid \text{prefix } c \right].$$

The formal recursion is defined from each prefix kernel, including unreachable prefixes, without conditioning on a zero-probability event. At the root and after prompt averaging this gives $\mathbb{E}[H_j(Y)^2] = \mathcal{E}_j$. The Lean theorems `local_ratio_residual_mean_zero`, `residual_square_step`, `trajectory_residual_second_moment`, and `response_residual_energy` verify these steps. `ratio_energy_eq_prefix_sum` identifies the recursive cost with the explicit sum of prefix-law expectations. No token independence is used.

Within an actual group, each leave-one-out baseline lies in $[-R_{\max}, R_{\max}]$. Thus $|A_i| \leq 2R_{\max}$ and the finite Cauchy–Schwarz inequality gives

$$\left(\frac{Z_j}{n} \right)^2 \leq \frac{1}{n} \sum_{i=1}^n A_i^2 H_j(y_i)^2 \leq 4R_{\max}^2 \frac{1}{n} \sum_{i=1}^n H_j(y_i)^2.$$

Expectation under the constructed conditional-i.i.d. group law proves $\mathbb{E}[(Z_j/n)^2] \leq 4R_{\max}^2 \mathcal{E}_j$. This inequality uses the actual shared-baseline statistic; it does not factor dependent sibling contributions. It is verified by `rloo_estimator_square_bound`, `rloo_estimator_second_moment_bound`, and `block_surrogate_second_moment`.

The common batch denominator cancels exactly:

$$\frac{D}{Kn} \left(\frac{\sum_b Z_{j,b}}{D} \right) = \frac{1}{K} \sum_b \frac{Z_{j,b}}{n}.$$

The right-hand side averages unbiased group contributions, independently across the K blocks, even when reward, length, and gate decisions are dependent within blocks. Its variance is at most $4R_{\max}^2 \mathcal{E}_j / K$. `corrected_objective_rescaling` combines the full ProMax objective with this cancellation, and `rescaled_surrogate_unbiased` proves its expectation using the actual block law. `rescaled_surrogate_uniform_tail`, `rescaled_performance_from_estimate`, and `rescaled_selected_false_improvement_bound` complete the simultaneous confidence, performance, and selection guarantees.

Finally, a conditional energy bound $e(c) \leq d_j^2$ implies $\mathcal{E}_j \leq T d_j^2$ by summing prefix expectations. A ratio bound $|q_j/p - 1| \leq d_j$ need only hold on positive- p actions; shared zero actions impose no artificial requirement $d_j \geq 1$. `local_ratio_energy_of_bound` and `family_ratio_energy_of_ratio_bound` prove this implication. After absorbing EOS, both kernels put mass one on EOS and the energy is zero. The resulting square-root horizon scaling concerns the statistical margin only, and does not remove the separate policy-distribution-shift cost from the reward bound.

C.5 Exact Shared-Baseline Group Variance

Fix a prompt and suppress its index. Write $X_i = R_i - \mathbb{E}[R_i]$, $Y_i = H_j(y_i)$, $\theta = \mathbb{E}[XY]$, $a = \mathbb{E}[X^2]$, $b = \mathbb{E}[Y^2]$, and $c = \mathbb{E}[X^2 Y^2]$. Both X and Y have mean zero. Let $S_X = \sum_i X_i$, $S_Y = \sum_i Y_i$, and $S_{XY} = \sum_i X_i Y_i$. For every realized group, expanding the leave-one-out sums gives

$$U = \frac{n S_{XY} - S_X S_Y}{n(n-1)}.$$

Reward translation leaves the RLOO statistic unchanged, so this equality uses the actual reward estimator. These statements are checked by `rloo_eq_sample_covariance` and `rloo_reward_translation`.

Direct calculation under the finite i.i.d. group law gives

$$\begin{aligned}\mathbb{E}[S_{XY}^2] &= nc + n(n-1)\theta^2, \\ \mathbb{E}[S_{XY}S_XS_Y] &= nc + n(n-1)\theta^2, \\ \mathbb{E}[S_X^2S_Y^2] &= nc + n(n-1)(ab + 2\theta^2).\end{aligned}$$

For example, add one independent pair (X, Y) to the running sums. Conditioning on the previous sums, expanding the fourth-degree polynomial, and using $\mathbb{E}[X] = \mathbb{E}[Y] = 0$ leaves the recurrence

$$\mathbb{E}[S_X^2S_Y^2]_{n+1} - \mathbb{E}[S_X^2S_Y^2]_n = c + 2nab + 4n\theta^2.$$

The mixed third-sum recurrence is $c + 2n\theta^2$. The finite product mass is decomposed exactly using `iid_expect_cons`, `iid_sample_sum_square`, `iid_sum_product_cross`, and `iid_centered_sums_mixed_square` prove the displayed moments by induction. In particular, the terms from overlapping sample indices are included, rather than discarded by an independence assumption.

Since $\mathbb{E}[U] = \theta$, expansion of $\mathbb{E}[(nS_{XY} - S_XS_Y)^2]/[n^2(n-1)^2] - \theta^2$ yields

$$\text{Var}(U) = \frac{c - \theta^2}{n} + \frac{ab + \theta^2}{n(n-1)}.$$

This is verified by `centered_rloo_variance` and `actual_rloo_variance`. The first coefficient is $\text{Var}(XY)$ and the second is nonnegative; their signs are checked by `covariance_variance_coefficients_nonneg`. Under $|R| \leq R_{\max}$, one has $|X| \leq 2R_{\max}$, $a \leq R_{\max}^2$, and $c \leq 4R_{\max}^2b$. The coefficient of θ^2 after collecting terms is $(2-n)/[n(n-1)] \leq 0$. Therefore

$$\text{Var}(U) \leq \left(\frac{4}{n} + \frac{1}{n(n-1)}\right) R_{\max}^2 b.$$

`actual_rloo_variance_bound` and `rloo_variance_factor_rate` verify the bound and $c_n \leq 5/n$.

For the autoregressive model, the conditional covariance target is exactly the prompt surrogate: $\theta_{j,x} = \mathcal{S}_{j,x}$, as verified by `response_centered_covariance`. The same history-tree ratio energy equals $b = \mathcal{E}_{j,x}$. Thus `response_rloo_variance_exact` and `response_rloo_variance_bound` instantiate the generic identity and bound on actual response groups. Expanding $U_j - \mathcal{S}_j = (U_j - \mathcal{S}_{j,x}) + (\mathcal{S}_{j,x} - \mathcal{S}_j)$ makes the cross term vanish under the conditional block law, giving

$$\text{Var}(U_j) = \tau_j^2 + \mathbb{E}_x[V_{j,x}(n)].$$

`block_surrogate_variance_exact` proves this equality directly from the prompt/response product mass. `group_surrogate_mse_exact` then proves the exact variance divided by K for the full empirical estimator; `group_surrogate_uniform_tail` and `group_selected_false_improvement_bound` supply the sharper confidence and selection guarantees. The bound taking the minimum with the earlier energy-only estimate is checked by `family_rloo_variance_combined_bound`.

C.6 Policy Fitting Within a Fixed Cover

Fix the policy class, centers and assignment map in Corollary 5.7. These objects may be constructed conditionally on independent training data, but not chosen after observing the evaluation blocks. Write $j = \nu(\theta)$ and let U_θ denote the actual RLOO group mean. For every realized group,

$$|U_\theta - U_{\theta_j}| \leq \frac{1}{n} \sum_i |R_i - b_i| |H_\theta(x, y_i) - H_{\theta_j}(x, y_i)|$$

$$\leq 2R_{\max}\zeta.$$

The leave-one-out baseline remains the mean of the other responses in this same group. No independence among the n summands is used in this deterministic bound. Averaging the K groups preserves the bound. On the population side, the previously proved RLOO/tree identity gives

$$|\mathcal{S}_\theta - \mathcal{S}_{\theta_j}| = |\mathbb{E}_p[R(H_\theta - H_{\theta_j})]| \leq R_{\max}\zeta.$$

Consequently, for every sample batch and every class member,

$$|\hat{\mathcal{S}}_\theta - \mathcal{S}_\theta| \leq |\hat{\mathcal{S}}_{\theta_j} - \mathcal{S}_{\theta_j}| + 3R_{\max}\zeta.$$

The event that any member exceeds its covered margin is therefore a subset of the finite union of center deviation events. Applying the exact group variance bound proves Eq. (31). If a fitted policy satisfies the positive-improvement threshold but has nonpositive true gain, the Adaptive lower bound forces it into that same deviation event. This proves the selection claim for any function of the evaluation data taking values in the fixed class, including continuous parameter fitting. The finite sampling space makes this event well-defined even for an infinite class index set.

The Lean proof in `formal/PolicyCover.lean` verifies the RLOO perturbation bound, population comparison, uniform event transfer, and reward lower bound. It also derives the sufficient radius Td from the finite token sum; the response-level cover is not inferred from an empirical check on the observed batch. The connection to the implemented objective is the same pointwise rescaling identity used in Corollary 5.5; applying it to a fitted policy does not require that policy to be independent of the batch.

C.7 Certificate Calibration from Normalized Weights and Logits

We prove Corollary 5.8 using finite normalized-weight inequalities. They supply sufficient numerical bounds for the existing certificate; the general reward and cover theorems retain their original conditions.

Normalized-weight energy. Let p be a finite probability mass function, let $0 < \ell \leq v_a \leq u$, and write $Z = \sum_a p_a v_a$, $q_a = p_a v_a / Z$. Then $\ell \leq Z \leq u$ and q is a probability mass function with exactly the support of p . Actions of zero p -mass contribute zero to all expectations below. The pointwise inequality $(v_a - \ell)(u - v_a) \geq 0$ implies

$$\mathbb{E}_p[v^2] \leq (\ell + u)Z - \ell u.$$

Since $((\ell + u)Z - 2\ell u)^2 \geq 0$, the right-hand side is at most $(\ell + u)^2 Z^2 / (4\ell u)$. Therefore

$$\chi^2(q||p) = \mathbb{E}_p[(v/Z - 1)^2] = \frac{\mathbb{E}_p[v^2]}{Z^2} - 1 \leq \frac{(u - \ell)^2}{4\ell u} =: C(\ell, u). \quad (36)$$

This energy bound is sharp: put probability $u/(\ell + u)$ on weight ℓ and probability $\ell/(\ell + u)$ on weight u . Then $Z = 2\ell u/(\ell + u)$ and equality holds.

Reverse KL and TV. Dividing the pointwise quadratic inequality by $v_a > 0$ and averaging gives $Z + \ell u \mathbb{E}_p[1/v] \leq \ell + u$. The nonnegative square $(Z - \ell u \mathbb{E}_p[1/v])^2$ now yields

$$Z \mathbb{E}_p[1/v] \leq \frac{(\ell + u)^2}{4\ell u}.$$

Using $\log x \leq x - 1$ for $x > 0$, we obtain

$$D_{\text{KL}}(p||q) = \mathbb{E}_p[\log(Z/v)] \leq Z\mathbb{E}_p[1/v] - 1 \leq C(\ell, u).$$

Finally, weighted Cauchy–Schwarz gives

$$\text{TV}(p, q) = \frac{1}{2}\mathbb{E}_p[|v/Z - 1|] \leq \frac{1}{2}\sqrt{\mathbb{E}_p[(v/Z - 1)^2]} \leq \frac{1}{2}\sqrt{C(\ell, u)}.$$

The sharpness claim above concerns the energy bound, not equality in these additional KL and TV inequalities.

Softmax and common shifts. For $p_a = e^{z_a} / \sum_b e^{z_b}$ and $q_a = e^{z'_a} / \sum_b e^{z'_b}$, direct cancellation gives

$$q_a = \frac{p_a e^{z'_a - z_a - c}}{\sum_b p_b e^{z'_b - z_b - c}} \quad \text{for every common shift } c \in \mathbb{R}.$$

If the logit increments lie in an interval $[c, c + \omega]$, their centered exponentials lie in $[1, e^\omega]$. Thus $C(1, e^\omega) = (e^\omega - 1)^2 / (4e^\omega)$. The shift may depend on the context. This argument retains the full softmax normalizer, so the bound is independent of the smallest positive token probability. A shared masked support can be treated as the finite action set at that row; weights on shared zero-probability actions may be set to one in the normalized-tilt formulation. The argument does not permit a change in sampling support.

History trees and the corrected ProMax statistic. Apply the local bounds at every prompt and history of length less than T . The additive energy identity gives $\mathcal{E}_j \leq TC_j$, and averaging local TV under any rollout occupancy preserves the bound $\epsilon_j = \sqrt{C_j}/2$. Substitution in the already proved Adaptive bound gives $B_j \leq \bar{B}_j$ from Eq. (34). Finite prompt averaging preserves both inequalities. Shared absorbing EOS transitions have zero local energy and divergence; the uniform horizon bound may overestimate their contribution.

For the actual RLOO group estimator, the previously proved second-moment bound becomes $\mathbb{E}[(Z_j/n)^2] \leq 4R_{\max}^2 TC_j$. Independence of complete groups, Chebyshev’s inequality, and a finite union bound give Eq. (35). The exact observed-token rescaling still identifies this estimator with $D(\Delta_j S_{\text{PM}} + \Delta_j G - \Delta_j N + \Delta_j C_{\text{clip}}) / (Kn)$. Combining its deviation bound with the population performance lower bound proves the stated reward certificate. All gate, normalization, clipping, random-length, and shared-baseline terms therefore retain their earlier definitions. A sharper valid group-variance bound can replace this second-moment estimate through Proposition 5.6.

A sufficient logit-cover radius. Let a center q' be a normalized tilt of rollout policy p with weights in $[\ell_0, u_0]$, and let a class member q be a normalized tilt of q' with weights in $[\ell_1, u_1]$. At each context, $q'_a/p_a \leq u_0/\ell_0$ on positive- p actions, and $|v_a/\mathbb{E}_{q'}[v] - 1| \leq (u_1 - \ell_1)/\ell_1$. Consequently,

$$\left| \frac{q_a}{p_a} - \frac{q'_a}{p_a} \right| \leq \frac{u_0}{\ell_0} \frac{u_1 - \ell_1}{\ell_1}.$$

At shared zero-probability actions both totalized ratios are zero. The triangle inequality over T positions bounds the difference of the additive response residuals by T times this quantity. Taking $(\ell_0, u_0) = (1, e^\omega)$ and $(\ell_1, u_1) = (1, e^r)$ proves the radius $Te^\omega(e^r - 1)$. This provides a sufficient metric for the fixed cover in Corollary 5.7; it does not establish how many centers suffice for a given neural policy class.

`formal/LogitBounds.lean` verifies the normalized-weight inequalities, the energy equality example, softmax normalization and shift invariance, the token-to-response cover bound, and the finite-prompt composition with the actual RLOO sampling and reward certificate.

C.8 Objective Consistency of $\gamma = 1.0$

Remark C.4 (On Setting $\gamma = 1.0$ for Sparse Terminal Rewards). For a response with T active tokens and shaped reward $\tilde{r}$ assigned only at the final token, the reward-to-go recurrence $G_t = r_t + \gamma G_{t+1}$ gives

$$G_t = \gamma^{T-t} \tilde{r}, \quad 1 \leq t \leq T.$$

Thus $\gamma = 1$ gives $G_t = \tilde{r}$ at every active position, as required by the undiscounted sequence-reward objective $J(\pi_\theta) = \mathbb{E}[R(x, y)]$ and the token expansion in Section 3.1. For nonzero $\tilde{r}$, requiring this equality at every suffix length also forces $\gamma = 1$. Both claims are checked by `sparse_terminal_return_gamma_one` and `gamma_one_iff_constant_sparse_return` in `formal/SparseReturn.lean`. Choosing $0 \leq \gamma < 1$ introduces position-dependent factors γ^{T-t} and down-weights earlier token contributions. Such an estimator generally no longer targets the undiscounted policy gradient; identifying it with a discounted-objective gradient additionally requires specifying the outer time weights in the policy-gradient formula. See Williams [1992] for background on REINFORCE.

C.9 The Normalization Mechanism for Actual RLOO Signals

Proof of Proposition 3.3. Let the group contain k rewards r_+ and m rewards r_- , with $n = k + m \geq 2$ and $\Delta = r_+ - r_- > 0$. The reward sum is $kr_+ + mr_-$. Subtracting the other-response mean gives

$$r_+ - \frac{(k-1)r_+ + mr_-}{n-1} = \frac{m\Delta}{n-1} = a_+, \quad r_- - \frac{kr_+ + (m-1)r_-}{n-1} = -\frac{k\Delta}{n-1} = -a_-.$$

Both magnitudes are strictly positive. Repeating each response advantage at its active tokens gives positive and negative first-moment magnitudes Pa_+ and Na_- , squared sums Pa_+^2 and Na_-^2 , and nonzero count $P + N$. Therefore

$$\kappa = \frac{Pa_+}{Na_-}, \quad D = Pa_+^2 + \kappa^2 Na_-^2 = a_+^2 \frac{P(P+N)}{N}.$$

Since $\alpha = \sqrt{(P+N)/D}$ and $\beta = \alpha\kappa$,

$$\alpha a_+ = \sqrt{N/P}, \quad \beta a_- = \sqrt{P/N}.$$

Substituting the RLOO magnitudes also gives

$$\kappa = \frac{Pm}{Nk} = \frac{P/k}{N/m} = \frac{\bar{T}_+}{\bar{T}_-}.$$

Multiplication by the token counts gives equal aggregate coefficient magnitudes $P\sqrt{N/P} = N\sqrt{P/N}$. All steps use the actual reward group and its response lengths. $\square$

The finite construction in `formal/AdaptiveMechanism.lean` indexes responses by the disjoint union of the higher- and lower-reward groups. It computes the reward sum and the RLOO signals, then forms the positive, negative and squared sums using arbitrary finite response-length weights. `finite_binary_rloo_normalization` composes these steps with the normalization map. The ratio and mass-balance identities are checked in the same module. The older equal-amplitude algebra remains a special case; it is not used to assume that binary RLOO advantages are ± 1 .

What is balanced. Before normalization, the ratio of positive to negative absolute token coefficient mass is

$$\frac{Pa_+}{Na_-} = \frac{\bar{T}_+}{\bar{T}_-}.$$

Thus unequal average lengths directly produce the scalar mass imbalance; class frequency alone does not, because RLOO already assigns different magnitudes to the two classes. After normalization, the two total coefficient magnitudes are equal. At the token level, $P \ll N$ gives $|\hat{A}_+| \gg |\hat{A}_-|$, but this is not the assertion $\beta < 1$. For one success and nine equally long failures with rewards 1 and 0, $\alpha = \beta = 3$; the negative amplitude grows from 1/9 to 1/3. The Lean counterexample `rare_success_does_not_imply_negative_attenuation` checks these factors. Importance weights, gates, and score vectors can change the final update, so coefficient balance is not a gradient-vector balance theorem.

Proof of Proposition 3.4. For positive c , scaling A to cA preserves the two sign sets and the nonzero-token count. The positive and negative first sums both acquire a factor c , and both squared sums acquire a factor c^2 . Consequently, κ is unchanged, D becomes c^2D , and the ideal factors become α/c and β/c . Their products with cA equal the original normalized advantages at every active position. The same calculation holds with arbitrary fixed nonnegative token or response-length weights.

For a group of size $n \geq 2$, direct rearrangement gives

$$r_i - \frac{1}{n} \sum_j r_j = \frac{n-1}{n} \left(r_i - \frac{\sum_j r_j - r_i}{n-1} \right).$$

Apply scale invariance with $c = (n-1)/n > 0$ to prove the baseline equivalence. $\square$

The same Lean module proves all five finite weighted-moment transformations, invariance of the resulting normalization map, and the equality obtained from the two actual baseline formulas. This equivalence assumes the same ideal normalization; it does not compare different normalization rules or different prompt groups. The independent-baseline identities and group-variance bounds elsewhere in the paper apply to the raw RLOO surrogate before this data-dependent map.

Proof of Corollary 3.5. Write $a = a_+ > 0$ and $b = a_- > 0$. The token-weighted mean and population variance are

$$\mu = \frac{Pa - Nb}{P + N}, \quad v = \frac{P(a - \mu)^2 + N(-b - \mu)^2}{P + N} = \frac{PN(a + b)^2}{(P + N)^2} > 0.$$

The centered positive and negative coefficients are $a - \mu = N(a + b)/(P + N)$ and $-b - \mu = -P(a + b)/(P + N)$. Dividing by $\sqrt{v}$ gives $\sqrt{N/P}$ and $-\sqrt{P/N}$, respectively, which are the asymmetric coefficients in Equation (9). These variance and square-root identities are checked by `two_level_token_zscore`. $\square$

A sign change with actual multivalued RLOO rewards. Consider three responses with rewards (0, 2/3, 1) and active-token lengths (1, 1, 4). Their RLOO advantages and token-weighted moments are

$$(A_1, A_2, A_3) = (-5/6, 1/6, 2/3), \quad \mu_{\text{tok}} = 1/3, \quad v_{\text{tok}} = 11/36.$$

Although $A_2 = 1/6 > 0$, its centered value is $A_2 - \mu_{\text{tok}} = -1/6$, so its ordinary token-standardized coefficient is negative. The ideal asymmetric coefficient is $\alpha A_2 > 0$ because $\alpha > 0$. Both sign

classes are present, and every reward lies in $[0, 1]$; the example uses the actual leave-one-out formula and includes no zero-advantage tokens. The finite RLOO values, weighted mean and variance, negative standardized coefficient, and positive asymmetric coefficient are checked in `formal/AdaptiveMechanism.lean`. This illustrates a distinction for multivalued rewards, consistent with the exact binary equivalence in Corollary 3.5.

Connection to the numerical implementation. If the normalization branch is taken and the product and factor clamps are inactive, the implementation replaces D by $D + \varepsilon_{\text{num}}$. The factor identity

$$\sqrt{\frac{P + N}{D + \varepsilon_{\text{num}}}} = \sqrt{\frac{D}{D + \varepsilon_{\text{num}}}} \sqrt{\frac{P + N}{D}}$$

shows that the two normalized amplitudes have the same multiplicative shrinkage; their relative factor ratio and zero-mean balance are preserved. This identity is checked by `stabilizer_common_shrinkage`. Under a reward rescaling, however, D changes while the fixed numerical stabilizer does not, so exact reward-scale invariance need not survive. Clamps and fallbacks have the separate explicit formulas in Proposition C.6.

C.10 Numerical Stability

The adaptive normalization (Section 3.2) includes several safeguards:

Remark C.5 (Numerical Thresholds vs. Divergence Bounds). The symbols ε_{num} and ε_{tol} in this appendix denote implementation hyperparameters used for numerical stability and uniform-reward detection. They are unrelated to the divergence upper bound ϵ in Definition 2.2.

1. **Sum threshold:** If $\sum_{(i,t) \in \mathcal{P}} A_{i,t} < \varepsilon_{\text{num}}$ or $-\sum_{(i,t) \in \mathcal{N}} A_{i,t} < \varepsilon_{\text{num}}$ (default $\varepsilon_{\text{num}} = 10^{-8}$), normalization is skipped and the original advantages are returned.
2. **Additive stabilizer:** The denominator inside the square root for α always includes $\varepsilon_{\text{num}} > 0$. This changes the second moment even when every clamp is inactive.
3. **Scale clamping:** α and β are clamped to $[\varepsilon_{\text{num}}, s_{\text{max}}]$ (default $s_{\text{max}} = 10$) to prevent extreme scaling.
4. **Intermediate overflow protection:** The quantity $\kappa^2 \sum_{(i,t) \in \mathcal{N}} A_{i,t}^2$ (where κ is defined in Definition 3.2) is clamped to 10^8 before computing α .
5. **Finiteness check:** If α or β is NaN or $\pm\infty$ after computation, normalization is skipped.

The following proposition quantifies how these safeguards change the ideal empirical constraints. A small deviation requires corresponding bounds on the stabilizer and clamp residuals; it is not automatic from the presence of these safeguards. The defaults remain hyperparameters whose empirical suitability must be evaluated separately.

Proposition C.6 (Moment Residuals of the Implemented Normalization). *Suppose both active token-sign groups are nonempty and all inputs are finite. Write*

$$P = \sum_{\mathcal{P}} A_{i,t}, \quad N = -\sum_{\mathcal{N}} A_{i,t}, \quad Q_+ = \sum_{\mathcal{P}} A_{i,t}^2, \quad Q_- = \sum_{\mathcal{N}} A_{i,t}^2,$$

and let $m = |\mathcal{P}| + |\mathcal{N}|$, $\kappa = P/N$, $D = Q_+ + \kappa^2 Q_-$, $\eta = \varepsilon_{\text{num}} > 0$. Before factor clamping, the implemented scales are

$$D_c = Q_+ + \min(\kappa^2 Q_-, C) + \eta, \quad \alpha_0 = \sqrt{m/D_c}, \quad \beta_0 = \alpha_0 \kappa,$$

where $C = 10^8$ is the intermediate cap. Their empirical mean is zero and their empirical second moment is D/D_c . In particular, if $\kappa^2 Q_- \leq C$, then the empirical variance v_0 satisfies

$$v_0 = \frac{D}{D + \eta} < 1, \quad 1 - v_0 = \frac{\eta}{D + \eta} \leq \frac{\eta}{D}. \quad (37)$$

In general,

$$\frac{D}{D_c} - 1 = \frac{\kappa^2 Q_- - \min(\kappa^2 Q_-, C) - \eta}{D_c},$$

so the product cap can change the second moment in either direction relative to one. Let a, b be the final clamped scales and let μ, v_2 be the resulting empirical mean and second moment on the m nonzero active tokens. Then

$$\begin{aligned} \mu &= \frac{aP - bN}{m}, & |\mu| &\leq \frac{|a - \alpha_0|P + |b - \beta_0|N}{m}, \\ v_2 &= \frac{a^2 Q_+ + b^2 Q_-}{m}, & \left| v_2 - \frac{D}{D_c} \right| &\leq \frac{|a - \alpha_0||a + \alpha_0|Q_+ + |b - \beta_0||b + \beta_0|Q_-}{m}. \end{aligned}$$

The empirical variance after factor clamping is $v_2 - \mu^2$. Positive clamping bounds preserve advantage signs and within-sign ordering. These statements describe the normalization branch in exact real arithmetic; when a fallback is taken, the active advantages are unchanged.

Proof. Partition the active index set into positive, negative, and zero advantages. The scaled first and second sums are $aP - bN$ and $a^2 Q_+ + b^2 Q_-$. Substituting $\beta_0 = \alpha_0 P/N$ gives zero first sum, and $\alpha_0^2 = m/D_c$ gives second moment D/D_c . The stabilizer deficit and product-cap identity follow by subtracting one. For the factor-clamp bounds, use

$$aP - bN = (a - \alpha_0)P - (b - \beta_0)N$$

and $a^2 - \alpha_0^2 = (a - \alpha_0)(a + \alpha_0)$, with the analogous identity for b , and apply the triangle inequality. Positivity of the clamped factors gives sign and within-sign order preservation. The finite-index moment identities, stabilizer and cap formulas, positive clamps, and residual bounds are verified in `formal/NormalizationSafeguards.lean`. $\square$

Fallback conditions. Normalization is skipped entirely when:

- One sign group is empty ($\mathcal{P} = \emptyset$ or $\mathcal{N} = \emptyset$). Assume, as in the token expansion convention above, that every sampled response contributes at least one active action token. Under the exact RLOO construction, the shaped rewards sum to zero within a prompt group, so this condition implies that every shaped reward is zero. The normalization is therefore skipped because its positive/negative constraints are infeasible; no gradient signal is created by this branch.
- The uniform-scale mode (Appendix C.11) is active and the implementation’s reward-uniformity tolerance is satisfied. In this branch the RLOO signal is intentionally replaced by the explicitly defined uniform-scale signal; the tolerance can include near-uniform rather than exactly equal rewards.

C.11 Uniform Scale Mode

The uniform scale mode is an optional rule for using homogeneous reward groups. When the implementation’s reward-uniformity test is satisfied (exact equality is the ideal sparse-reward case; the configured tolerance can also include near-uniform rewards), the branch replaces the leave-one-out signal by the explicit rule below and skips adaptive normalization. For exactly identical rewards, the ordinary leave-one-out baseline would yield $\tilde{r}_i = 0$ for all i . The replacement gives nonzero coefficients only when the common reward is nonzero; an all-zero group remains zero. A nonzero coefficient alone does not establish a useful expected update. The population result below identifies a sufficient reward convention and its limitations. This statement is for the KL-free sparse-reward configuration: if an additional per-token KL or other shaping term is enabled, the post-shaping returns need not remain uniform and the branch must be analyzed with those augmented returns.

Definition C.7 (Uniform Scale). For prompt groups where $\text{std}(\{r_1, \dots, r_n\}) < \varepsilon_{\text{tol}}$, the implementation classifies the group as uniform-scale eligible, sets the shaped reward to $\tilde{r}_i = r_i/n$, and skips adaptive normalization. Exact equality of the rewards is a special case, not an additional premise of the scalar result.

Proposition C.8 (Signed coefficient in uniform-scale mode). *In the uniform-scale branch, a tolerance-selected reward group uses $A_i = r_i/n$ before the actual ProMax loss is evaluated. For an active token, write $a_t \geq 0$ for the action-mask/reduction weight, $M_t \in \{0, 1\}$ for the detached prefix gate, $w_t \geq 0$ for the detached rollout-to-snapshot weight, and $u_t = \exp(\log \pi_\theta - \log \pi_{\text{old}})$ for the differentiable PPO ratio. Away from the two PPO clipping kinks, the derivative of the scalar loss with respect to the current log-probability is*

$$-a_t M_t w_t u_t \frac{r_i}{n} I_t,$$

where $I_t = 1$ on the branch through which the current ratio differentiates and $I_t = 0$ when clipping or the detached gate removes that derivative. In particular, whenever $a_t, M_t, w_t, u_t > 0$ and the unclipped branch is active, the coefficient has the same sign as $-r_i$ (the sign for gradient descent therefore reinforces positive rewards and suppresses negative rewards). For $n > 0$, r_i/n is positive, zero, or negative exactly when r_i is positive, zero, or negative. This is a statement about each retained scalar score coefficient; it does not claim a coordinate-wise or full-vector gradient-direction guarantee.

Proof. With $\tilde{r}_i = r_i/n$, the implemented scalar token loss is the masked, importance-weighted PPO term, so its derivative must retain the prefix gate, detached IS coefficient, and clipping branch. The finite real-arithmetic derivative in the non-kink regions is checked by `ppoTerm_hasDerivAt` and `gated_ppo_hasFDDerivAt` in `formal/PPODerivative.lean`; the finite sum over action positions is checked by `finite_gated_ppo_hasFDDerivAt`. The simpler sign fact for the uniform coefficient is `uniform_scale_token_coefficient_sign`. Thus a positive reward reinforces the retained sampled score contribution and a negative reward suppresses it, subject to the gate and clipping branch. The exact effect on individual parameter coordinates depends on the score vectors. $\square$

Proposition C.9 (Population effect of the optional uniform-scale branch). *Fix a prompt and a differentiable finite response policy p_θ , with rewards in $\{r_-, r_+\}$, $r_- < r_+$, and $n \geq 2$ independent responses. Write $J_x(\theta) = \mathbb{E}_{p_\theta}[R]$, $s(\theta) = \Pr_{p_\theta}(R = r_+)$ and $u_\theta(y) = \nabla_\theta \log p_\theta(y)$, with the zero-mass score convention of the finite model. Assume the uniformity test selects exactly the homogeneous groups. At the on-policy point, with all prefix gates accepted and PPO clipping inactive, compare the*

sequence-score statistics

$$g = \frac{1}{n} \sum_{i=1}^n \hat{A}_i u_\theta(y_i)$$

with uniform scale on and off. The mixed-group coefficients are identical in both cases and may include the actual asymmetric normalization and safeguards; coefficients are held fixed when computing the score update. Then

$$\begin{aligned} \mathbb{E}[g_{\text{on}} - g_{\text{off}}] &= \lambda_n(s) \nabla_\theta s, \\ \lambda_n(s) &:= \frac{r_+ s^{n-1} - r_- (1-s)^{n-1}}{n}. \end{aligned} \tag{38}$$

Since $\nabla_\theta J_x = (r_+ - r_-) \nabla_\theta s$, the additional expected term is aligned with the per-prompt reward gradient whenever $r_+ s^{n-1} \geq r_- (1-s)^{n-1}$. In particular, $r_- \leq 0 \leq r_+$ is sufficient. For 0/1 rewards, the extra term is $s^{n-1} \nabla_\theta J_x / n$.

Proof. RLOO is zero on a homogeneous group, including after its ordinary normalization fallback. Turning uniform scale on replaces that zero by r_+/n on the all-high event and r_-/n on the all-low event; other groups are unchanged. The independent response law gives

$$\begin{aligned} \mathbb{E} \left[\mathbb{I}_{\text{all high}} \sum_i u_\theta(y_i) \right] &= n s^{n-1} \nabla_\theta s, \\ \mathbb{E} \left[\mathbb{I}_{\text{all low}} \sum_i u_\theta(y_i) \right] &= -n (1-s)^{n-1} \nabla_\theta s. \end{aligned}$$

These identities follow by separating each response from its $n-1$ siblings in the finite product law. The success-restricted score expectation equals $\nabla_\theta s$ by finite policy differentiation; the failure-restricted expectation is its negative because the complete score has mean zero. Multiplying by the reward and the two factors $1/n$ proves Eq. (38). $\square$

This is a fixed-prompt, on-policy mechanism statement. The statistic sums each response’s token scores before any random total-token reduction. If every response has the same deterministic active length L , a global token mean multiplies both sides by $1/L$. A random denominator, off-policy weights, action-dependent masks, or active PPO clipping require the corrections in the general objective and reward certificate. Moreover, different prompts generally have different $\lambda_n(s)$, so the fixed-prompt result does not establish alignment after summing gradients of shared parameters across prompts. The result concerns the extra term; it does not assert that every finite stochastic optimizer step improves reward. When $0 < s < 1$, the probability of an all-success group is s^n , and all-failure 0/1 groups still produce zero coefficients.

Proposition C.10 (Complete binary update under class-dependent lengths). *Consider the same finite on-policy model with $n = m + 1 \geq 2$ iid responses, but now include the actual token reduction. Assume that every response with label $q \in \{0, 1\}$ has a deterministic active length L_q satisfying $0 < L_q \leq L_{\max}$. Let the positive/negative factors be computed from the length-weighted RLOO advantages using the implemented additive stabilizer, intermediate product cap, factor clamps and identity fallback, with numerical lower clamp $\eta \in (0, 1]$. Let H_i be the resulting coefficient of the response score after division by the sampled total active length. On-policy, with every prefix gate accepted and PPO clipping inactive, the complete expected sequence-score update satisfies*

$$\mathbb{E} \left[\sum_{i=1}^n H_i(Y_{1:n}) u_\theta(Y_i) \right] = \Lambda \nabla_\theta s, \quad \Lambda \geq \frac{\eta}{L_{\max}} > 0,$$

where $s = \Pr_{p_\theta}(R = 1)$. This holds both with uniform-scale disabled and with the optional 0/1 uniform-scale replacement on homogeneous groups. The multiplier may depend on p_θ , the sibling count and the lengths, but it is common to every parameter direction.

Proof. For a fixed coordinate i , condition on the m sibling labels. The implemented factors and the total token denominator are then fixed for the two possible labels of Y_i . The finite product law and the zero-mean score identity give the exact coordinate factorization

$$\mathbb{E}[H_i(Y_{1:n})u_\theta(Y_i)] = \left(\sum_z p_\theta^{\otimes m}(z) [H_i(1, z) - H_i(0, z)] \right) \mathbb{E}[\mathbf{1}_{\{R=1\}} u_\theta(Y)].$$

For the binary RLOO signal, the sibling mean lies in $[0, 1]$. Positive factor clamps are at least η , while every sampled total length is at most $nL_{\max}$. Therefore each contrast is at least $\eta/(nL_{\max})$. The n coordinates give $\Lambda \geq \eta/L_{\max}$. On a homogeneous all-success group, the uniform replacement adds $1/(n \sum_i L_{Y_i})$ and has nonnegative contrast; on an all-failure 0/1 group it adds zero. Thus the lower bound is preserved. The finite conditioning, factorization, safeguard lower bound and the resulting common multiplier are checked by `coordinate_expectation`, `complete_update_expectation`, and `implementation_policy_direction` in `formal/BinaryUpdate.lean`. The label-dependent-length premise is explicit: if response length also depends on within-class content, the coefficient is no longer a function of the sibling labels alone and an additional covariance/error term is needed. $\square$

Proposition C.11 (Within-class length covariance decomposition). *Drop the class-dependent-length premise and let $C_i(y_{1:n})$ be the actual response-score coefficient produced by the implementation, including its length-weighted normalization, total-token denominator and optional uniform-scale term. For a fixed direction v , write $f_v(y) = \langle v, u_\theta(y) \rangle$. Assume both binary response classes have positive finite probability. Define Λ_C as the sum over coordinates and sibling draws of the difference between the conditional class means of C_i , and define $\mathcal{E}_v$ as the corresponding weighted within-class covariance between C_i and f_v . Then the exact identity is*

$$\mathbb{E} \left[\sum_i C_i(Y_{1:n}) f_v(Y_i) \right] = \Lambda_C \mathbb{E}[\mathbf{1}_{\{R=1\}} f_v(Y)] + \mathcal{E}_v,$$

and

$$|\mathcal{E}_v| \leq \sqrt{V_C n V_{f,v}},$$

Here, writing Z for the $n - 1$ independent sibling responses and inserting Y at coordinate i , define $\bar{C}_i(q, Z) := \mathbb{E}[C_i(Y, Z) \mid R(Y) = q, Z]$ and $\bar{f}_v(q) := \mathbb{E}[f_v(Y) \mid R(Y) = q]$. The variances are

$$V_C := \sum_i \mathbb{E}_{Z,Y} [(C_i(Y, Z) - \bar{C}_i(R(Y), Z))^2],$$

$$V_{f,v} := \mathbb{E}_Y [(f_v(Y) - \bar{f}_v(R(Y)))^2].$$

Therefore the complete update is guaranteed to have the same directional sign as the reward gradient whenever the positive main term dominates this explicit covariance margin. The bound does not assume independence between length, normalization and the response score.

Proof. Condition on the sibling responses and split the current response support into the two reward classes. Subtracting and adding the conditional class means of both C_i and f_v gives the displayed identity; the terms with one centered factor vanish within each class. Applying the finite Cauchy–Schwarz inequality to the remaining product gives the square-root bound. The exact decomposition is checked by `general_update_decomposition`, its finite covariance bound

by `covariance_error_abs_bound`, and the implementation-specific coefficient is connected by `response_update_decomposition` in `formal/BinaryLength.lean`. If lengths are deterministic within each class, the coefficient is sibling-label measurable and $\mathcal{E}_v = 0$, recovering Proposition C.10. $\square$

Corollary C.12 (Local true-reward improvement of the complete binary update). *Fix a prompt, binary rewards $R \in \{0, 1\}$, and $n \geq 2$ iid responses from the finite differentiable policy above, with parameters in a real Hilbert space and a policy that is nonnegative and normalized in a neighborhood of θ . Assume $0 < s(\theta) < 1$, where $s(\theta) = \Pr_{p_\theta}(R = 1)$. At the on-policy point, let every prefix gate accept and PPO clipping be inactive. Allow arbitrary content-dependent active lengths $0 < L(y) \leq L_{\max}$ and retain the actual stabilizer, product cap, factor clamps and identity fallback. The lower factor clamp is $0 < \eta \leq 1$ and the upper clamp is at least η . The optional uniform-scale branch may be enabled or disabled; when enabled, assume its uniformity test selects exactly homogeneous groups. Using the actual response-score coefficients C_i and variances in Proposition C.11, set*

$$g := \nabla s(\theta), \quad U := \mathbb{E} \left[\sum_{i=1}^n C_i(Y_{1:n}) u_\theta(Y_i) \right], \quad f_g(y) := \langle g, u_\theta(y) \rangle.$$

The actual class multiplier satisfies $\Lambda_C \geq \eta/L_{\max} > 0$, and

$$\begin{aligned} \langle g, U \rangle &\geq \Lambda_C \|g\|^2 - \sqrt{V_C n V_{f,g}} \\ &\geq \frac{\eta}{L_{\max}} \|g\|^2 - \sqrt{V_C n V_{f,g}}. \end{aligned}$$

If $\sqrt{V_C n V_{f,g}} < \Lambda_C \|g\|^2$, there exists $r > 0$ such that every $0 < t < r$ satisfies

$$s(\theta + tU) > s(\theta).$$

The simpler condition $\sqrt{V_C n V_{f,g}} < \eta \|g\|^2 / L_{\max}$ is sufficient but can be more conservative than using the actual Λ_C . Under the class-dependent-length premise of Proposition C.10, $g \neq 0$ alone suffices for this local improvement: $U = \Lambda g$ with $\Lambda \geq \eta/L_{\max} > 0$.

Proof. For each fixed sibling draw, the sibling mean reward lies in $[0, 1]$. Positive normalization factors are bounded below by η , and the sampled total length is at most $nL_{\max}$. Averaging the resulting success/failure coefficient contrast within each class gives $\Lambda_C \geq \eta/L_{\max}$ even when length varies with content. The optional all-success replacement adds a nonnegative contrast. Apply Proposition C.11 with $v = g$ and use the finite-policy identity $\mathbb{E}[\mathbf{1}_{\{R=1\}} f_g(Y)] = \|g\|^2$ to obtain the displayed lower bound. A positive right-hand side implies $\left. \frac{d}{dt} s(\theta + tU) \right|_{t=0} = \langle g, U \rangle > 0$. By differentiability, the difference quotient is positive for every sufficiently small positive t , proving strict improvement. For class-dependent lengths, Proposition C.10 directly gives $\langle g, U \rangle = \Lambda \|g\|^2 > 0$ whenever $g \neq 0$. The multiplier bound is checked by `response_class_multiplier_lower_bound` in `formal/BinaryLength.lean`; the gradient connection, directional bound and both local improvement statements are checked by `success_gradient_correct`, `population_ascent_bound`, `population_local_improvement`, `response_local_improvement` and `class_length_local_improvement` in `formal/BinaryAscent.lean`. $\square$

This result concerns a sufficiently small step along the complete population update at a fixed prompt. It supplies no numerical step-size radius and does not imply improvement for every stochastic or Adam step. Arbitrary off-policy updates, active gates and clipping remain covered by the explicit correction terms in Theorem 5.4.

Reward translation and a complete counterexample. Adding a common offset c to both reward levels preserves the reward gradient and all raw RLOO signals, but changes the extra multiplier by

$$\lambda_n^{\text{shifted}}(s) - \lambda_n(s) = \frac{c}{n}(s^{n-1} - (1-s)^{n-1}).$$

For an explicit example, use two one-token responses per group, $p_\theta(\text{high}) = \theta$, $p_\theta(\text{low}) = 1 - \theta$, and rewards 4 and 3. At $\theta = 1/10$, $J(\theta) = 3 + \theta$, so $J'(\theta) = 1$. Mixed-group RLOO coefficients are $+1, -1$; their ideal normalization leaves them unchanged. The implemented real-arithmetic safeguards instead multiply both by

$$a_\varepsilon = \min \left\{ 10, \max \left\{ \varepsilon, \sqrt{\frac{2}{2+\varepsilon}} \right\} \right\}, \quad 0 < \varepsilon \leq 1,$$

so $0 < a_\varepsilon \leq 1$; the product cap is inactive. The mixed rewards have sample variance $1/2$, hence their standard deviation exceeds the implementation’s 10^{-8} uniformity tolerance, while homogeneous groups have zero standard deviation. Thus this example follows the actual branch selection. Direct expectation under the two-response product law gives

$$\mathbb{E}[g_{\text{off}}] = a_\varepsilon > 0, \quad \mathbb{E}[g_{\text{on}}] = a_\varepsilon - \frac{23}{20} \leq -\frac{3}{20} < 0. \quad (39)$$

This is an expected ascent-direction reversal in a valid bounded-reward model, at an on-policy point with accepted gates and inactive clipping. It does not require a floating-point failure. With rewards 0, 1 in the same model, the corresponding expectation is $a_\varepsilon + \theta/2 > 0$ for $0 < \theta < 1$. The optional rule is therefore tied to the reward origin; these two reward conventions cannot be treated as equivalent ablations.

A constant reward carries no population signal. If every response in the policy support has reward c , uniform scale produces $g = (c/n^2) \sum_i u_\theta(y_i)$. The expected score is zero, so $\mathbb{E}[g] = 0$, as required by the constant reward objective. For every parameter direction v , independence gives

$$\text{Var}(\langle v, g \rangle) = \frac{c^2}{n^3} \mathbb{E}_p[\langle v, u_\theta(Y) \rangle^2].$$

Ordinary RLOO is identically zero in this setting. The optional branch can therefore introduce noise without useful population signal, while $c = 0$ leaves the entire update zero. The finite product-law expectation, differential connection, offset counterexample with stabilizer/clamp, and directional variance are checked in `formal/UniformScale.lean`.

D REINFORCE Pro: Proofs and Variants

D.1 Proof of Theorem 4.7

We provide a complete proof of Theorem 4.7 under the definitions in Section 4.3. For convenience, write the log-threshold interval as

$$[\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})] = [-\tau_-, \tau_+], \quad \tau_+ := \log(1 + \epsilon_{\text{high}}), \quad \tau_- := -\log(1 - \epsilon_{\text{low}}),$$

so that $\rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ is equivalent to $\log \rho_t \in [-\tau_-, \tau_+]$.

Mask Definitions The three masks compared in Theorem 4.7 are:

- **Token-level (IcePop / token-level ratio mask):**

$$M_t^{\text{tok}} := \mathbb{I}[\rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]] = \mathbb{I}[\log \rho_t \in [-\tau_-, \tau_+]].$$

- **Sequence-level (GSPO / sequence-level ratio gate):**

$$M_t^{\text{seq}} := \mathbb{I}[\log \rho^{\text{seq}} \in [-\tau_-, \tau_+]], \quad \log \rho^{\text{seq}} := \frac{1}{T} \sum_{s=1}^T \log \rho_s,$$

for all t .

- **Prefix-level (Causal Trust Region proxy):**

$$M_t^{\text{pre}} := \mathbb{I}[\rho_t^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]] = \mathbb{I}[\log \rho_t^{\text{pre}} \in [-\tau_-, \tau_+]],$$

where ρ_t^{pre} is defined in Eq. (12).

Proof of Theorem 4.7. Part (a): Early deviation. Assume $\rho_1 \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ and $|\log \rho_t| \leq \varepsilon$ for all $t > 1$, with $0 \leq \varepsilon \leq \min(\tau_+, \tau_-)$. Then for every $t > 1$,

$$\log \rho_t \in [-\varepsilon, \varepsilon] \subseteq [-\tau_-, \tau_+],$$

so $M_t^{\text{tok}} = 1$ for all $t > 1$, while $M_1^{\text{tok}} = 0$ (since $\rho_1 \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$). Hence token-level IcePop masks only position $t = 1$.

For the prefix mask, note first that $M_1^{\text{pre}} = 0$ because $\rho_1^{\text{pre}} = \rho_1 \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$. To quantify how long the rejection can persist, use Eq. (12):

$$\log \rho_t^{\text{pre}} = \frac{1}{t} \sum_{s=1}^t \log \rho_s = \frac{1}{t} \left(\log \rho_1 + \sum_{s=2}^t \log \rho_s \right).$$

If $\log \rho_1 > \tau_+$, then $\log \rho_s \geq -\varepsilon$ for all $s \geq 2$, hence $\sum_{s=2}^t \log \rho_s \geq -(t-1)\varepsilon$ and therefore

$$\log \rho_t^{\text{pre}} \geq \frac{1}{t} (\log \rho_1 - (t-1)\varepsilon) = \frac{\log \rho_1 + \varepsilon}{t} - \varepsilon.$$

Thus, if $t < (\log \rho_1 + \varepsilon)/(\tau_+ + \varepsilon)$, then $\log \rho_t^{\text{pre}} > \tau_+$ and hence $M_t^{\text{pre}} = 0$. The lower-violation case $\log \rho_1 < -\tau_-$ is symmetric: since $\log \rho_s \leq \varepsilon$ for $s \geq 2$, we have $\sum_{s=2}^t \log \rho_s \leq \varepsilon(t-1)$ and therefore

$$\log \rho_t^{\text{pre}} \leq \frac{1}{t} (\log \rho_1 + \varepsilon(t-1)) = \frac{\log \rho_1 - \varepsilon}{t} + \varepsilon.$$

If $t < (-\log \rho_1 + \varepsilon)/(\tau_- + \varepsilon)$, then $\log \rho_t^{\text{pre}} < -\tau_-$ and hence $M_t^{\text{pre}} = 0$. This proves the sufficient conditions in Equations (14) and (15).

Part (b): Late deviation. Assume there exists $t^* \in \{1, \dots, T\}$ such that $\rho_{t^*} \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ and $|\log \rho_t| \leq \varepsilon$ for all $t \neq t^*$, with $0 \leq \varepsilon \leq \min(\tau_+, \tau_-)$. For any $t < t^*$, all active log-ratios in the prefix satisfy $\log \rho_s \in [-\tau_-, \tau_+]$. Since $\log \rho_t^{\text{pre}} = \frac{1}{t} \sum_{s=1}^t \log \rho_s$ is an average of values in $[-\tau_-, \tau_+]$, it follows that $\log \rho_t^{\text{pre}} \in [-\tau_-, \tau_+]$, and hence $M_t^{\text{pre}} = 1$ for all $t < t^*$. Therefore prefix masking can only occur for positions $t \geq t^*$, which implies

$$|\{t : M_t^{\text{pre}} = 0\}| \leq T - t^* + 1.$$

The sequence-level IS mask M_t^{seq} is constant across positions by definition, so it either rejects all tokens or rejects none.

For the claimed regime where prefix rejects but sequence-level accepts, consider the pure single-spike case where $\rho_t = 1$ for $t \neq t^*$ and $\rho_{t^*} > 1$. Then

$$\log \rho^{\text{seq}} = \frac{1}{T} \log \rho_{t^*}, \quad \log \rho_{t^*}^{\text{pre}} = \frac{1}{t^*} \log \rho_{t^*}.$$

Since $t^* \leq T$ and $\log \rho_{t^*} > 0$, we have $\log \rho_{t^*}^{\text{pre}} \geq \log \rho^{\text{seq}}$. Thus whenever

$$\frac{1}{t^*} \log \rho_{t^*} > \tau_+ \quad \text{and} \quad \frac{1}{T} \log \rho_{t^*} \leq \tau_+,$$

we obtain $M_{t^*}^{\text{pre}} = 0$ while $M_t^{\text{seq}} = 1$ for all t . For every $t^* < T$, the explicit choice $\log \rho_{t^*} = \tau_+(t^* + T)/2$ satisfies these inequalities, since

$$\tau_+ < \tau_+ \frac{t^* + T}{2t^*}, \quad 0 < \tau_+ \frac{t^* + T}{2T} < \tau_+.$$

For the lower threshold, choose $\log \rho_{t^*} = -\tau_-(t^* + T)/2$. Then

$$-\tau_- \frac{t^* + T}{2t^*} < -\tau_-, \quad -\tau_- < -\tau_- \frac{t^* + T}{2T} < 0,$$

which again rejects the spike under the prefix mask and accepts the whole sequence under the sequence mask. Both constructions have only one out-of-bound token. When $t^* = T$, the two statistics at that position are identical for every trajectory, so strict separation there is impossible.

Part (c): Sustained one-sided violation. If $\log \rho_s > \tau_+$ for all s , then for every t the prefix average satisfies $\log \rho_t^{\text{pre}} > \tau_+$, since it is an average of values each strictly larger than τ_+ . Each token mask also rejects directly by its defining interval test. Hence $M_t^{\text{tok}} = M_t^{\text{pre}} = 0$ for all t . The sequence-level average $\log \rho^{\text{seq}}$ is also strictly larger than τ_+ , so $M_t^{\text{seq}} = 0$ for all t . The lower-violation case $\log \rho_s < -\tau_-$ is symmetric. $\square$

D.2 Complete Finite-Tree Adaptive Bound

Proof of Theorem 2.10. Fix a prompt and write $p = \pi_{\text{roll}}$, $q = \pi_{\theta}$. For a history h with n actions remaining, let $V_n^p(h)$ and $V_n^q(h)$ be the respective expected terminal rewards. Let $S_n(h)$ be the rollout-occupancy surrogate for this continuation, and set $E_n(h) = V_n^q(h) - V_n^p(h) - S_n(h)$. Its local surrogate gain is $\sum_a (q(a | h) - p(a | h)) V_n^p(ha)$. Expanding the normalized policy recursions and collecting terms gives the exact identity

$$\begin{aligned} E_{n+1}(h) &= \sum_a p(a | h) E_n(ha) + I_n(h), \\ I_n(h) &:= \sum_a (q(a | h) - p(a | h)) (V_n^q(ha) - V_n^p(ha)). \end{aligned}$$

Here $E_0(h) = 0$. In particular, the recursion uses rollout probabilities in its outer expectation; no policy-occupancy substitution is assumed.

For either continuation starting at ha , bounded reward gives

$$|V_n^q(ha) - V_n^p(ha)| \leq 2R_{\max} D_{\text{TV}}(P^p(\cdot | ha), P^q(\cdot | ha)).$$

The continuation TV is at most 1 by normalization. Applying the expected-token-TV sum bound to this re-rooted tree gives $n\epsilon$. The KL chain rule on the same tree gives continuation reverse KL at

most $n\delta$, so Pinsker gives the third bound $\sqrt{n\delta/2}$. All required local contexts have length less than T ; there is no assumption on divergences beyond the generation horizon. Define $b_n = \min(1, n\epsilon, \sqrt{n\delta/2})$. Then

$$|I_n(h)| \leq 2R_{\max} b_n \sum_a |q(a | h) - p(a | h)| = 4R_{\max} b_n D_{\text{TV}}^{\text{tok}}(h; p, q).$$

Unrolling the error recursion from $E_T(\emptyset)$ yields a sum of these local interactions, averaged over rollout histories. At position t , the continuation has $T - t$ actions. Taking absolute values and then averaging over the finite prompt distribution therefore gives

$$|\text{Error}(\pi_\theta)| \leq 4R_{\max} \sum_{t=1}^T \bar{D}_t b_{T-t}.$$

The performance lower bound follows by subtracting this error allowance from the surrogate. The trajectory identity in Remark E.8 identifies E_T with the negative reward-weighted remainder, so the same absolute bound applies to that remainder. Shared absorbing EOS transitions have zero local TV and zero interaction. $\square$

The exact recursion is checked by `surrogate_error_future_identity`. The three future-TV bounds are combined in `futureTV_le_adaptive`, and `surrogate_error_adaptive_root` and `trajectory_remainder_adaptive` prove the root error and actual trajectory remainder bounds. `family_remainder_adaptive` and `family_performance_lower_bound_adaptive` include the finite prompt law and the paper’s expected per-position TV, in `formal/AdaptiveBound.lean`.

D.3 Prefix Causal Trust Region Mechanism Bounds

This subsection proves Lemmas 4.9 and 4.10 and Theorem 4.11, which formalize a per-position mechanism interpretation of prefix Causal Trust Region masking.

Proof of Lemma 4.9. Fix $t \in \{1, \dots, T\}$ and a reachable context $c_{t+1} = (x, y_{\leq t})$ with $d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) > 0$. By the definition of the context visitation distribution, for any policy π ,

$$\begin{aligned} d_{t+1}^\pi(c_{t+1}) &= P(x) P^\pi(y_{\leq t} | x) \\ &= P(x) \prod_{s=1}^t \pi(y_s | c_s), \end{aligned}$$

where $c_s = (x, y_{<s})$ and the second line is the autoregressive factorization. Under the mutual-support Assumption 2.3, the ratio $d_{t+1}^{\pi_\theta}(c_{t+1})/d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})$ is well-defined for all such contexts. Therefore,

$$\begin{aligned} \frac{d_{t+1}^{\pi_\theta}(c_{t+1})}{d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})} &= \prod_{s=1}^t \frac{\pi_\theta(y_s | c_s)}{\pi_{\text{roll}}(y_s | c_s)} \\ &= \prod_{s=1}^t \rho_s = W_t, \end{aligned}$$

as claimed. $\square$

Proof of Lemma 4.10. Fix $t \in \{1, \dots, T\}$. By definition of total variation distance,

$$\|d_{t+1}^{\pi_\theta} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} = \frac{1}{2} \sum_{c_{t+1}} |d_{t+1}^{\pi_\theta}(c_{t+1}) - d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})|.$$

By Lemma 4.9, for every reachable c_{t+1} with $d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) > 0$,

$$d_{t+1}^{\pi_{\theta}}(c_{t+1}) = d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) W_t.$$

Substituting this into the previous display yields

$$\begin{aligned} \|d_{t+1}^{\pi_{\theta}} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} &= \frac{1}{2} \sum_{c_{t+1}} d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) |W_t - 1| \\ &= \frac{1}{2} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [|W_t - 1|], \end{aligned}$$

which proves the claim. $\square$

Proof of Theorem 4.11. Fix $t \in \{1, \dots, T\}$. Recall that $\tau_+ := \log(1 + \epsilon_{\text{high}})$ and $\tau_- := -\log(1 - \epsilon_{\text{low}})$.

Step 1: Rewrite the acceptance event. By definition of the prefix mask in Eq. (13),

$$M_t^{\text{pre}} = 1 \iff \log \rho_t^{\text{pre}} \in [-\tau_-, \tau_+].$$

Using Eq. (12), this is equivalent to the cumulative log-ratio constraint

$$-t\tau_- \leq \sum_{s=1}^t \log \rho_s \leq t\tau_+.$$

Step 2: Convert to a bound on the prefix density ratio. Recall that $W_t = \prod_{s=1}^t \rho_s$. Equivalently,

$$W_t = \exp\left(\sum_{s=1}^t \log \rho_s\right).$$

Therefore, the previous display implies that, on $\{M_t^{\text{pre}} = 1\}$,

$$e^{-t\tau_-} \leq W_t \leq e^{t\tau_+}.$$

Step 3: Bound the ratio residual and take expectations. If a scalar w lies in an interval $[a, b]$ with $0 < a \leq b$, then $|w - 1| \leq \max\{b - 1, 1 - a\}$. Applying this with $a = e^{-t\tau_-}$ and $b = e^{t\tau_+}$ yields that on $\{M_t^{\text{pre}} = 1\}$,

$$|W_t - 1| \leq \max\{e^{t\tau_+} - 1, 1 - e^{-t\tau_-}\}.$$

Multiplying by M_t^{pre} and taking expectations under $c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}$ gives

$$\begin{aligned} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [M_t^{\text{pre}} |W_t - 1|] &\leq \max\{e^{t\tau_+} - 1, 1 - e^{-t\tau_-}\} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [M_t^{\text{pre}}] \\ &\leq \max\{e^{t\tau_+} - 1, 1 - e^{-t\tau_-}\}, \end{aligned}$$

which completes the proof. $\square$

The complete finite-history statement is checked in `formal/TreePrefixGate.lean`. The gate is defined by the actual cumulative log-ratio divided by the active prefix length. `accepted_prefix_density_envelope` proves the density-ratio and residual bounds on positive-mass accepted prefixes; `prefix_gate_residual_bound` and `prefix_gate_residual_le_envelope` prove both expectation inequalities under the normalized rollout tree. `prefix_gate_tv_bound` also identifies the corresponding bound on the half-L1 distance of the filtered prefix masses. These statements allow an action-dependent gate and common zero-mass tokens. They concern filtered probability mass and do not assert an unbiased masked policy gradient.

D.4 Sub-Gaussian Strengthening: Per-Position Bounds via Prefix Density Ratios

This subsection proves Lemmas D.2 and D.6, Corollaries D.3, D.7 and D.8, and Theorem D.4, which strengthen the prefix density-ratio mechanism view with explicit per-position bounds under a conditional concentration assumption.

D.4.1 Statements and Interpretation

The mechanism in Section 4.4 is distributional: it identifies W_t as the change-of-measure object that controls context shift. To turn Lemma 4.10 into an explicit per-position bound, we introduce a quantitative concentration assumption on the per-token log-ratio increments.

Assumption D.1 (Conditionally sub-Gaussian log-ratio increments). Under the rollout distribution $y_t \sim \pi_{\text{roll}}(\cdot | c_t)$, define the centered increment

$$X_t := \log \rho_t - \mathbb{E}[\log \rho_t | c_t].$$

Assume that $\log \rho_t$ is conditionally integrable and that $D_{\text{KL}}(\pi_{\text{roll}}(\cdot | c_t) \parallel \pi_{\theta}(\cdot | c_t)) < \infty$ on every reachable context. There exists a constant $\sigma^2 > 0$ such that for all $t \in \{1, \dots, T\}$, all reachable contexts c_t , and all $\lambda \in \mathbb{R}$,

$$\mathbb{E}[\exp(\lambda X_t) | c_t] \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right).$$

Lemma D.2 (Second-moment control of the prefix density ratio). *In the finite-vocabulary history-tree model, assume the mutual-support Assumption 2.3 and Assumption D.1. Then for any $t \in \{1, \dots, T\}$,*

$$\mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}}[W_t^2] \leq \exp(2\sigma^2 t).$$

Corollary D.3 (An explicit per-position context-shift bound). *In the same finite-vocabulary history-tree model, assume the mutual-support Assumption 2.3 and Assumption D.1. Then for any $t \in \{1, \dots, T\}$,*

$$\|d_{t+1}^{\pi_{\theta}} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\exp(2\sigma^2 t) - 1}.$$

Theorem D.4 (A per-position surrogate-error bound under sub-Gaussian prefix drift). *In the same finite-vocabulary history-tree model, assume the mutual-support Assumption 2.3, Assumption D.1, and bounded rewards (i.e., $|R(x, y)| \leq R_{\max}$). Then the surrogate error in Eq. (5) satisfies*

$$|\text{Error}(\pi_{\theta})| \leq 2R_{\max} \sum_{t=1}^T \sqrt{\exp(2\sigma^2(t-1)) - 1}.$$

On the finite history tree, the proof expands the cumulative advantage signal into its expectations under each prefix distribution, applies the concentration bound to their differences, and uses $|g_t| \leq 2R_{\max}$. This complete finite-model argument is verified by `surrogate_error_of_mgf` in `formal/PrefixConcentration.lean`. General measurable prompt spaces, almost-sure conditional assumptions, and the general filtration formulation remain separate formalization obligations.

Corollary D.5 (A concentration-based improvement certificate). *For each fixed prompt in the finite-vocabulary model, under the assumptions of Theorem D.4, define*

$$B_T(\sigma) := 2R_{\max} \sum_{t=1}^T \sqrt{\exp(2\sigma^2(t-1)) - 1}.$$

For this fixed prompt x , write $J_x(\pi) := \mathbb{E}_{y \sim P^\pi(\cdot|x)}[R(x, y)]$ and let $\mathcal{L}_{x, \pi_{\text{roll}}}^{\text{LLM}}(\pi_\theta)$ denote the corresponding rollout surrogate with the prompt held fixed. Then

$$J_x(\pi_\theta) - J_x(\pi_{\text{roll}}) \geq \mathcal{L}_{x, \pi_{\text{roll}}}^{\text{LLM}}(\pi_\theta) - B_T(\sigma).$$

In particular, $\mathcal{L}_{x, \pi_{\text{roll}}}^{\text{LLM}}(\pi_\theta) > B_T(\sigma)$ implies strict improvement in expected reward.

Proof. Apply the lower side of the absolute surrogate-error bound in Theorem D.4. The lower bound and strict-improvement implication are checked by `performance_lower_bound_of_mgf` and `performance_improves_of_mgf` in `formal/PrefixConcentration.lean`. $\square$

The certificate concerns the raw population surrogate. Theorem 5.4 connects the actual ProMax objective to that surrogate using explicit gate, normalization, clipping and sampling terms. Its estimation argument also accepts the concentration error bound above in place of the Adaptive cost whenever those assumptions hold.

Acceptance-coupled filtered drift bounds. The Pro Max loss (Eq. (18)) multiplies each per-token surrogate term by the prefix gate M_t^{pre} , so gradient signal is received only from *accepted* prefixes. To quantify policy drift on this filtered subset, define the accepted-prefix context measure (a sub-probability measure)

$$\bar{d}_{t+1}^\pi(c_{t+1}) := d_{t+1}^\pi(c_{t+1}) M_t^{\text{pre}}(c_{t+1}), \quad t \in \{1, \dots, T\},$$

where we view M_t^{pre} as a deterministic function of the length- t prefix $c_{t+1} = (x, y_{\leq t})$ under the fixed policy pair $(\pi_{\text{roll}}, \pi_\theta)$. For these possibly unequal-mass measures, we explicitly use $\|\mu - \nu\|_{\text{TV}} := \frac{1}{2} \sum_c |\mu(c) - \nu(c)|$, the half- L^1 convention; this need not equal $\sup_B |\mu(B) - \nu(B)|$ when the total masses differ. No renormalization is applied. Its total mass under rollout is the acceptance rate

$$\alpha_t := \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}}] = \sum_{c_{t+1}} \bar{d}_{t+1}^{\pi_{\text{roll}}}(c_{t+1}).$$

Lemma D.6 (TV identity for accepted-prefix context measures). *In the finite-vocabulary, normalized history-tree model used above, assume the mutual-support Assumption 2.3. Then for any $t \in \{1, \dots, T\}$,*

$$\|\bar{d}_{t+1}^{\pi_\theta} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} = \frac{1}{2} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}} | W_t - 1|].$$

Corollary D.7 (Filtered drift bound with acceptance rate). *In the same finite-vocabulary, normalized history-tree model, assume the mutual-support Assumption 2.3 and Assumption D.1. Then for any $t \in \{1, \dots, T\}$,*

$$\mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}} | W_t - 1|] \leq \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1}.$$

Equivalently,

$$\|\bar{d}_{t+1}^{\pi_\theta} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1}.$$

Corollary D.8 (Gate-induced effective horizon). *In the same finite-vocabulary, normalized history-tree model, assume the mutual-support Assumption 2.3 and Assumption D.1. Define the effective horizon*

$$H_{\text{eff}} := \sum_{t=1}^T \alpha_t = \sum_{t=1}^T \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}}].$$

Then the cumulative accepted-prefix drift satisfies

$$\begin{aligned} \sum_{t=1}^T \|\bar{d}_{t+1}^{\pi_\theta} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} &\leq \frac{1}{2} \sum_{t=1}^T \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1} \\ &\leq \frac{1}{2} \sqrt{H_{\text{eff}}} \sqrt{\sum_{t=1}^T (\exp(2\sigma^2 t) - 1)}. \end{aligned}$$

In particular, Corollary D.8 makes explicit the $\sqrt{\alpha_t}$ per-position attenuation and summarizes its aggregate effect through H_{eff} , formalizing a “gate + concentration” effective-horizon view. The attenuation includes the effect of reduced accepted mass; it does not assert that the renormalized conditional distributions become closer.

These accepted-prefix quantities are a filtered drift diagnostic for the policy pair that supplies the prefix ratio (the implementation pair is $(\pi_{\text{roll}}, \pi_{\text{old}})$). They coincide with the behavior/current-policy pair at the start of an update when $\pi_{\text{old}} = \pi_\theta$. Because M_t^{pre} depends on the sampled prefix including y_t , it is measurable with respect to c_{t+1} rather than c_t ; therefore the displayed TV identities do not by themselves assert an unbiased policy-gradient identity for the masked objective. The mask is a causal sample-level filter, and a direct bound on the masked gradient would require an additional change-of-measure theorem on the action-extended prefix space.

The remaining proofs follow below.

For a finite vocabulary and horizon, the concentration argument is verified on the full autoregressive history tree in `formal/PrefixConcentration.lean`. The construction multiplies normalized, history-dependent token kernels to obtain prefix masses, proves their normalization and density-ratio identity, and connects the resulting finite expectations to the recursive second moment. Common zero-probability actions, including shared absorbing EOS transitions, are allowed under mutual support. No independence across tokens is assumed. The centered-MGF condition is needed only at $\lambda = 2$ and only at rollout-reachable histories shorter than the chosen horizon; zero-probability branches vanish in the second-moment recursion. The all- λ assumption below implies this condition. The conditional mean is proved non-positive from the local policy probabilities, rather than supplied as an additional hypothesis. The same development proves the prefix and accepted-prefix TV bounds, their effective-horizon aggregation, and the resulting surrogate-error bound in Lemmas D.2 and D.6, Corollaries D.3, D.7 and D.8, and Theorem D.4. The exact theorem names and remaining general probability-space interfaces are recorded in `formal/PROOF_AUDIT.md`.

D.4.2 Proofs

Proof of Lemma D.2. Fix $t \in \{1, \dots, T\}$. Throughout this proof, expectations are taken under the rollout distribution $c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}$ (equivalently, $(x, y_{\leq t})$ generated by π_{roll}). Recall $W_t = \prod_{s=1}^t \rho_s$.

Step 1: Decompose log-ratios into mean + centered increments. For each $s \in \{1, \dots, t\}$, define

$$\mu_s := \mathbb{E}[\log \rho_s \mid c_s], \quad X_s := \log \rho_s - \mu_s.$$

Under mutual support and the conditional integrability in Assumption D.1, the conditional expectation is well-defined, and moreover

$$\mu_s = \mathbb{E}_{y_s \sim \pi_{\text{roll}}(\cdot \mid c_s)} \left[\log \frac{\pi_\theta(y_s \mid c_s)}{\pi_{\text{roll}}(y_s \mid c_s)} \right] = -D_{\text{KL}}(\pi_{\text{roll}}(\cdot \mid c_s) \parallel \pi_\theta(\cdot \mid c_s)) \leq 0.$$

Therefore,

$$W_t^2 = \exp\left(2 \sum_{s=1}^t \log \rho_s\right)$$

$$\begin{aligned}
&= \exp\left(2 \sum_{s=1}^t \mu_s\right) \cdot \exp\left(2 \sum_{s=1}^t X_s\right) \\
&\leq \exp\left(2 \sum_{s=1}^t X_s\right),
\end{aligned}$$

and hence

$$\mathbb{E}[W_t^2] \leq \mathbb{E}\left[\exp\left(2 \sum_{s=1}^t X_s\right)\right].$$

Step 2: Control the MGF of the adapted sum by tower property. Let $\mathcal{F}_s := \sigma(x, y_{\leq s})$ denote the natural filtration. Since $c_s = (x, y_{< s})$ is $\mathcal{F}_{s-1}$ -measurable and X_s depends only on (c_s, y_s) , we have

$$\mathbb{E}[\exp(2X_s) \mid \mathcal{F}_{s-1}] = \mathbb{E}[\exp(2X_s) \mid c_s].$$

By Assumption D.1 with $\lambda = 2$,

$$\mathbb{E}[\exp(2X_s) \mid c_s] \leq \exp\left(\frac{2^2 \sigma^2}{2}\right) = \exp(2\sigma^2).$$

Therefore, applying the tower property yields

$$\begin{aligned}
\mathbb{E}\left[\exp\left(2 \sum_{s=1}^t X_s\right)\right] &= \mathbb{E}\left[\exp\left(2 \sum_{s=1}^{t-1} X_s\right) \cdot \mathbb{E}[\exp(2X_t) \mid \mathcal{F}_{t-1}]\right] \\
&\leq \exp(2\sigma^2) \mathbb{E}\left[\exp\left(2 \sum_{s=1}^{t-1} X_s\right)\right].
\end{aligned}$$

Iterating this inequality gives

$$\mathbb{E}\left[\exp\left(2 \sum_{s=1}^t X_s\right)\right] \leq \exp(2\sigma^2 t).$$

Combining with Step 1 completes the proof. $\square$

Proof of Corollary D.3. Fix $t \in \{1, \dots, T\}$. By Lemma 4.10,

$$\|d_{t+1}^{\pi_\theta} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} = \frac{1}{2} \mathbb{E}[|W_t - 1|].$$

By Cauchy–Schwarz, $\mathbb{E}[|Z|] \leq \sqrt{\mathbb{E}[Z^2]}$, so

$$\mathbb{E}[|W_t - 1|] \leq \sqrt{\mathbb{E}[(W_t - 1)^2]}.$$

Expanding the square yields

$$\mathbb{E}[(W_t - 1)^2] = \mathbb{E}[W_t^2] - 2\mathbb{E}[W_t] + 1.$$

By Lemma 4.9, $W_t = d_{t+1}^{\pi_\theta}(c_{t+1})/d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})$, so

$$\mathbb{E}[W_t] = \sum_{c_{t+1}} d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) \frac{d_{t+1}^{\pi_\theta}(c_{t+1})}{d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})} = \sum_{c_{t+1}} d_{t+1}^{\pi_\theta}(c_{t+1}) = 1.$$

Together with Lemma D.2, this implies

$$\mathbb{E}[(W_t - 1)^2] \leq \exp(2\sigma^2 t) - 1.$$

Substituting into the first display gives the claim. $\square$

Proof of Theorem D.4. Recall the error decomposition in Eq. (5):

$$|\text{Error}(\pi_\theta)| \leq \sum_{t=1}^T 2\|g_t\|_\infty \cdot \|d_t^{\pi_\theta} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}}.$$

Step 1: Bound the advantage factor by bounded rewards. Fix t and a context c_t . By definition,

$$g_t(c_t) := \mathbb{E}_{y_t \sim \pi_\theta(\cdot | c_t)}[A_t^{\pi_{\text{roll}}}(c_t, y_t)], \quad A_t^{\pi_{\text{roll}}}(c_t, y_t) := \mathbb{E}_{\pi_{\text{roll}}}[R(x, y) | c_t, y_t] - \mathbb{E}_{\pi_{\text{roll}}}[R(x, y) | c_t].$$

Under bounded rewards $|R(x, y)| \leq R_{\max}$, both conditional expectations lie in $[-R_{\max}, R_{\max}]$, hence $|A_t^{\pi_{\text{roll}}}(c_t, y_t)| \leq 2R_{\max}$ and consequently $|g_t(c_t)| \leq 2R_{\max}$. Therefore $\|g_t\|_\infty \leq 2R_{\max}$.

Step 2: Apply the per-position TV bound. For each $t \in \{1, \dots, T\}$, observe that d_t is the context distribution indexed by token position t , i.e., it corresponds to a length- $(t-1)$ prefix. Thus Corollary D.3 with index $(t-1)$ gives

$$\|d_t^{\pi_\theta} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\exp(2\sigma^2(t-1)) - 1}.$$

Step 3: Combine. Substituting the bounds from Steps 1-2 into the error decomposition yields

$$\begin{aligned} |\text{Error}(\pi_\theta)| &\leq \sum_{t=1}^T 2(2R_{\max}) \cdot \frac{1}{2} \sqrt{\exp(2\sigma^2(t-1)) - 1} \\ &= 2R_{\max} \sum_{t=1}^T \sqrt{\exp(2\sigma^2(t-1)) - 1}, \end{aligned}$$

which completes the proof. $\square$

Proof of Lemma D.6. Fix $t \in \{1, \dots, T\}$. Recall the accepted-prefix context measure

$$\bar{d}_{t+1}^{\pi}(c_{t+1}) := d_{t+1}^{\pi}(c_{t+1}) M_t^{\text{pre}}(c_{t+1}).$$

By definition of total variation distance,

$$\begin{aligned} \|\bar{d}_{t+1}^{\pi_\theta} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} &= \frac{1}{2} \sum_{c_{t+1}} |\bar{d}_{t+1}^{\pi_\theta}(c_{t+1}) - \bar{d}_{t+1}^{\pi_{\text{roll}}}(c_{t+1})| \\ &= \frac{1}{2} \sum_{c_{t+1}} M_t^{\text{pre}}(c_{t+1}) |d_{t+1}^{\pi_\theta}(c_{t+1}) - d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})|. \end{aligned}$$

By Lemma 4.9, for every reachable c_{t+1} with $d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) > 0$ one has $d_{t+1}^{\pi_\theta}(c_{t+1}) = d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})W_t$. Substituting yields

$$\begin{aligned} \|\bar{d}_{t+1}^{\pi_\theta} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} &= \frac{1}{2} \sum_{c_{t+1}} d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) M_t^{\text{pre}}(c_{t+1}) |W_t - 1| \\ &= \frac{1}{2} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [M_t^{\text{pre}} |W_t - 1|], \end{aligned}$$

which proves the claim. $\square$

Proof of Corollary D.7. Fix $t \in \{1, \dots, T\}$ and write $\alpha_t := \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}}]$. By Cauchy–Schwarz,

$$\mathbb{E}[M_t^{\text{pre}} | W_t - 1|] \leq \sqrt{\mathbb{E}[M_t^{\text{pre}}]} \sqrt{\mathbb{E}[(W_t - 1)^2]} = \sqrt{\alpha_t} \sqrt{\mathbb{E}[(W_t - 1)^2]}.$$

As in the proof of Corollary D.3, one has $\mathbb{E}[W_t] = 1$ and hence

$$\mathbb{E}[(W_t - 1)^2] = \mathbb{E}[W_t^2] - 1.$$

By Lemma D.2, $\mathbb{E}[W_t^2] \leq \exp(2\sigma^2 t)$, so $\mathbb{E}[(W_t - 1)^2] \leq \exp(2\sigma^2 t) - 1$. Substituting into the previous display gives

$$\mathbb{E}[M_t^{\text{pre}} | W_t - 1|] \leq \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1}.$$

The equivalent TV bound follows immediately from Lemma D.6. $\square$

Proof of Corollary D.8. Define $\alpha_t := \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}}]$ and $H_{\text{eff}} := \sum_{t=1}^T \alpha_t$. Applying Corollary D.7 for each t and summing yields

$$\sum_{t=1}^T \|\bar{d}_{t+1}^{\pi_{\theta}} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \frac{1}{2} \sum_{t=1}^T \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1}.$$

For the compressed effective-horizon form, apply Cauchy–Schwarz to the right-hand side with $b_t := \sqrt{\exp(2\sigma^2 t) - 1}$:

$$\begin{aligned} \sum_{t=1}^T \sqrt{\alpha_t} b_t &\leq \sqrt{\sum_{t=1}^T \alpha_t} \sqrt{\sum_{t=1}^T b_t^2} \\ &= \sqrt{H_{\text{eff}}} \sqrt{\sum_{t=1}^T (\exp(2\sigma^2 t) - 1)}, \end{aligned}$$

which proves the claim. $\square$

D.5 Alternative Trust-Region Masking Variants

This appendix collects several masking variants that are compatible with the ratio setup in the main text (Section 4.1). Throughout, samples are generated by the behavior policy π_{roll} and gradients are computed under the training policy π_{θ} . We write the per-token importance ratio as

$$\rho_t := \frac{\pi_{\theta}(y_t | c_t)}{\pi_{\text{roll}}(y_t | c_t)}.$$

We also reuse the prefix statistic ρ_t^{pre} from Definition 4.4. For thresholds $\epsilon_{\text{low}} \in (0, 1)$ and $\epsilon_{\text{high}} > 0$, define the log-threshold interval

$$[\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})] = [-\tau_-, \tau_+], \quad \tau_+ := \log(1 + \epsilon_{\text{high}}), \quad \tau_- := -\log(1 - \epsilon_{\text{low}}).$$

Variant I: Absorbing Prefix-Average Mask The main text uses a pointwise prefix gate at each position t . A more rejection-style variant enforces an *absorbing* semantics: once a prefix violates the threshold, all subsequent positions are rejected.

$$\widetilde{M}_t^{\text{pre}} := \mathbb{I}[\rho_t^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]], \quad M_t^{\text{abs-pre}} := \prod_{u=1}^t \widetilde{M}_u^{\text{pre}}.$$

Variant II: Prefix-Max Mask / Max-Filtering Max-based trust-region criteria can be expressed via a sample-based proxy $k_2(\rho)$:

$$k_2(\rho) := \frac{1}{2}(\log \rho)^2.$$

The corresponding prefix-max statistic and mask are

$$S_t^{\max} := \max_{1 \leq s \leq t} k_2(\rho_s), \quad M_t^{\text{pre-max}} := \mathbb{I}[S_t^{\max} \leq \delta_{\max}].$$

Equivalently, $M_t^{\text{pre-max}} = 1$ iff $\max_{s \leq t} |\log \rho_s| \leq \sqrt{2\delta_{\max}}$.

Variant III: Prefix-Score Mask / Bound-Driven Score-based masking controls a prefix-level aggregate of per-token ratio deviations. Define a nonnegative per-token deviation

$$a_t := |\rho_t - 1|$$

and deterministic nonnegative weights $(w_t)_{t \geq 1}$ with $\sum_{s \leq t} w_s > 0$. The prefix score is

$$\widetilde{W}_t := \frac{\sum_{s=1}^t w_s a_s}{\sum_{s=1}^t w_s},$$

and the corresponding mask is

$$M_t^{\text{pre-score}} := \mathbb{I}[\widetilde{W}_t \leq \delta_W].$$

A special case is the uniform-weight choice $w_s \equiv 1$, which yields $\widetilde{W}_T = \frac{1}{T} \sum_{t=1}^T |\rho_t - 1|$ at the full horizon.

Variant IV: Prefix Abnormal-Token Rate / TRM-Style TRM [Li et al., 2025] uses per-token divergence estimates (or their sample-based proxies) to decide whether a trajectory lies inside a trust region. We can adapt this idea to a prefix-level gate by counting *abnormal* tokens in the prefix. Concretely, define the abnormal-token indicator

$$B_t := \mathbb{I}[D_{\text{KL}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta}) > \delta_{\text{TRM}}],$$

and the prefix abnormal count and rate

$$N_t := \sum_{s=1}^t B_s, \quad \widehat{p}_t := \frac{N_t}{t}.$$

Given a rate threshold $\xi \in [0, 1]$, the corresponding prefix mask is

$$M_t^{\text{pre-anom}} := \mathbb{I}[\widehat{p}_t \leq \xi].$$

In practice, one may instantiate B_t using TRM's computation method for $D_{\text{KL}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta})$; for example, via the sample-based proxy $k_3(\rho) := \rho - 1 - \log \rho$. Under the mutual-support Assumption 2.3, the ratio definition $\rho_t = \pi_{\theta}/\pi_{\text{roll}}$ gives

$$\mathbb{E}_{\pi_{\text{roll}}}[k_3(\rho_t) \mid c_t] = D_{\text{KL}}(\pi_{\text{roll}}(\cdot \mid c_t) \parallel \pi_{\theta}(\cdot \mid c_t)).$$

The finite-support normalization algebra behind this equality is checked by the Lean theorem `finite_k3_proxy_eq_reverse_kl`; its general conditional-expectation version still requires the probability-space formalization listed in the audit report. The following inverse certificate connects a sample-level k_3 threshold to the inlier condition used below. A threshold on the conditional mean KL does not give this pointwise implication for each sampled token.

Lemma D.9 (A Log-Ratio Interval from the Sample k_3 Proxy). Write $F(z) = e^z - 1 - z$, so $k_3(\rho) = F(\log \rho)$ for $\rho > 0$. For $\tau_-, \tau_+ \geq 0$, if

$$k_3(\rho) \leq \delta \quad \text{and} \quad \delta \leq \min\{F(-\tau_-), F(\tau_+)\},$$

then $-\tau_- \leq \log \rho \leq \tau_+$. In particular, for any $\tau_0 > 0$, the threshold

$$\delta_0(\tau_0) := \min\{e^{-\tau_0} - 1 + \tau_0, e^{\tau_0} - 1 - \tau_0\} > 0$$

ensures that $B_s = \mathbb{I}[k_3(\rho_s) > \delta]$ with $0 \leq \delta \leq \delta_0(\tau_0)$ satisfies $B_s = 0 \Rightarrow |\log \rho_s| \leq \tau_0$.

Proof. F is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, \infty)$, with $F(0) = 0$. If $\log \rho < -\tau_-$ or $\log \rho > \tau_+$, then $F(\log \rho) > \delta$, a contradiction. The same strict monotonicity gives positivity of both terms defining $\delta_0(\tau_0)$. The interval certificate, strict monotonicity, and positive threshold are checked in `formal/K3Inlier.lean`. $\square$

Lemma D.10 (From Abnormal-Token Rate to a Prefix-Average Constraint). Fix a position $t \in \{1, \dots, T\}$ and write $z_s := \log \rho_s$ for $s \leq t$. Assume there exist finite constants $\Lambda_+, \Lambda_- > 0$ and an inlier threshold $\tau_0 \geq 0$ such that

$$-\Lambda_- \leq z_s \leq \Lambda_+, \quad \forall s \leq t.$$

Let $B_s \in \{0, 1\}$ be an abnormal-token indicator satisfying the explicit inlier condition

$$B_s = 0 \implies |z_s| \leq \tau_0, \quad \Lambda_+ \geq \tau_0, \quad \Lambda_- \geq \tau_0.$$

Define its prefix rate by

$$\hat{p}_t := \frac{1}{t} \sum_{s=1}^t B_s,$$

For any threshold $\xi \in [0, 1]$, if $\hat{p}_t \leq \xi$, then the prefix-average log-ratio obeys

$$\frac{1}{t} \sum_{s=1}^t \log \rho_s \in \left[-\tau_0 - \xi(\Lambda_- - \tau_0), \tau_0 + \xi(\Lambda_+ - \tau_0) \right].$$

In particular, choosing log-thresholds (τ_-, τ_+) that satisfy

$$\tau_- \geq \tau_0 + \xi(\Lambda_- - \tau_0), \quad \tau_+ \geq \tau_0 + \xi(\Lambda_+ - \tau_0)$$

ensures $\hat{p}_t \leq \xi$ implies the prefix-average constraint $\frac{1}{t} \sum_{s \leq t} \log \rho_s \in [-\tau_-, \tau_+]$, and hence $\rho_t^{\text{pre}} \in [e^{-\tau_-}, e^{\tau_+}]$.

Proof. Let $N_t := \sum_{s=1}^t B_s$ so that $\hat{p}_t = N_t/t \leq \xi$. For the upper bound, split the sum into inliers and outliers:

$$\sum_{s=1}^t z_s \leq (t - N_t) \tau_0 + N_t \Lambda_+,$$

since inliers satisfy $z_s \leq \tau_0$ and outliers satisfy $z_s \leq \Lambda_+$. Dividing by t and using $N_t/t \leq \xi$ yields

$$\frac{1}{t} \sum_{s=1}^t z_s \leq \tau_0 + \xi(\Lambda_+ - \tau_0).$$

Similarly, using $z_s \geq -\tau_0$ for inliers and $z_s \geq -\Lambda_-$ for outliers gives

$$\frac{1}{t} \sum_{s=1}^t z_s \geq -\tau_0 - \xi(\Lambda_- - \tau_0).$$

The final implication to ρ_t^{pre} follows from $\log \rho_t^{\text{pre}} = \frac{1}{t} \sum_{s \leq t} \log \rho_s$. $\square$

For the sample-based indicator in Lemma D.9, the composition of the inverse certificate with this prefix-rate bound is checked by `k3_abnormal_prefix_average_bound` in `formal/K3Inlier.lean`. This supplies the inlier condition from the actual proxy threshold while retaining the stated bounds $[-\Lambda_-, \Lambda_+]$ on all log-ratios.

E Supporting Identities and Comparison Details

The results in this section support the main method and reward certificate. They collect the population surrogate connection, standard divergence identities, and comparisons with alternative correction granularities.

E.1 Centered Surrogate Equivalence and Error Decomposition

Proof of Proposition 2.8. Write $Q_t(c_t, a) = \mathbb{E}_{\pi_{\text{roll}}}[R \mid c_t, a]$, $V_t(c_t) = \mathbb{E}_{\pi_{\text{roll}}}[R \mid c_t]$, and $\bar{b}(x) = \mathbb{E}[b \mid x]$. Conditional independence of b and policy normalization give

$$\mathbb{E}_{\pi_{\text{roll}}}[(R - b)\rho_t \mid c_t] = \sum_a \pi_\theta(a \mid c_t)(Q_t(c_t, a) - V_t(c_t)) + V_t(c_t) - \bar{b}(x).$$

The last two terms are independent of π_θ . At $\pi_\theta = \pi_{\text{roll}}$ the advantage sum is zero. Multiplying by the predictable mask $m_t(c_t)$, averaging over rollout contexts, and summing over t proves the first equality. The second follows by replacing b by zero and using $\mathbb{E}_{\pi_{\text{roll}}}[R(\rho_t - 1) \mid c_t] = \sum_a \pi_\theta(a \mid c_t)(Q_t(c_t, a) - V_t(c_t))$. On post-EOS padding states the advantage is zero and the ratio is one. $\square$

The finite conditional-law proof, including rollout continuation sums and fixed context/action-mask weights, is checked by `finite_surrogate_equivalence` and `finite_context_surrogate_equivalence` in `formal/SurrogateBridge.lean`; `finite_independent_baseline_average` integrates an explicitly independent baseline sampling law, and `finite_surrogate_derivative_equivalence` transfers the derivative through the constant offset. This gives an exact population connection for the stated finite conditional model; it does not equate individual Monte Carlo estimates. Random-length batch reductions are treated separately in Proposition 5.3 and Section 5.3; a general measure-theoretic interface remains outside this formalization.

The finite-state TV step behind Eq. (5) is the elementary bound $|\mathbb{E}_{d_t^{\pi_\theta}}[g_t] - \mathbb{E}_{d_t^{\pi_{\text{roll}}}}[g_t]| \leq 2\|g_t\|_\infty \|d_t^{\pi_\theta} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}}$; its finite-sum form is checked by the Lean theorem `finite_tv_expectation_bound`, and summing this step over a finite horizon is checked by `finite_error_decomposition_bound` in `formal/ErrorDecomp.lean`. The finite history-tree modules construct the context-visitation laws, conditional token kernels, and corresponding autoregressive identities explicitly. The abstract finite-sum inequality above is checked independently by `finite_error_decomposition_bound`; a general measure-theoretic conditional-expectation interface remains outside the present formalization.

E.2 Cited Supporting Results for IS Correction

This subsection records several standard identities and bounds used in our comparisons of importance sampling (IS) correction mechanisms. All statements are adapted from existing trust-region analyses for LLM-RL (e.g., Qi et al. [2026], Li et al. [2025]). We include them for completeness, with the finite-model Lean proof locations recorded below.

Remark E.1 (TV as an Expectation of the Ratio Residual). Under the mutual-support Assumption 2.3, for any context c_t ,

$$D_{\text{TV}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta}) = \frac{1}{2} \mathbb{E}_{y_t \sim \pi_{\text{roll}}(\cdot|c_t)}[|\rho_t - 1|], \quad \rho_t = \frac{\pi_{\theta}(y_t|c_t)}{\pi_{\text{roll}}(y_t|c_t)}.$$

See, e.g., Qi et al. [2026]. For finite token support, the ratio identity is checked by `tv_ratio_identity_of_reverse_support` in `formal/ProMax.lean`; the normalized autoregressive history-tree instantiation is checked by the path-mass results in `formal/TreeSurrogate.lean` and `formal/PrefixConcentration.lean`. Only the general measurable-space version remains outside the present formalization.

Remark E.2 (Coarse-Graining Lower Bounds). Let $\mathcal{V}$ be the vocabulary and let $\mathcal{C} = \{C_1, \dots, C_m\}$ be any partition of $\mathcal{V}$. For a policy π , define the coarse-grained distribution $\pi^{\mathcal{C}}(\cdot|c_t)$ by $\pi^{\mathcal{C}}(C_j|c_t) := \sum_{v \in C_j} \pi(v|c_t)$. Then for any policies π, π' and any context c_t ,

$$D_{\text{TV}}(\pi(\cdot|c_t) \parallel \pi'(\cdot|c_t)) \geq D_{\text{TV}}(\pi^{\mathcal{C}}(\cdot|c_t) \parallel \pi'^{\mathcal{C}}(\cdot|c_t)),$$

and, if $\pi(\cdot|c_t)$ is absolutely continuous with respect to $\pi'(\cdot|c_t)$,

$$D_{\text{KL}}(\pi(\cdot|c_t) \parallel \pi'(\cdot|c_t)) \geq D_{\text{KL}}(\pi^{\mathcal{C}}(\cdot|c_t) \parallel \pi'^{\mathcal{C}}(\cdot|c_t)).$$

See, e.g., Qi et al. [2026]. For a finite vocabulary and finite partition map, the TV contraction in this remark is checked by `REINFORCEProMax.finite_coarse_tv_lower_bound` in `formal/CoarseGrain.lean`. The KL contraction and its full equality condition, including zero-mass cells, are checked by `finite_coarse_kl_contraction` and `finite_coarse_kl_equality_iff_proportional` in `formal/CoarseKL.lean`; only the stated one-sided absolute continuity is required.

Remark E.3 (Binary TV Lower Bound from a Sampled Token). Fix a context c_t and a token $y_t \in \mathcal{V}$ such that $\pi_{\text{roll}}(y_t|c_t) > 0$. Consider the binary partition $\mathcal{C} = \{\{y_t\}, \mathcal{V} \setminus \{y_t\}\}$. Then

$$D_{\text{TV}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta}) \geq |\pi_{\text{roll}}(y_t|c_t) - \pi_{\theta}(y_t|c_t)| = \pi_{\text{roll}}(y_t|c_t) |\rho_t - 1|.$$

For a finite token support, this binary-partition instance is checked by `binary_tv_lower_bound` in `formal/CoarseGrain.lean`; the arbitrary finite-partition TV and KL statements are verified in that module and `formal/CoarseKL.lean`, respectively.

Remark E.4 (Sequence-Level TV Bound by Expected Token-Level TV). Fix a prompt x , a finite vocabulary, and normalized autoregressive policies that may depend on the complete prefix context. Write $P^{\pi}(\cdot|x)$ for the resulting distribution over length- T responses. Then

$$D_{\text{TV}}(P^{\pi_{\text{roll}}}(\cdot|x) \parallel P^{\pi_{\theta}}(\cdot|x)) \leq \sum_{t=1}^T \mathbb{E}_{c_t \sim d_t^{\pi_{\text{roll}}}}[D_{\text{TV}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta})].$$

Here d_t^{π} is the distribution of the length- $(t-1)$ prefix context before token t (so the initial context is d_1^{π} and the terminal length- T response distribution is d_{T+1}^{π}). The one-step estimate is

$$\|d_{t+1}^{\pi_{\theta}} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \|d_t^{\pi_{\theta}} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}} + \bar{D}_t.$$

Summing from the common initial context proves the displayed bound. The weighting by rollout visitation is retained; a uniform maximum over contexts is not required. Both statements are proved directly for the generated path probabilities by `prefixTV_step_weighted` and `sequenceTV_le_expected_tokenTV` in `formal/TreeTV.lean`. No policy-support condition is needed. Shared absorbing EOS padding is allowed. See also Qi et al. [2026].

Remark E.5 (Coarse-Graining Gap for TV). Fix a context c_t and a partition $\mathcal{C} = \{C_1, \dots, C_m\}$. For policies π, π' , define the cell-wise slack

$$s_j(c_t) := \sum_{v \in C_j} |\pi(v|c_t) - \pi'(v|c_t)| - |\pi(C_j|c_t) - \pi'(C_j|c_t)|.$$

Then $s_j(c_t) \geq 0$ and the gap between the true and coarse-grained TV divergences satisfies

$$D_{\text{TV}}(\pi(\cdot|c_t) \parallel \pi'(\cdot|c_t)) - D_{\text{TV}}(\pi^{\mathcal{C}}(\cdot|c_t) \parallel \pi'^{\mathcal{C}}(\cdot|c_t)) = \frac{1}{2} \sum_{j=1}^m s_j(c_t).$$

Moreover, each slack term is bounded by the total mass of its cell:

$$0 \leq s_j(c_t) \leq \pi(C_j|c_t) + \pi'(C_j|c_t).$$

See, e.g., Qi et al. [2026]. For finite supports, the displayed slack decomposition, its non-negativity, and the cell-mass bound are checked by `finite_tv_gap_slack_decomposition`, `finite_tv_gap_slack_nonneg`, and `finite_tv_gap_slack_mass_bound` in `formal/CoarseGrain.lean`.

Remark E.6 (Equality Condition for Coarse-Grained KL). Under the absolute continuity condition in Remark E.2, equality in the KL coarse-graining inequality holds if and only if, for every cell C_j with $\pi(C_j|c_t) > 0$, there is a constant $\lambda_j > 0$ such that

$$\pi(v|c_t) = \lambda_j \pi'(v|c_t) \quad \text{for every } v \in C_j.$$

Equivalently, the two conditional distributions inside each positive-mass cell agree after normalization. The condition must include tokens with $\pi(v|c_t) = 0$ but $\pi'(v|c_t) > 0$; restricting the ratio test to the positive support of π is insufficient under one-sided absolute continuity. For arbitrary finite cells, including zero-probability tokens, the full partition characterization is checked by `finite_coarse_kl_equality_iff_proportional` in `formal/CoarseKL.lean`. The proportionality constant is necessarily $\lambda_j = \pi(C_j|c_t)/\pi'(C_j|c_t)$. Cells with $\pi(C_j|c_t) = 0$ contribute zero to both KL expressions and impose no condition on π' . The concrete two-token counterexample `coarse_grain_kl_positive_support_condition_counterexample` in `formal/KLEquality.lean` records why the weaker wording fails.

Definition E.7 (A Finite-Horizon Surrogate for Sequence Rewards). For this finite-horizon statement, fix a deterministic horizon T . If responses terminate at different lengths, we pad them with an absorbing end-of-sequence token and include the corresponding action mask. The post-termination transition is deterministic and shared by both policies (hence its ratio is 1), so the identities below apply to the padded horizon without adding spurious likelihood-ratio terms. Define the rollout-based, sequence-reward surrogate

$$\mathcal{L}_{\pi_{\text{roll}}}^{\text{LLM}}(\pi_{\theta}) := \mathbb{E}_{x \sim P(x), y \sim P^{\pi_{\text{roll}}}(\cdot|x)} \left[R(x, y) \cdot \sum_{t=1}^T (\rho_t - 1) \right],$$

and the remainder term

$$\Delta^{\text{LLM}}(\pi_{\text{roll}}, \pi_{\theta}) := \mathbb{E}_{x \sim P(x), y \sim P^{\pi_{\text{roll}}}(\cdot|x)} \left[R(x, y) \cdot \sum_{t=1}^T (\rho_t - 1) \left(1 - \prod_{j=t+1}^T \rho_j \right) \right],$$

where $\rho_t = \pi_{\theta}(y_t|c_t)/\pi_{\text{roll}}(y_t|c_t)$ and $P^{\pi_{\text{roll}}}(\cdot|x)$ denotes the rollout trajectory distribution. See, e.g., Qi et al. [2026].

Interpretation of the surrogate–remainder split. For a fixed prompt, write $p(y) := P^{\pi_{\text{roll}}}(y|x)$ and $q(y) := P^{\pi_{\theta}}(y|x)$. Under the finite-horizon mutual-support convention, $q(y)/p(y) = \prod_{t=1}^T \rho_t(y)$. Thus the left-hand side of the performance difference identity is $\sum_y q(y)R(x, y) - \sum_y p(y)R(x, y)$, the change in expected sequence reward. The product telescoping identity gives

$$\prod_{t=1}^T \rho_t - 1 = \sum_{t=1}^T (\rho_t - 1) \prod_{j=t+1}^T \rho_j = \underbrace{\sum_{t=1}^T (\rho_t - 1)}_{\text{Surrogate}(y)} - \underbrace{\sum_{t=1}^T (\rho_t - 1) \left(1 - \prod_{j=t+1}^T \rho_j\right)}_{\text{Remainder}(y)}.$$

Multiplying by $p(y)R(x, y)$ and summing over trajectories yields

$$\sum_y q(y)R(x, y) - \sum_y p(y)R(x, y) = \sum_y p(y)R(x, y) \text{Surrogate}(y) - \sum_y p(y)R(x, y) \text{Remainder}(y).$$

The surrogate is therefore the additive, rollout-estimable part, while the remainder records the interaction with ratios at later positions. This is an exact finite-horizon algebraic identity, not a claim that the remainder is always nonnegative or negligible. It vanishes for $T = 1$; for longer responses it is small only under additional policy-coupling or trust-region conditions. By Proposition 2.8, this sequence-reward surrogate equals the standard per-step surrogate, and also equals the unclipped trajectory-level surrogate after subtracting its rollout-policy value. Independent baselines do not break this equality. The constructed finite-history-tree version of this identity is checked by `trajectory_surrogate_remainder_identity` in `formal/TreeSurrogate.lean`; the abstract finite-support algebra is checked by `finite_horizon_surrogate_identity`. The additional effects of the actual ProMax gate, normalization, clipping, and batch reduction are identified separately in Proposition 5.3.

Remark E.8 (A Finite-Horizon Performance Difference Bound). Under the mutual-support Assumption 2.3 and the fixed/EOS-padded horizon convention in Definition E.7 (with finite-length Assumption 2.5), one has the exact identity

$$J(\pi_{\theta}) - J(\pi_{\text{roll}}) = \mathcal{L}_{\pi_{\text{roll}}}^{\text{LLM}}(\pi_{\theta}) - \Delta^{\text{LLM}}(\pi_{\text{roll}}, \pi_{\theta}).$$

This display is conditional on the shown expectations being well-defined and on the conditional change-of-measure identity being integrable under the prompt law $P(x)$. Here the Lean-certified statement is for a finite-vocabulary, normalized autoregressive history tree, pointwise in a fixed prompt. It extends to a finite empirical prompt distribution by a finite weighted sum; an arbitrary prompt measure requires the usual integrability and change-of-measure hypotheses in addition to the finite conditional model. The displayed identity is the finite-horizon change-of-measure statement; its telescoping proof uses mutual support so that $\mathbb{E}_{\pi_{\text{roll}}}[R \prod_{s=1}^T \rho_s] = \mathbb{E}_{\pi_{\theta}}[R]$. For each fixed prompt, a finite vocabulary and a uniform bound $D_{\text{TV}}^{\text{tok}}(c_t) \leq \epsilon$ at every history, the finite autoregressive-tree recursion gives the following quadratic estimate without an additional ratio-coupling premise:

$$|\Delta^{\text{LLM}}(\pi_{\text{roll}}, \pi_{\theta})| \leq 2R_{\text{max}} T(T-1) \epsilon^2,$$

This estimate is checked by `surrogate_error_quadratic` in `formal/AutoregressiveTree.lean`; the proof allows history-dependent policies and uses rollout occupancy for the surrogate. The same finite-tree recursion also gives the expected-sum-TV estimate

$$|\Delta^{\text{LLM}}(\pi_{\text{roll}}, \pi_{\theta})| \leq 4R_{\text{max}} \sum_{t=1}^T \bar{D}_t.$$

Consequently, whenever the quadratic estimate holds, and also whenever the separate linear estimate holds,

$$J(\pi_\theta) - J(\pi_{\text{roll}}) \geq \mathcal{L}_{\pi_{\text{roll}}}^{\text{LLM}}(\pi_\theta) - \min\left(2R_{\max} T(T-1) \epsilon^2, 4R_{\max} \sum_{t=1}^T \bar{D}_t\right).$$

Both estimates are formalized for the finite autoregressive tree under the stated bounded-reward and policy-normalization conditions, and the finite prompt mixture is handled by linearity. The exact change-of-measure identity remains valid independently of either bound. Theorem 2.10 further supplies the complete position-weighted Adaptive bound on this same remainder. The random-length reduction identities and finite sampling margins are handled separately in Proposition 5.3 and Section 5.3.

E.3 Comparison with Other Advantage Estimation Methods

We compare REINFORCE Pro Max with existing critic-free LLM-RL methods along four dimensions: baseline type, advantage normalization strategy, IS correction granularity, and whether the IS correction preserves the causal structure of autoregressive generation. We organize the comparison into two parts: (i) baseline and normalization choices (this subsection) and (ii) IS correction mechanisms (the next subsection). These dimensions correspond to the key design choices that affect the two factors in the error decomposition (Equation (5)): the advantage factor (targeted by baseline and normalization) and the context shift factor (targeted by IS correction).

Baseline and normalization. REINFORCE++ [Hu et al., 2025] and GRPO [Shao et al., 2024] include the current reward in their group mean baselines. RLOO [Ahmadian et al., 2024] supplies a baseline conditionally independent of each sample’s own response given the prompt, with the exact $(n-1)/n$ relation to the mean-baseline estimator described in Appendix C.1. Normalization is a separate choice: a common positive scale preserves relative magnitudes, while centering can change signs. ProMax’s asymmetric scales solve the token-weighted mean and second-moment constraints while preserving advantage signs, including when response lengths differ.

E.4 Comparison with Other Importance Sampling Correction Methods

We adopt a trust-region masking viewpoint: when mismatch between π_{roll} and π_θ is large, one should attenuate or reject unreliable gradient contributions. Existing methods differ mainly in the granularity of this attenuation (see Section 2.6 and Table 1 for a unified notation).

Single-sample ratio statistics vs. distributional drift. Although the per-token ratio residual $|\rho_t - 1|$ is a single-sample quantity, it admits a clean divergence interpretation. Conditional on the context c_t , Remark E.1 implies that $|\rho_t - 1|$ has expectation $2D_{\text{TV}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_\theta)$ under $y_t \sim \pi_{\text{roll}}(\cdot|c_t)$, so thresholding ρ_t can be viewed as thresholding a noisy proxy for a TV-based trust region. At the same time, the binary lower bound in Remark E.3 shows that the effect of a large ratio depends on the behavior-policy mass at the sampled token: for low-probability tokens, even a large ρ_t can correspond to a small absolute probability shift. Finally, policy drift at the level of full response distributions admits an additive upper bound by the expected token-level TVs along the trajectory (Remark E.4), which motivates per-position and prefix-structured filtering mechanisms.

Finite-horizon remainder bounds. For a finite vocabulary, the autoregressive-tree recursion gives both a quadratic max-TV bound and a linear expected-sum-TV bound for the higher-order term in the rollout-based performance-difference identity (recorded in Remark E.8). These bounds use the rollout occupancy for the surrogate and do not require a ratio bound. The linear expression $4R_{\max} \sum_{t=1}^T \bar{D}_t$ makes the per-position structure explicit and clarifies why long-horizon stability can benefit from controlling divergence along the trajectory rather than only

via a single global trust-region statistic. The finite-tree instances of both inequalities are Lean-verified in ‘formal/AutoregressiveTree.lean’; the finite prompt-mixture extension is checked by `family_remainder_quadratic` and `family_remainder_linear` in `formal/PromptMixture.lean`. The EOS/action-mask interface is checked for a context-predictable absorbing continuation by `trajectory_masked_surrogate_eq` in `formal/TreeSurrogate.lean`. The full Adaptive refinement is checked by `trajectory_remainder_adaptive` and `family_remainder_adaptive` in `formal/AdaptiveBound.lean`. It retains the expected local TV at each position and multiplies it by a bound on the remaining continuation drift (Theorem 2.10). Random-length reduction changes the estimator’s population target as quantified in Section 5.3.

Coarse-grained divergence proxies as controllable trust-region gates. Full next-token divergences (TV/KL) are expensive to compute over the entire vocabulary, motivating coarse-grained approximations such as Binary or Top- K schemes. Formally, divergences computed on any partition are lower bounds on the true divergences (Remark E.2). Therefore, a gate that masks whenever a coarse-grained proxy exceeds a fixed threshold is *conservative*: it does not reject updates that are provably within the trust region defined by the corresponding full-vocabulary divergence. The remaining question is when such proxies may under-detect drift.

For TV divergence, the coarse-graining gap admits an explicit slack decomposition and a mass-based upper bound (Remark E.5). In particular, for Top- K style partitions that isolate head tokens and aggregate the tail into a single “other” cell, the approximation error is controlled by the tail probability mass. When this aggregated tail mass is small, Top- K TV can serve as a quantitatively controlled proxy for per-position divergence gating, and such per-position proxies can be aggregated along prefixes to yield a prefix-structured gate that respects the causal order of generation. For KL divergence, equality of coarse-grained KL holds exactly when, in every cell with positive p -mass, the two distributions are proportional throughout that cell (Remark E.6). The finite identity in `formal/CoarseKL.lean` expresses the full-minus-coarse KL as a sum of nonnegative cell-wise Gibbs gaps; consequently, a quantitative near-equality claim should be stated as small total Gibbs gap unless an additional ratio metric is specified. This identifies when KL computed on a reduced support faithfully represents the full-vocabulary KL.

As a concrete causal instantiation, Appendix D.5 includes a prefix abnormal-token rate gate (Variant IV) based on TRM-style per-token divergence proxies. Under mild boundedness conditions on per-token log-ratios, controlling the prefix abnormal-token rate yields a deterministic prefix-average log-ratio bound (Lemma D.10), making the coarse-grained gate a controllable proxy for prefix-level ratio filtering.

Token-level truncation and masking. Under this view, TIS and IcePop are token-level analogues of trust-region masking: they operate directly on the per-token ratio ρ_t . TIS clamps ρ_t to a fixed interval $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$, retaining a biased but bounded weight at every position. IcePop instead masks out tokens whose ratio falls outside the interval, removing extreme sampled coefficients at the cost of possible selection bias and discarded gradient signal. Both apply a local decision at each position and therefore do not account for the downstream context shift caused by an early deviation (Remark 4.1).

The implementation of `seq-mask-tis` and `reinforce_pro` uses a different separation: the sequence or prefix statistic is a detached gate, while the retained mismatch coefficient is the detached, *unclipped* token weight $\exp(\log \pi_{\text{old}} - \log \pi_{\text{roll}})$. Thus these modes should not be described as TIS clipping. The prefix gate controls where the coefficient is applied; it does not by itself bound the magnitude of an accepted coefficient, so the threshold choice and numerical stability remain part of the method’s implementation contract.

Sequence-level aggregation and gating. Sequence-level corrections aggregate ratios over the whole trajectory. GSPO [Zheng et al., 2025] clips the length-normalized sequence likelihood ratio

ρ^{seq} (a geometric mean of token ratios) and applies a single coefficient to all tokens, which aligns with sequence-level rewards but loses positional information and can dilute isolated deviations at long horizons (Remark 4.2). TRM [Li et al., 2025] applies a binary sequence gate (e.g., a max-KL trust-region criterion) and rejects entire trajectories, which is theoretically principled but sacrifices per-position granularity.

Prefix-level (causal) filtering. Token- and sequence-level mechanisms leave a gap for long-horizon generation: token-level methods ignore how an early deviation propagates to future contexts (Remark 4.1), while sequence-level gates share a single decision across all positions. Our prefix mask M_t^{pre} (defined in Section 4.3) provides a causal, per-position gate constructed from prefix ratio statistics. Compared with TRM’s sequence-level gate $M^{\text{TRM}}(x, y)$, it preserves positional granularity by masking only the token-level loss terms whose prefixes violate the threshold, while still respecting the autoregressive order. The resulting filtering can propagate early-deviation rejection and detect sustained same-side threshold violations (Theorem 4.7).

REINFORCE Pro Max. Taken together with the advantage-estimation choices discussed in this appendix, REINFORCE Pro Max targets all four dimensions discussed in this appendix. The RLOO baseline provides statistical independence (Remark C.1); adaptive normalization preserves advantage sign and, in its nondegenerate ideal form, enforces zero empirical mean and unit empirical variance on non-zero tokens, with explicit moment residuals for the implemented safeguards (Equation (8) and Proposition C.6, Section 3.2); and prefix cumulative IS provides causal, per-position masking with conditional early/late comparisons and sustained-violation guarantees (Theorem 4.7). Their combined effect on the actual batch objective is given by Proposition 5.3: empirical advantage reweighting and prefix filtering contribute separate, explicitly measurable terms.